First Progress Review

Chapter 2: Literature Review

Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings

KH Viraj Sameera · PhD in Computer Science (full-time, 36 months, started 1 October 2026)

Department of Computer Science, Faculty of Applied Sciences, University of Sri Jayewardenepura

Supervisors: Dr. WMKS Ilmini (Principal) · Dr. WGCW Kumara · Dr. AM Deepal K Attanayake (Clinical)

Draft version 1 · 10 November 2026

---

2. Literature Review

This chapter reviews the literature on which the framework builds and identifies, for each objective, the point at which existing work stops. It updates the review in the accepted proposal with the Research Paper Bank assembled during the first months of candidature. The chapter first describes how the literature was searched, selected and recorded (Section 2.1), then reviews the clinical and data context (Section 2.2) and each objective in turn: multi-agent orchestration (O1, Section 2.3), uncertainty and deferral (O2, Section 2.4), data-efficient, multi-source and federated models (O3, Section 2.5) and clinical evaluation of human-AI teams (O4, Section 2.6). Section 2.7 draws out themes that cut across the objectives, Section 2.8 synthesises what is established and what remains open, and Section 2.9 gives evidence tables for each objective.

2.1 Review Method

2.1.1 Search and selection

The review is a structured narrative review organised by objective. Searches were run in PubMed and PubMed Central, arXiv, Google Scholar, IEEE Xplore and the ACM Digital Library, and the proceedings of MICCAI, NeurIPS, ICML, ICLR, CVPR and AAAI were browsed directly. Search strings combined a domain term (brain tumour, glioma, neuro-oncology, medical imaging, computational pathology, clinical decision support) with a

method term for each objective: multi-agent, orchestrator, agentic, tool use (O1); calibration, uncertainty, selective prediction, learning to defer, conformal prediction, abstention, out-of-distribution (O2); self-supervised, transfer learning, missing modality, multiple-instance learning, foundation model, federated learning, differential privacy (O3); and human-AI collaboration, reader study, automation bias, workload, usability, trust, reporting guideline (O4). Reference lists of key papers were followed backwards and forwards.

Inclusion criteria were: (i) relevance to at least one objective; (ii) publication from 2021 onwards, except for seminal methods that define a field (for example temperature scaling, deep ensembles, selective classification and learning to defer); (iii) peer-reviewed venue or a preprint from an established group, with predatory journals excluded; (iv) an open-access full text, so that every paper can be read and re-checked; and (v) enough methodological detail to extract the study design, data and evaluation. As advised at the supervision meeting of 5 November 2026, priority was given to papers that describe their methodology in detail, to studies with external validation, and to studies reporting negative results.

The collection grew in two rounds. By 9 November 2026, 314 papers had been collected and their methodology extracted. A second round on 10 November 2026 added 313 open-access papers and removed 68 that had no open-access full text, were duplicates, or no longer matched the objectives, giving the current bank of **559 papers** (Table 2.1). Of these, 524 (94%) were published in 2021 or later, 299 appeared in journals or conference proceedings and 260 are arXiv or medRxiv preprints; 13 were designated *base papers* whose design the framework adopts or extends directly.

Table 2.1: The Research Paper Bank by objective (as of 10 November 2026).

Objective	Theme	Papers	Base papers
O1	Multi-agent architecture, orchestration, uncertainty hand-off and audit trail	143	4
O2	Calibrated uncertainty and escalation (when to defer to the clinician)	109	3
O3	Data-efficient MRI and pathology models, fusion and federated learning	146	4
O4	Clinical evaluation: performance, escalation appropriateness, workload, usability, trust and safety	161	2
Total		559	13

Figure 2.1 summarises how the collection was assembled. Because the bank was built for a doctoral programme rather than a single systematic review question, it is not a PRISMA systematic review; the flow diagram is given so that the reader can see how many papers

entered, how many were removed and why, and how the final set is divided between the objectives.

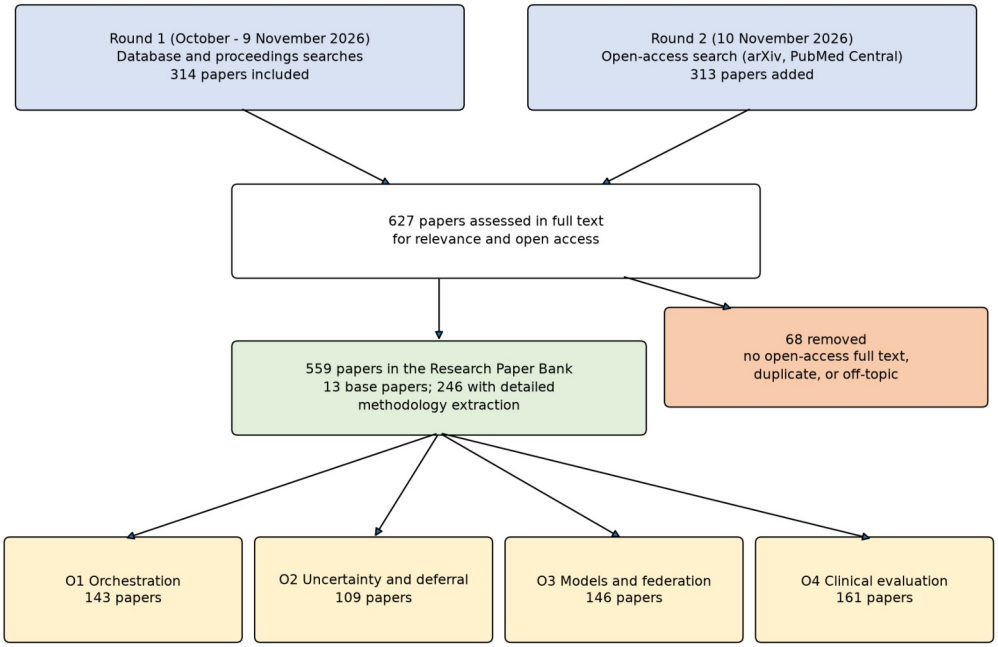


Figure 2.1: Flow of papers into the Research Paper Bank (two collection rounds, October-November 2026).

Table 2.2: Search terms used for each objective (combined with the domain terms brain tumour, glioma, neuro-oncology, medical imaging, computational pathology or clinical decision support).

Objective	Method terms
O1	multi-agent; orchestrator; agentic; LLM agent; tool use; ReAct; tumour board; clinical workflow; audit trail; accountability; model cards
O2	uncertainty quantification; calibration; temperature scaling; Monte Carlo dropout; deep ensemble; selective prediction; reject option; learning to defer; triage; conformal prediction; risk control; out-of-distribution; failure detection; abstention
O3	segmentation; nnU-Net; transformer; missing modality; self-supervised; transfer learning; semi-supervised; domain generalisation; foundation model; multiple-instance learning; whole-slide image; fusion; radiogenomics; federated learning; differential privacy; harmonisation
O4	reader study; human-AI collaboration; automation bias; reliance; explainability; workload; usability; acceptance; trust; reporting guideline; randomised trial; low- and middle-income countries; neurosurgery; tractography; surgical planning

2.1.2 Appraisal of the evidence

Each study was appraised for the features that matter most for this research rather than with a single scoring instrument. Five questions were asked: (i) was the evaluation internal only, or was there an external or prospective test; (ii) were the data clinical or benchmark data, and from which populations; (iii) was uncertainty or confidence reported, and was it calibrated; (iv) was a human expert part of the evaluation, and how; and (v) were code and

data available. These questions recur in the evidence tables at the end of this chapter (Tables 2.6 to 2.9). Two features of the evidence base should be noted at the outset. First, 260 of the 559 papers are preprints, mostly in the fast-moving areas of language-model agents and conformal prediction; their findings are treated as provisional and are cited for ideas and design rather than as established results. Second, very few studies come from South Asia, and none from Sri Lanka, which is itself a finding that motivates the local validation in Objective 3.

2.1.3 Data extraction

For every paper a structured record was completed in a review sheet per objective: aim, method, main result, tools, research gap, relation to this study and limitations. For papers with full methodological reporting (246 so far), a detailed extraction was added covering study design and institutions, datasets and sample sizes, preprocessing, model architecture, training settings, baselines, evaluation metrics, statistical tests and availability of code and data. These records are published in the Research Paper Bank on the progress website, where each paper links to its source. The extraction directly informs the methodology in Chapter 3: baselines, metrics and statistical tests were chosen from what comparable studies used.

2.2 Clinical and Data Context

Brain and nervous-system tumours are uncommon but carry disproportionate mortality (Price et al., 2024). Sri Lanka sees roughly 700 new cases and 587 deaths a year (International Agency for Research on Cancer, 2024; National Cancer Control Programme, 2024). The most consequential recent change for computational work is the 2021 WHO classification, which integrates histology with molecular markers to define tumour type and grade (Louis et al., 2021). A model trained on a pre-2021 label is therefore not predicting the same target as one aligned to the current classification, and single-source systems can only support a diagnosis, never produce one of record. Surgery is constrained by the trade-off between extent of resection and preservation of eloquent function (Sanai & Berger, 2008; Molinaro et al., 2020).

The data on which most published models are built differ from the Sri Lankan setting. BraTS 2021 provided about 2,000 multi-institutional glioma cases, but mostly from Western centres and in pre-processed, co-registered form (Baid et al., 2021). BraTS-Africa was created to test whether such models work on routine-care MRI from Sub-Saharan

Africa, acquired on older or low-field scanners with poorer contrast and resolution in patients who present later (Adewole et al., 2026); the winning team reached a lesion-wise Dice of 0.86 only after segmentation-aware augmentation and ensembling of three architectures on the small local dataset (Ankomah et al., 2025). Low- and middle-income settings also face shortages of scanners, specialists and pathology services, delayed presentation and poor incidence data (Yevudza et al., 2024), and AI evaluations there are few and rarely real-world (Ciecierski-Holmes et al., 2022). These observations motivate the data-efficient, locally validated and federated approach of O3.

2.2.1 Tumour types, imaging and the integrated diagnosis

The tumours the registry will see most often are gliomas, meningiomas and brain metastases, with a smaller number of lymphomas, paediatric tumours and other entities. Gliomas are diagnosed under the 2021 WHO classification by combining histology with molecular markers, chiefly IDH mutation, 1p/19q co-deletion, ATRX loss and, for glioblastoma, features such as TERT promoter mutation; MGMT promoter methylation informs treatment (Louis et al., 2021). Standard pre-operative MRI includes T1-weighted, contrast-enhanced T1, T2-weighted and FLAIR sequences, often with diffusion and perfusion imaging and, for surgical planning, diffusion tensor imaging for tractography. In routine practice, sequences are frequently missing or acquired with different protocols, which is why robustness to missing sequences is a recurring theme below.

Imaging-only prediction of molecular status is an active field but not a substitute for tissue. A meta-analysis of 104 studies found pooled test-cohort sensitivity of 0.80 and specificity of 0.85 for IDH, and lower figures for 1p/19q (Farahani et al., 2026). A careful evaluation of MGMT promoter methylation prediction from MRI found all models close to chance (AUC 0.49-0.63) (Saeed et al., 2023), a negative result that cautions against claims of 'virtual biopsy'. Public datasets that link MRI to molecular markers, such as the UT Southwestern glioma dataset, are now appearing (Reddy et al., 2026), but the definitive diagnosis remains integrated and multi-source, which is why the framework places the diagnosis at a dedicated fusion agent with a clinician validating every diagnosis of record.

2.2.2 Public datasets available for development

Table 2.3 lists the public datasets most relevant to development and external testing. They allow every method to be built and benchmarked before local data are available, but all

differ from Sri Lankan routine care in scanners, protocols, pre-processing and case mix (Chan et al., 2025).

Table 2.3: Public datasets relevant to this research.

Dataset	Content	Use in this study
BraTS 2021 (Baid et al., 2021)	About 2,000 multi-institutional glioma MRI cases (T1, T1c, T2, FLAIR) with expert segmentations; MGMT labels for a subset	Training and benchmarking of MRI segmentation (WP2)
BraTS-Africa (Adewole et al., 2026)	Routine-care glioma MRI from Sub-Saharan Africa, often on older or low-field scanners	External test under scanner and population shift
BraTS-PEDs (Kazerooni et al., 2024)	Paediatric brain-tumour MRI with segmentations	Robustness check for paediatric cases
BraTS-MEN-RT (LaBella et al., 2026)	750 meningioma radiotherapy-planning MRIs with gross tumour volume labels	Meningioma segmentation baseline
FeTS (Zenk et al., 2025; Linardos et al., 2025)	Glioma data partitioned by institution for federated benchmarking	Simulated federation (WP5)
UTSW glioma (Reddy et al., 2026)	MRI with molecular markers and segmentations	Imaging-marker experiments
TCGA glioma slides (Chen et al., 2022b; Wang et al., 2023)	Whole-slide H&E images with genomic data for adult gliomas	Pathology baselines and fusion (WP4)
CoRe-BT (Rivera et al., 2026)	Multimodal radiology-pathology-text benchmark for brain-tumour typing	Fusion benchmarking

2.2.3 Neuro-oncology care in low-resource settings

Evidence from comparable health systems describes the conditions the framework must work in. Neuro-oncology in Sub-Saharan Africa is limited by scarce scanners, neurosurgeons, neuropathology and radiotherapy, delayed presentation and poor incidence data (Yevudza et al., 2024). A systematic review of AI in low- and middle-income countries found only ten studies that met its criteria, half without details of algorithms or training data, and repeated problems of reliability, workflow fit and usability (Ciecierski-Holmes et al., 2022). Deployment studies in rural clinics found AI decision support misaligned with short consultations and heavy workloads (Wang et al., 2021a). Where AI has been evaluated carefully in such settings, results are encouraging: AI grading of diabetic retinopathy in a randomised trial raised referral adherence from 39.6% to 51.5% (Mathenge et al., 2022), and AI auto-segmentation reduced radiotherapy contouring from about an hour to ten minutes of review (Kibudde et al., 2024). Reviews of AI for diabetic retinopathy and tuberculosis in these countries point to reduced cost and workload but uneven reporting (Onyeze et al., 2025; Vijayan et al., 2026), and a responsible-AI framework for low-resource settings emphasises governance and local ownership (Alzyood, 2026). These studies show that benefit in low-resource care depends on fitting the workflow and on local validation, both of which this study builds in.

2.3 Multi-Agent Orchestration in Clinical AI (O1)

2.3.1 From single models to agents

Agentic AI combines a reasoning core with tools, memory and the ability to act over several steps. Foundational work showed that interleaving reasoning with tool calls improves reliability (Yao et al., 2023), and general frameworks such as AutoGen and MetaGPT made multi-agent conversation and role assignment easy to build (Wu et al., 2023; Hong et al., 2023). A recent architecture for healthcare agents describes four components (planning, action, reflection and memory) and proposes evaluation on recommendation accuracy, task execution, adaptation and explainability (Liu et al., 2025a).

2.3.2 Medical multi-agent systems

Medical multi-agent systems fall into three groups. The first simulates a multidisciplinary team with language-model agents. MDAgents lets a moderator agent rate case complexity and assign a single agent, a team or an integrated care team, achieving the best accuracy on seven of ten medical benchmarks (Kim et al., 2024); MedAgents and the Multi-Agent Conversation framework showed similar gains from structured discussion, the latter on 302 rare-disease cases (Tang et al., 2024; Chen et al., 2025a). The second orchestrates specialised tools. An oncology agent that called vision transformers, MedSAM, OncoKB and guideline retrieval raised decision accuracy on 20 multimodal cases from about 30% to 87% (Ferber et al., 2025); MedRAX and CXRAgent coordinate chest X-ray tools, CXRAgent adding an evidence validator that rates each tool claim as high, medium or low confidence (Fallahpour et al., 2025; Lou et al., 2026). The third is closest to this study: orchestrated clinical workflows with a defined pathway. An orchestrator with seven red-flag agents supported secondary-headache diagnosis in primary care (Wu et al., 2026b); an emergency triage system used a weighted confidence score below 0.7 to raise acuity, keeping the clinician as final decision-maker (Sharma et al., 2026); a multi-agent system automated BT-RADS scoring for glioma follow-up MRI with 76.0% agreement with an expert neuroradiologist, against 57.5% for the original clinical reports (Jabal et al., 2026); and TumorBoard combined radiology, neuropathology, molecular and guideline agents with a claim-evidence ledger and a safety governor that deferred 84.2% of unsafe cases under evidence deletion (Liu et al., 2026c).

At the most autonomous end, an agent operating in a simulated, FHIR-based electronic health record managed 574 emergency cases with 88.9% diagnostic accuracy and

outperformed board-certified physicians on matched cases (Ferber et al., 2026). Under clinical-scale workloads, orchestrated multi-agent designs sustained accuracy far better than a single agent handling all tasks (65.3% versus 16.6% at 80 concurrent tasks) and were cheaper to run (Klang et al., 2025).

2.3.3 Failure, consensus and accountability

The same literature documents how such systems fail. An analysis of 1,642 execution traces from seven frameworks found failure rates of 41% to 86.7%, caused by system design (44.2%), inter-agent misalignment (32.3%) and weak verification (23.5%) (Cemri et al., 2025). An audit of medical multi-agent systems over 14,400 cases found that unsupported observations entered 16.6% of cases and propagated downstream, that agents simply repeated their initial view in 98.4% of discussions, and that final answers showed authority bias in 28.8% (Zhu et al., 2025a). Optimising each component separately does not optimise the system: a best-of-breed system with excellent information-gathering scored lower diagnostic accuracy (67.7%) than an integrated multi-agent design (77.4%) (Bedi et al., 2025). Language models alone also remain unreliable for clinical decisions, misreading laboratory results and failing to follow guidelines (Hager et al., 2024). Governance work provides documentation and audit frameworks—model cards (Mitchell et al., 2019), internal algorithmic auditing (Raji et al., 2020) and the FUTURE-AI consensus (Lekadir et al., 2025)—but these are organisational processes rather than run-time mechanisms.

2.3.4 Foundations of language-model agents and collaboration

Several general mechanisms underpin current medical agents. Communicative agents with assigned roles (CAMEL) produced outputs preferred by human evaluators in 76% of comparisons (Li et al., 2023); multi-agent debate improved factual accuracy and reasoning over a single model, for example from 77.0% to 85.0% on grade-school mathematics (Du et al., 2023); and a mixture of open-source agents with an aggregator exceeded a frontier model on an instruction-following benchmark (Wang et al., 2024b). Surveys of language-model-based autonomous agents and multi-agent systems map these designs and identify the recurrent problems of hallucination, coordination and evaluation (Wang et al., 2024c; Guo et al., 2024a; Xi et al., 2025). A review of trust, risk and security management for agentic multi-agent systems, covering 180 primary studies, argues that governance,

explainability, privacy and model operations must be designed in from the start (Raza et al., 2026).

2.3.5 Benchmarks and simulated clinical environments

Because real clinical deployment is rare, agents are mostly evaluated in simulations. Agent Hospital simulated an entire hospital in which doctor agents improved by treating simulated patients (Li et al., 2024b). AgentClinic showed that models that score highly on static examination questions lose much of their accuracy when information must be gathered sequentially from a simulated patient (Schmidgall et al., 2024), and AI Hospital and ClinicalLab reached similar conclusions for multi-department diagnosis (Fan et al., 2024; Yan et al., 2025). In a virtual electronic health record, the best model completed only 69.7% of clinical tasks, with action tasks harder than queries (Jiang et al., 2025). MedAgentBoard compared multi-agent collaboration with conventional methods across medical tasks and found that collaboration does not always help (Zhu et al., 2025b). An evaluation of language models on 2,400 real abdominal cases concluded that they were not ready for autonomous decisions, misreading laboratory values and ignoring guidelines (Hager et al., 2024). For neurology, a multi-agent system with retrieval reached 94.6% accuracy on clinical-reasoning questions, while retrieval alone reduced the accuracy of one medical model from 52.9% to 41.2% (Sorka et al., 2025), showing that design choices can harm as well as help.

2.3.6 Imaging and pathology agents

Agents that coordinate imaging tools are most relevant to the MRI and Pathology Agents of this framework. MMedAgent learned to select medical tools across modalities (Li et al., 2024a); MedAgent-Pro followed an evidence-based workflow that raised balanced accuracy for glaucoma from 56.4% (GPT-4o alone) to 90.4% (Wang et al., 2025b); and VILA-M3 combined a vision-language model with expert models to outperform Med-Gemini (Nath et al., 2025). For brain imaging specifically, an agentic language model performed training-free neuroradiological image analysis by calling tools (Erdur et al., 2026), and LLM assistance raised junior radiologists' AUC for pre-operative glioma classification from 0.77 to 0.83 (Li et al., 2025a). In a usability study of LLM-assisted brain MRI differential diagnosis, assisted readers were correct in 61.4% of cases against 46.5% unassisted (Kim et al., 2025b). In pathology, multi-agent systems analyse whole-slide images with task allocation and verification (Lyu et al., 2025), a multi-agent copilot

supported evidence-based reasoning in human pathology (Weishaupt et al., 2025), and a language model with retrieval diagnosed neuropathological tumour subtypes correctly in 90% of tested cases, but not without retrieval (Hewitt et al., 2024). A review of agentic systems in computational pathology concluded that most are retrospective prototypes at low autonomy levels with no proven clinical benefit and no regulatory clearance (Lu et al., 2026).

2.3.7 Tumour boards and neuro-oncology decision support

Tumour boards are the clinical analogue of a multi-agent system: specialists contribute evidence and a decision emerges. Language models have been compared with breast, sarcoma, melanoma and molecular tumour boards with mixed agreement (Sorin et al., 2023; Dehdab et al., 2026; Babic et al., 2026; Schmutz et al., 2025), and in neuro-oncology radiotherapy a prospective comparison found that the model omitted indicated treatments in straightforward cases and suggested unnecessary therapy in borderline ones (Tini et al., 2025). A review of language models as decision aids in neuro-oncology highlights shared decision-making but cautions about reliability (Lawson McLean et al., 2024). Multi-agent designs for oncology add structure: EvoMDT outperformed single models on cancer question benchmarks (Liu et al., 2026b), a neuro-symbolic multi-agent system for trial matching reached F1 0.82 against 0.47 for GPT-4 zero-shot (Loaiza-Bonilla et al., 2026), and the Healthcare Agent Orchestrator summarised patients for molecular tumour boards (Blondeel et al., 2025). A thoracic tumour-board system was developed, evaluated and deployed, with an LLM judge agreeing with physicians in 89% of ratings (Ellis-Caleo et al., 2026). Closest to this study, TumorBoard addresses longitudinal neuro-oncology with a claim-evidence ledger and a safety governor (Liu et al., 2026c), and a multi-agent system automated BT-RADS scoring for glioma follow-up (Jabal et al., 2026). None of these, however, attaches calibrated uncertainty from imaging and pathology models to the board's reasoning.

2.3.8 Uncertainty, abstention and routing in agents

A growing body of work asks how language-model agents can know when they are unsure. Semantic entropy, which measures uncertainty over meanings rather than word sequences, predicted correctness better than token-level entropy (Kuhn et al., 2023) and detected hallucinations across 30 task-model combinations with AUROC 0.79 (Farquhar et al., 2024). Black-box uncertainty methods for closed models (Lin et al., 2024b), conformal

factuality guarantees that raised correctness from about 30% to 80% by removing unsupported claims (Mohri & Hashimoto, 2024), and surveys of calibration in language models (Liu et al., 2025b) provide tools, but a position paper argues that the aleatoric-epistemic split does not fit interactive agents (Kirchhof et al., 2025), and a review of uncertainty in medical language models proposes a broader framework (Atf et al., 2025). KnowNo used conformal prediction to make a robot planner ask for help only when needed, cutting human help by 10-24% at a guaranteed success rate (Ren et al., 2023), an idea transferable to clinical escalation.

In medicine, uncertainty-aware abstention improved selective performance on clinical texts (Vazhentsev et al., 2026), selective prediction for extracting data from clinical notes abstained on 43% of notes and cut total cost by 51% (Swaminathan et al., 2023), and a review of abstention in clinical language models proposed a decision-theoretic framework with abstention operating characteristics (Presacan et al., 2026). Confidence-aware collaboration between doctor and pharmacist agents improved diagnosis and medication accuracy by about 7 percentage points (Zheng et al., 2025), and confidence estimation has been proposed for trustworthy agents working on electronic health records (Song et al., 2025). Two recent preprints bring calibrated uncertainty into agent routing: Monte Carlo dropout uncertainty helped a decision-support agent only when communicated as a binary error-risk flag, cutting confident misdiagnoses from 8.5% to 2.7% (Hauke et al., 2026); and CRC-Router applied conformal risk control to routing on chest radiographs (Li et al., 2026), as did an uncertainty-aware agentic triage system for chest X-rays (Li et al., 2025b). An explainable agentic framework for stroke imaging with an abstention agent shows the same direction for neuroimaging (Islam, 2026). Learning-to-defer methods that consider which human expert to ask (Mozannar et al., 2023a; Hemmer et al., 2023) and benchmarks for deferral with realistic synthetic experts (Alves et al., 2025) provide the theoretical basis.

2.3.9 Governance, monitoring and audit

Accountability for a composite decision requires records of what each component did. Model cards document intended use and evaluation conditions (Mitchell et al., 2019), internal algorithmic auditing defines an end-to-end audit process (Raji et al., 2020), and FUTURE-AI sets 30 consensus recommendations including traceability (Lekadir et al., 2025). For deployed imaging AI, ReclAIm used a multi-agent system to monitor performance and trigger fine-tuning when decline was detected, restoring performance to

within 2% of baseline after drops of up to 40.6% (Tzanis & Klontzas, 2025), and hospital 'AI quality improvement' units have been proposed to monitor performance and distribution shift (Feng et al., 2022). Safety-focused architectures such as Polaris, a constellation of specialist models with safety checks, performed on par with nurses on safety dimensions (Mukherjee et al., 2024), and MedGuards uses agents to detect and correct medical errors (Ma et al., 2026). Orchestration of explainability for multimodal, longitudinal imaging is still immature: of 15 explainability libraries, only one supported joint multimodal input (Pahud de Mortanges et al., 2024). These works show the components of an auditable system but not an architecture in which audit is part of every hand-off.

2.3.10 Gap for O1

Existing medical multi-agent systems route by question difficulty, role or language-model self-assessment; confidence, where present, is qualitative or heuristic (Lou et al., 2026; Sharma et al., 2026). Only a handful of recent preprints attach statistically grounded uncertainty to routing (Li et al., 2026; Hauke et al., 2026), and they address a single routing step on chest radiographs. No published architecture for the brain-tumour pathway makes calibrated uncertainty part of the contract between agents, propagates it across dependent hand-offs, and records every AI output and clinician validation in an auditable trail. This is the contribution addressed by O1 and RQ1.

Table 2.4: Representative multi-agent clinical systems and what they lack relative to this study.

Study	Domain and design	Main result	Missing for this study
Kim et al. (2024)	LLM agents; moderator assigns team size by complexity	Best on 7/10 medical QA benchmarks	No calibrated confidence; no clinician in the loop
Ferber et al. (2025)	Oncology; GPT-4 calling vision and retrieval tools	Decision accuracy ~30% to ~87% (20 cases)	No deferral rule; tool errors pass silently
Lou et al. (2026)	Chest X-ray; director with evidence validator	67.0% CheXbench; preferred by radiologists	Confidence is High/Medium/Low, not calibrated
Sharma et al. (2026)	Emergency triage; orchestrator, four agents	80% exact ESI match, 0% undertriage (60 vignettes)	Heuristic weights; simulated cases only
Jabal et al. (2026)	Glioma follow-up MRI; extractor and scorer agents	76.0% vs 57.5% BT-RADS agreement (492 exams)	No uncertainty or escalation
Liu et al. (2026c)	Neuro-oncology tumour board; evidence ledger, safety governor	Deferred 84.2% of unsafe cases	Preprint; synthetic benchmark; no imaging models
Li et al. (2026)	Chest X-ray routing with conformal risk control	Best risk-coverage trade-off	Single step; no dependent chain or clinician cost

2.4 Uncertainty, Calibration and Deferral (O2)

2.4.1 Estimating uncertainty

The premise of the framework is that an agent must know when it does not know. Monte Carlo dropout approximates Bayesian inference in deep networks (Gal & Ghahramani, 2016) and deep ensembles provide a simple, strong alternative (Lakshminarayanan et al., 2017); separating aleatoric from epistemic uncertainty clarifies which part more data can reduce (Hüllermeier & Waegeman, 2021). Reviews of uncertainty quantification in medical deep learning group methods into Bayesian, ensemble, test-time augmentation, deterministic single-pass and post-hoc families (Lambert et al., 2024a; Loftus et al., 2022). In brain-tumour segmentation, the QU-BraTS challenge ranked uncertainty methods and found ensembles and test-time augmentation common among the best teams (Mehta et al., 2022), segment-level uncertainty tracked Dice and could flag poor segmentations (Mehrtash et al., 2020), and conformal prediction produced voxel-wise uncertainty that correlated strongly with Dice ($r = -0.83$) on internal and external glioma data (Elyassirad et al., 2025). For IDH prediction from MRI, a confidence-informed network trained on 2,481 gliomas from eight institutions reached 96% accuracy with a Brier score of 0.013 (Bangalore Yogananda et al., 2025). In pathology, uncertainty-thresholded predictions raised external AUROC from 0.93 to 0.99 (Dolezal et al., 2022), and an uncertainty-aware ensemble separated glioblastoma from lymphoma and flagged 67 unseen tumour types with AUROC 0.92 where Monte Carlo dropout reached only 0.67 (Zhao et al., 2025).

2.4.2 Calibration and its fragility

Modern networks are over-confident, and temperature scaling corrects most of this on in-distribution data (Guo et al., 2017). The central negative result is that calibration degrades under dataset shift, exactly where it is most needed; temperature scaling fitted on clean validation data does not hold, and deep ensembles are the most robust option (Ovadia et al., 2019). Medical benchmarks confirm the problem: no confidence-scoring function reliably prevented silent failures in medical image classification, and none consistently beat the plain maximum softmax response (Bungert et al., 2023; Bernhardt et al., 2022; Jaeger et al., 2023). In pathology, ensembles gave the best uncertainty under domain shift and label noise (Mehrtens et al., 2023b). Multi-domain temperature scaling, which uses heterogeneity across sites, improves robustness to shift (Yu et al., 2022), a result directly relevant to a federation of a few heterogeneous hospitals.

2.4.3 Acting on uncertainty: selective prediction, deferral and conformal guarantees

Selective classification gives a model a reject option and guarantees the error on accepted cases: on ImageNet, 2% top-5 error was guaranteed at about 60% coverage (Geifman & El-Yaniv, 2017). Calibrated selective classification further requires the accepted cases to be well calibrated, including under a new hospital (Fisch et al., 2022). Learning to defer goes further by modelling the human expert: the system learns a predictor and a rejector jointly with a consistent surrogate loss (Mozannar & Sontag, 2020; Madras et al., 2018), later made calibrated with one-vs-all classifiers (Verma & Nalisnick, 2022) and extended to multiple experts, cost-sensitivity and workload constraints (Verma et al., 2023; Alves et al., 2024). Triage formulations allocate scarce human effort explicitly (Raghu et al., 2019; Okati et al., 2021), and sequential deferral has been studied in simulated treatment settings (Joshi et al., 2022). In medicine, combining deferral with uncertainty reached the same F1 while deferring far fewer patients (for example 36% versus 69% for pleural effusion) (Liu et al., 2022), and complementarity-driven deferral to clinicians reduced false positives and clinician reads in real screening programmes (Dvijotham et al., 2023).

Conformal prediction provides distribution-free coverage guarantees for any model (Angelopoulos & Bates, 2021), and conformal risk control extends them to arbitrary monotone losses such as the false-negative rate (Angelopoulos et al., 2024a). Medical applications have flagged unreliable predictions in prostate pathology (one error in 794 predictions versus fourteen without conformal prediction) (Olsson et al., 2022), reduced false positives of a head-CT model at a new hospital from 45% to 5% while abstaining on 14% of examinations (Angelopoulos et al., 2024b), and combined conformal selection with cost-aware deferral for sepsis triage under shift (Kwon & Kim, 2026). In brain-tumour MRI classification, withholding 1.8% of predictions cut selective risk by a third (Alshamlan, 2026). The guarantees, however, are marginal rather than per patient, assume exchangeable data, and can fail for subgroups and under shift (Mehrtens et al., 2023a); federated conformal prediction is only beginning to address distributed sites (Lu et al., 2023).

2.4.4 Measuring calibration and the limits of calibration metrics

Calibration is usually summarised by expected calibration error, but the measure has known weaknesses. Calibration tests for multi-class problems provide a unifying statistical framework (Widmann et al., 2019), and information-theoretic analysis shows how the

binning in expected calibration error affects its bias (Futami & Fujisawa, 2024). Revisiting calibration across modern architectures found that vision transformers and MLP-Mixers are among the best calibrated and degrade less under shift (Minderer et al., 2021), and foundation models change the picture further: post-hoc calibration helps them in distribution but can become counterproductive under severe shift (Hekler et al., 2025). Importantly, calibration and failure detection are different goals: most calibration methods were useless or harmful for predicting individual failures (Zhu et al., 2022), and in medical image classification self-supervised pre-training improved calibration (Sambyal et al., 2023). For this study, calibration error will therefore be reported together with failure-detection measures, not as a proxy for them.

2.4.5 Single-pass and large-model uncertainty

Ensembles and Monte Carlo dropout require many forward passes, which matters for 3D MRI. Deep deterministic uncertainty matched five-member ensembles on out-of-distribution detection with a single pass (Mukhoti et al., 2023), and large pre-trained models with reliability extensions improved uncertainty across tasks as model size increased (Tran et al., 2022). Evidential deep learning gives uncertainty in one pass; a region-based evidential method for brain-tumour segmentation achieved the best uncertainty-error overlap among evidential variants (Li et al., 2022). Out-of-distribution detection is surveyed in general (Yang et al., 2024) and for medical imaging specifically (Hong et al., 2024), and the OpenMIBOOD benchmark provides medical out-of-distribution test sets (Gutbrod et al., 2025). These options let the framework trade computational cost against uncertainty quality for each agent.

2.4.6 Uncertainty and failure detection in brain-tumour imaging

Brain-tumour studies show both the value and the limits of uncertainty. Exploiting uncertainty in a segmentation network raised enhancing-tumour Dice by 3.15 points (Lee et al., 2022); uncertainty maps for post-operative glioblastoma segmentation were judged useful, with 72% of segmentations rated acceptable by three physicians (Holtzman Gazit et al., 2022); and uncertainty-aware segmentation of meningioma radiotherapy targets improved Dice and Hausdorff distance while adding a map that a standard network lacks (Wang et al., 2025a). In radiogenomics, the lowest-uncertainty EGFR predictions were 83% accurate against 48% for the most uncertain (Hu et al., 2021a), and a calibrated craniopharyngioma classifier revealed that an earlier published model had been over-fitted

(Prince et al., 2023). Uncertainty estimates for segmentation can be unreliable at the subject level (Jungo & Reyes, 2019), and how voxel uncertainty is aggregated to a case-level score matters: spatially aware aggregation outperformed simple averaging (Guarino et al., 2026), soft Dice confidence was near-optimal for selective segmentation including on BraTS (Borges et al., 2026), and pairwise Dice between ensemble members was a strong and simple failure detector across segmentation tasks (Zenk et al., 2024). Quality-control frameworks that flag failed segmentations raised mean Dice on accepted cases from 0.82 to 0.84 (Williams et al., 2021), and counterfactual auditing of glioma segmentations detected failures across MRI datasets (Le et al., 2026). A task-aware evaluation of uncertainty for glioma diagnosis found that a composite trust score did not consistently beat classification uncertainty for selective prediction (Rojas et al., 2026).

2.4.7 Learning to defer: theory and extensions

Learning to defer has developed rapidly since the first consistent estimator (Mozannar & Sontag, 2020). Joint learning of predictor and rejector is more sample-efficient than staged learning when expert labels are scarce (Charusaie et al., 2022), and good deferral is possible with only a few expert predictions per class—six per class reached 99% of the full-label accuracy in a radiology task (Hemmer et al., 2023). Softmax parametrisation can be made both calibrated and consistent (Cao et al., 2023), realizable consistent losses give stronger guarantees (Mao et al., 2024b), and fine-tuning improves deferral (Raman & Yee, 2021). Extensions handle multiple experts and populations of experts (Mao et al., 2024a; Tailor et al., 2024; Keswani et al., 2021), top-k experts (Montreuil et al., 2025), multiple users who may be complemented or deferred to (Zhang et al., 2024b), coverage constraints (Zhang et al., 2024c), and multiple objectives through post-processing (Charusaie & Samadi, 2024). Combining human labels with model probabilities through confusion matrices roughly halved error relative to either alone on image benchmarks (Kerrigan et al., 2021). Asymmetric abstention, which rejects differently for the two error types, improved biomedical decisions without rejecting more cases (Gandouz et al., 2021)—the closest existing idea to the cost-asymmetric threshold of this study.

2.4.8 Conformal prediction in medical imaging

Conformal methods are increasingly applied to medical images. Learn-then-test calibrates algorithms to control risks such as false-discovery rate (Angelopoulos et al., 2025), and selective conformal risk control combines selection with risk guarantees (Xu et al., 2025).

Group-conditional conformal prediction reduced coverage gaps between skin types (Lu et al., 2022), reliable-region conformal prediction met very low error targets where classical methods failed (Zhang et al., 2023b), and conformal prediction under ambiguous ground truth showed that voted labels can under-cover expert disagreement (Stutz et al., 2023). For segmentation, conformal methods give pixel-wise sets (Mossina et al., 2024), control false positives (Mossina & Friedrich, 2025), respect anatomy (Gaillochet et al., 2026), give asymmetric volumetric risk (Alijani et al., 2026) and produce robust volume estimates under covariate shift (Lambert et al., 2024b; Cheung et al., 2025); in prostate MRI, conformal prediction improved volume agreement with the reference (Gade et al., 2024). Reviews of clinical conformal prediction note that studies are isolated and rarely involve practising clinicians (Vazquez & Facelli, 2022; Angelopoulos & Bates, 2023). In pathology, uncertainty estimates detected 70-90% of mispredictions (Poceviciute et al., 2022), and benchmarks under domain shift favour ensembles (Mehrtens et al., 2023b; Fayjie et al., 2024).

2.4.9 Deferral that reduces workload in practice

Workload reduction is the practical promise of deferral. In mammography, AI identification of normal examinations could exclude 19.1% of examinations without missing a cancer (Lang et al., 2021), and entropy-guided review of automatic sleep scoring improved agreement with the gold standard while sending about 60% of epochs to review (Kang et al., 2021). Self-supervised learning with abstention improved diabetic retinopathy screening safety (Chopra et al., 2026), and a benchmarking study of Bayesian deep learning for diabetic retinopathy showed that uncertainty-based referral improves accuracy on retained cases but degrades under shift (Band et al., 2022; Filos et al., 2019). These results define the operating points that a clinical escalation policy should aim for, and the shift problem that it must monitor.

2.4.10 Gap for O2

Three limitations define the gap. These formulations address one model making one decision, not a chain in which uncertainty must be composed across agents. They assume a fixed, homogeneous cost of error, whereas clinical costs are asymmetric and differ by decision type. And they model the expert as always available, whereas specialist time is the binding constraint (Geifman & El-Yaniv, 2017; Mozannar & Sontag, 2020; Alves et al., 2024). Large-language-model agents add a further difficulty: the usual aleatoric/epistemic

split does not transfer to interactive agents (Kirchhof et al., 2025), and how uncertainty is *communicated* to a downstream agent changes whether it is used (Hauke et al., 2026). A calibrated escalation policy over a dependent chain, with clinically elicited asymmetric costs and finite capacity, is the contribution of O2 and RQ2.

2.5 Data-Efficient, Multi-Source and Federated Models (O3)

2.5.1 MRI segmentation and classification

U-Net-based architectures remain the backbone of brain-tumour segmentation. nnU-Net configures itself to each dataset and matched or beat specialised methods on 23 public datasets (Isensee et al., 2021); a later re-evaluation found that many claimed improvements over it disappear under rigorous validation (Isensee et al., 2024). Transformer designs such as Swin UNETR and TransBTS (Hatamizadeh et al., 2022; Wang et al., 2021b) and scaled convolutional networks such as MedNeXt (Roy et al., 2023) are competitive, and promptable foundation models (MedSAM, MedSAM2) generalise across modalities but need prompts and struggle with small or unclear boundaries (Ma et al., 2024; Ma et al., 2025). On real-world clinical data with missing sequences, nnU-Net remained within or above the range of human inter-rater agreement (Dice 0.86 internal, 0.93 external) (Pemberton et al., 2023). Routine scans often lack sequences; dedicated models learn shared representations that tolerate missing modalities, with the largest gains when few sequences are available (Zhang et al., 2022b; Liu et al., 2023).

2.5.2 Learning with few labels

Because annotated local data will be scarce, label efficiency is essential. Self-supervised pre-training of Swin transformers on unlabelled 3D volumes improved downstream segmentation after fine-tuning (Tang et al., 2022), and stepwise transfer learning achieved expert-level paediatric tumour segmentation from limited data, with AI contours judged acceptable more often than expert contours (80.2% versus 65.4%) (Boyd et al., 2024). Even limited target-domain adaptation yields rapid gains in cross-dataset brain-tumour segmentation (Grande et al., 2026b). What remains unestablished is whether representations learned this way produce confidence estimates fit to drive an autonomy decision; most evaluations report accuracy only.

2.5.3 Computational pathology and fusion

Attention-based multiple-instance learning lets a single slide-level label supervise attention over thousands of tiles (Ilse et al., 2018), and CLAM made this data-efficient and interpretable (Lu et al., 2021). Pathology foundation models such as UNI now serve as general-purpose encoders with strong few-shot transfer (Chen et al., 2024c), although clinical benchmarks show that their performance varies by task and site (Campanella et al., 2025) and that they do not remove site-specific batch effects (Kömen et al., 2024). In brain tumours, deep learning reached neuropathologist-level integrated classification of adult diffuse gliomas (AUC above 0.90 for all major types) (Wang et al., 2023), and intra-operative molecular classification has been achieved with stimulated Raman histology and ultra-fast sequencing, the latter abstaining rather than making confident errors (Hollon et al., 2023; Vermeulen et al., 2023). Fusion of histology with genomic features outperformed either source and WHO 2016 grading (Chen et al., 2022b), and a mixture-of-experts fusion of pathology and radiology foundation models reached AUC 0.94 on an independent test set for WHO 2021 subtyping while still working when only one modality was present (Saueressig et al., 2026). Multimodal radiology-pathology-text benchmarks are beginning to appear (Rivera et al., 2026).

2.5.4 Federated and privacy-preserving learning

Federated averaging trains a shared model by exchanging parameters rather than data (McMahan et al., 2017); FedProx and SCAFFOLD correct for heterogeneous clients (Li et al., 2020; Karimireddy et al., 2020), and FedBN keeps batch normalisation local to absorb scanner differences (Li et al., 2021b). Early medical studies showed feasibility without sharing patient data (Sheller et al., 2020), and the decisive demonstration trained a glioblastoma boundary model across 71 sites on six continents (Pati et al., 2022). The FeTS challenges benchmarked aggregation methods and exposed worst-case sites hidden by good average Dice (Zenk et al., 2025; Linardos et al., 2025). An extensive benchmark found that plain FedAvg already performs very well for brain-tumour segmentation, with other methods giving only small gains (Manthe et al., 2024). An international paediatric platform lost less than 1.5% in classification and 3% in segmentation relative to centralised training (Lee et al., 2024a), and federated evaluation platforms such as MedPerf allow benchmarking at sites without moving data (Karargyris et al., 2023). Personalised methods address appearance shift and modality mismatch between hospitals (Jiang et al., 2022;

Wang et al., 2022; Dai et al., 2024), and differentially private federated learning has been shown to be viable for medical images (Adnan et al., 2022; Abadi et al., 2016).

2.5.5 Architectures and foundation models for brain MRI

Beyond nnU-Net and Swin UNETR, transformer hybrids such as TransBTS, NestedFormer and CKD-TransBTS model inter-sequence relations explicitly (Wang et al., 2021b; Xing et al., 2022; Lin et al., 2023), and MedNeXt scales convolutional networks with transformer-inspired design (Roy et al., 2023). Foundation models are reaching neuroimaging: BrainSegFounder, pre-trained on large unlabelled neuroimaging collections, beat previous challenge winners on BraTS and stroke-lesion data, with larger models and more pre-training data helping (Cox et al., 2024); MedSAM2 extended promptable segmentation to 3D and cut annotation cost by more than 85% in large annotation studies (Ma et al., 2025); and adapters make the general Segment Anything model usable for medical images (Wu et al., 2025), although an experimental study found its zero-shot medical performance highly variable (Mazurowski et al., 2023). Foundation models for glioma genotyping from multi-parametric MRI are also emerging (Farahani et al., 2025b; Farahani et al., 2025a). For the registry, promptable models are attractive as annotation assistants, with clinicians correcting rather than drawing contours.

2.5.6 Missing sequences, metastases and real-world data

Real data rarely include every sequence. Adversarial co-training kept Dice usable when only one or two sequences were present, where earlier methods fell to 10-20% (Wang et al., 2021d); latent correlation learning, mmFormer, M3AE and alignment-based methods train a single model for all combinations (Zhou et al., 2021; Zhang et al., 2022b; Liu et al., 2023; Liu et al., 2024a). In a multicentre study of brain metastases, a model trained to handle missing sequences performed as well as one requiring all sequences (Grovik et al., 2021), and nnU-Net remained robust to missing sequences on real-world glioma data (Pemberton et al., 2023). Metastasis detection reached 95.8% sensitivity with 0.39 false positives per patient and raised radiologists' sensitivity (Yin et al., 2022); in a randomised crossover multi-reader study, AI assistance raised contour Dice from 0.87 to 0.92 and shortened contouring time (Luo et al., 2024); and 2.5D and 3D models on multinational data showed the trade-off between sensitivity and false positives across centres (Ottesen et al., 2022). For paediatric tumours, a single-centre pathologically diverse dataset gave Dice 0.85 for whole tumour, close to human variability, but only 0.26 for cystic components

(Piffer et al., 2026). These studies show what clinically useful performance looks like and where it fails.

2.5.7 Semi-supervised learning and domain adaptation

Semi-supervised learning uses unlabelled cases directly: uncertainty-guided mutual consistency selects reliable predictions for consistency training (Zhang et al., 2021), a teacher-student approach addressed annotation scarcity in brain-tumour segmentation with the student surpassing the teacher on subregions (Liu et al., 2026a), and semi-supervised contrastive learning improved label efficiency (Hu et al., 2021b). Self-supervised methods for 3D MRI include 3D SimCLR with Monte Carlo dropout (Ali et al., 2021), imbalance-aware self-supervised radiomics (Li et al., 2021a), contrastive few-shot tumour detection (Chin et al., 2024) and a self-supervised vision transformer for tumour classification (Karagoz et al., 2024). Domain shift is addressed by unsupervised cross-modality adaptation (He et al., 2021), causality-inspired single-source generalisation (Ouyang et al., 2021), generalisation across heterogeneous sequences (Zhang et al., 2025; Grande et al., 2026b), self-training with tumour-aware deformations (Grande et al., 2026a) and annotation-efficient target-modality adaptation (d'Assier et al., 2022). Active learning that combines transfer learning with selective annotation can reduce labelling for tumour classification (Hao et al., 2021). These methods are the toolkit for WP2, where local labels will be few and local scanners different.

2.5.8 Computational pathology for brain tumours

Multiple-instance learning dominates slide-level analysis (Gadermayr & Tschuchnig, 2022): TransMIL models correlations between tiles and reached AUC 0.93 on CAMELYON16 against 0.88 for CLAM (Shao et al., 2021), CAMIL adds tissue context (Fourkioti et al., 2023), and hierarchical self-supervised vision transformers scale to gigapixel images (Chen et al., 2022a; Chen & Krishnan, 2022). Self-supervised pre-training reaches baseline performance with 40% of the labels (Schirris et al., 2021), generalises biomarker prediction across cohorts (Niehues et al., 2023; Saldanha et al., 2023) and produces phenotype representations (Ye et al., 2024; Hou et al., 2024), while lightweight self-supervised models offer domain generalisation with lower computing cost (Shafique et al., 2025). Vision-language pathology foundation models reach strong zero-shot subtyping (Lu et al., 2024); benchmarks compare adaptation strategies (Lee et al., 2024b), and a neuropathology-specific foundation model captured brain-tissue features

better than general models (Verma et al., 2025). Batch effects survive in foundation-model features and are not removed by stain normalisation (Kömen et al., 2024), which motivates stain augmentation methods such as RandStainNA (Shen et al., 2022).

For gliomas specifically, a multi-centre assessment reached 0.79 accuracy internally and 0.73 on an external test set for five WHO CNS5 types (Jin et al., 2023), transfer learning classified diffuse glioma subtypes with balanced accuracy 0.87 (Im et al., 2021), a single-centre weakly supervised model inferred IDH status with AUC 0.95 on TCGA (Zuo et al., 2025), deep learning on pathology images stratified survival risk (Chunduru et al., 2022; Luo et al., 2023), low-rank contrastive learning improved grading on both FFPE and frozen sections (Zhang et al., 2022a), and integrating molecular data with pathology improved classification (Pei et al., 2021). Review work sets out the challenges of segmentation and classification in glioma pathology (Kurc et al., 2020), and a subtype study combined in-domain and out-of-domain transfer learning (Despotovic et al., 2023). For paediatric tumours, UNI with attention-based MIL was best but performance fell from in-site to out-of-site testing (Tampu et al., 2026). Intra-operative work includes frozen-section discrimination of lymphoma from glioma that outperformed several pathologists (Zhang et al., 2024a), label-free optical imaging for molecular classification (Hollon et al., 2023), ultra-fast sequencing-based classification during surgery that abstained rather than misclassified (Vermeulen et al., 2023), and methylation classification from sparse data (Brändl et al., 2025) concordant with final neuropathology in 89% of cases (Djirackor et al., 2021).

2.5.9 Radiology-pathology fusion

Fusion of imaging and histology predates the 2021 classification: a pipeline combining radiology and histology data reached balanced accuracy 0.86 for glioma classification (Hamidinekoo et al., 2020), and weakly supervised fusion of whole-slide images and multi-parametric MRI ranked third in the CPM-RadPath challenge (Hsu et al., 2022). Pathomic Fusion combined histology with genomics (Chen et al., 2022b), and the latest mixture-of-experts fusion of foundation-model features aligned to WHO 2021 subtypes still worked when only one modality was present (Saueressig et al., 2026). The CoRe-BT benchmark now supports robust multimodal typing with radiology, pathology and text (Rivera et al., 2026). The gap is that these studies assume curated, complete cohorts and report no calibrated confidence for the fused decision.

2.5.10 Federated learning: heterogeneity, pathology and privacy

Data heterogeneity is the main technical obstacle in medical federation. FedDG learns domain-general features by sharing amplitude spectra and improved Dice on unseen sites (Liu et al., 2021); personalised methods adapt the shared model to each site (Jiang et al., 2023; Wang et al., 2024a; Wang et al., 2022; Lin et al., 2024a); and federated evidential active learning selects which local cases to annotate under domain shift (Chen et al., 2024a). Federation with missing modalities has been addressed with modality-specific encoders (Dai et al., 2024). For brain tumours, early feasibility work showed multi-institutional training without data sharing (Sheller et al., 2018), an international paediatric platform lost little accuracy relative to central training (Lee et al., 2024a), federated IDH prediction from MRI was feasible but incurred a cost relative to central training (Bas & Ozturk-Isik, 2026), federated radiomics generalised to an external dataset of 466 patients (Ren et al., 2025), and Byzantine-robust aggregation protected heterogeneous brain-MRI federation from corrupted updates (Karami et al., 2026). Federated learning was also feasible for multiple-sclerosis lesion segmentation across real sites despite heterogeneity (Hindawi et al., 2025). Studies that apply federation to brain-tumour classification on public MRI collections (Albalawi et al., 2024; Islam et al., 2022) are less informative than the multi-institutional studies above, because their sites are simulated.

Computational pathology has its own federated literature: federated learning on gigapixel slides (Lu et al., 2020), swarm learning across three centres (Saldanha et al., 2022), nationwide deployment behind hospital firewalls in Germany (Babendererde et al., 2026), dropout-based and cross-assessment aggregation for histopathology (Gunesli et al., 2021; Zhang et al., 2023a), and a systematic review of federated histopathology classification (Touhami et al., 2026). Privacy threats in federated medical learning are real—gradients can leak information (Das et al., 2024)—so differential privacy (Adnan et al., 2022; Singh et al., 2020; Nampalle et al., 2023), privacy-preserving harmonisation (Galada et al., 2024), few-shot synthetic data generation for sharing (Shi & Wang, 2025) and broader privacy reviews (Zhu et al., 2024) are relevant. Surveys and benchmarks of federated medical image analysis summarise methods and show that results depend strongly on how data are split (Guan et al., 2023; Zhou et al., 2025), and federated foundation models are beginning to be explored (Liu et al., 2024b). Finally, the intersection of federated learning and uncertainty quantification is identified as a gap by two reviews (Koutsoubis et al., 2024;

Koutsoubis et al., 2025), with federated conformal prediction an early answer (Lu et al., 2023).

2.5.11 Gap for O3

Three gaps remain. Most fusion methods assume complete, high-quality inputs, whereas in practice histology lags imaging, slide quality varies and molecular markers may be unaffordable. Label-efficient and federated methods are validated almost entirely on Western, curated cohorts and large, well-resourced sites; the regime of a handful of small, heterogeneous hospitals is little characterised. And federation is studied as a training-time issue in isolation: how aggregation changes calibration, and therefore the behaviour of a deferral policy validated on an earlier model, is essentially unexamined (Koutsoubis et al., 2024; Koutsoubis et al., 2025). These define O3 and RQ3.

2.6 Clinical Evaluation of Human-AI Teams (O4)

2.6.1 Reporting and validation standards

Reporting guidelines now cover each stage of clinical AI evaluation: DECIDE-AI for early-stage live evaluation of decision support, with explicit items on human factors and safety (Vasey et al., 2022); CONSORT-AI for randomised trials (Liu et al., 2020); TRIPOD+AI for prediction models (Collins et al., 2024); and STARD-AI for diagnostic accuracy studies (Sounderajah et al., 2025). Metrics Reloaded recommends combining overlap and boundary metrics for segmentation and calibration metrics when probabilities are used (Maier-Hein et al., 2024). A scoping review of 86 randomised trials of AI found mostly single-centre studies with diagnostic-yield endpoints and poor demographic reporting (Han et al., 2024).

2.6.2 Does the team beat its parts?

The evidence is mixed and instructive. On average, human-AI combinations underperformed the better of human or AI alone in decision tasks (Malone et al., 2024). The effect of AI assistance varied greatly between radiologists, could not be predicted from experience, and incorrect AI predictions harmed performance (Yu et al., 2024); clinicians were susceptible to inaccurate advice whatever its claimed source (Gaube et al., 2021). Explanations increased reliance on wrong advice (Bansal et al., 2021) and had variable effects across clinicians (Nicolson et al., 2025). Positive results come where the human can

weigh the AI's reliability case by case: endoscopists followed AI more when it was right than when it was wrong, and the team beat both (Reverberi et al., 2022). Cognitive forcing functions reduced over-reliance at a cost in perceived usability (Bucinca et al., 2021). Most relevant to this study, selective prediction reduced the overall harm of automation bias but changed the error pattern: when told the AI was abstaining, clinicians missed more diagnoses and treatments (Jabbour et al., 2025). Automation bias also appears under time pressure in computational pathology (Rosbach et al., 2024) and in neuroradiology reading (Kim et al., 2025a).

2.6.3 Workload, usability and trust

A meta-analysis found that concurrent AI assistance reduced reading time by 27%, and AI pre-screening reduced reading volume by 62%, with sensitivity maintained or improved (Chen et al., 2024b). In mammography, automatic identification of normal examinations cut workload by 17% while excluding 1% of true positives (Rodriguez-Ruiz et al., 2019), and AI-integrated screening pathways improved single-reader performance (Frazer et al., 2024). In a low- and middle-income radiotherapy setting, AI auto-segmentation reduced contouring from about an hour to ten minutes of human review (Kibudde et al., 2024). Usability and trust are measured with established instruments—the System Usability Scale, NASA Task Load Index and Likert trust scales (Brooke, 1996; Hart & Staveland, 1988)—and trust is shaped by clinical experience, perceived risk and how uncertainty is presented (Cao et al., 2024).

2.6.4 Surgical planning and neurosurgery

AI in neurosurgery is reviewed broadly (Awuah et al., 2024; Baker et al., 2024), and machine-learning-based trajectory planning to the tumour has been demonstrated (Dundar et al., 2022). Navigated tractography is accurate for motor and language pathways in glioma surgery (Ille et al., 2021), and automated post-operative segmentation can standardise extent-of-resection assessment (Luque et al., 2024; Zanier et al., 2023). These works evaluate geometric accuracy rather than surgeon acceptability, and none presents the uncertainty of the underlying anatomy alongside a recommended route.

2.6.5 Reliance, explanation and decision-making

The human-AI decision literature shows that appropriate reliance—following the AI when it is right and not when it is wrong—is the goal, not trust as such. A decision-theoretic

framework separates loss due to mis-reliance from loss due to poor information (Guo et al., 2024b), and measurement constructs for appropriate reliance are being standardised (Raees & Papangelis, 2026). People overestimate their own competence in ways that reduce appropriate reliance (He et al., 2023); learning from feedback improves it (Schemmer et al., 2023); onboarding with learned rules raised team accuracy by 5.2% (Mozannar et al., 2023b); and time pressure changes the accuracy-time trade-off and reliance (Swaroop et al., 2023). Explanations rarely enable complementary performance because most tasks do not allow the explanation to be verified (Fok & Weld, 2023), proxy tasks and subjective measures can mislead when evaluating explainable AI (Buçinca et al., 2020), and interface design shapes collaboration in high-stakes decisions (Chen et al., 2025b). Presenting AI uncertainty affects reliance, moderated by the user's initial decision and background (Cao et al., 2024); adjusting for trust can reduce inappropriate reliance (Srinivasan & Thomason, 2025); and majority voting among doctors improved the appropriateness of reliance in pathology (Gu et al., 2024). These findings shape the clinician interface in Objective 1: the system shows calibrated confidence and the reason for escalation, and the study measures appropriate reliance rather than raw agreement.

2.6.6 Automation bias in imaging and pathology

Automation bias—over-reliance on automated advice—is well documented. AI assistance shortened reading time for cerebral aneurysm detection at all experience levels but exposed readers to automation bias (Kim et al., 2025a); in computational pathology, time pressure increased the severity of automation bias (Rosbach et al., 2024), and anchoring on suggestions persisted across pathologists (Rosbach et al., 2026). In mammography, automation bias degraded performance in multi-reader settings but improved it for single readers (Frazer et al., 2024), and reader bias changed with cancer prevalence and AI support (Al-Bazzaz et al., 2024). Explainable AI did not remove automation bias (Schemmer et al., 2022), and reviews describe deskilling and cognitive risks of generative AI (Al-Anezi, 2026; Bittencourt, 2025). The most relevant finding is that selective prediction reduced the overall harm of automation bias but increased missed diagnoses and treatments when clinicians were told that the AI had abstained (Jabbour et al., 2025). An escalation system must therefore be evaluated on how clinicians behave on escalated cases, not only on how accurate the AI is on the others.

2.6.7 Reader studies in pathology and radiology

Reader studies give the design template for WP7. In pathology, AI assistance improved Gleason grading so that assisted pathologists outperformed both unassisted pathologists and the AI alone (Bulten et al., 2021); a liver-cancer assistant improved accuracy when correct but reduced it when wrong (Kiani et al., 2020); AI improved efficiency in prostate biopsies (Eloy et al., 2023), PD-L1 scoring (Wang et al., 2021c) and mitotic counting reproducibility (Bertram et al., 2022; Graham et al., 2026); and the mode of assistance mattered, with one mode robust to false-positive AI predictions (Chuang et al., 2026). In radiology, AI support improved fracture detection most for non-obvious fractures (Bachmann et al., 2024), improved diagnostic confidence and sensitivity especially for junior clinicians in emergency imaging (Novak et al., 2024), gave modest AUROC gains and shorter interpretation in breast ultrasound (Kwon et al., 2026), and improved early lung-cancer detection on radiographs (Dissez et al., 2022). Dual suggestions helped some specialties more than others (Wu et al., 2026a). In neuro-oncology, AI predictions raised physicians' glioma-grading accuracy from 82.5% to 87.7%, with explanations adding no further significant gain (Jin et al., 2024), and LLM assistance improved radiologists' brain-tumour diagnosis in a constrained setting (Song et al., 2026). A large public dataset of radiologist-AI interactions now supports secondary analysis (Moehring et al., 2025).

2.6.8 Workload, triage and safety evidence

Beyond the meta-analysis of workload reduction (Chen et al., 2024b), triage studies quantify potential savings: AI as an additional reader suggested a 69.5% workload reduction in a hybrid triage scenario (Seker et al., 2024), simulated breast-screening triage reduced workload but missed some microcalcification cancers (Riccardi et al., 2026), and a narrative review summarises workload effects across specialties (Jeong et al., 2025). Safety evidence is less developed: a systematic review linked AI to patient-safety outcomes (Choudhury & Asan, 2020), an analysis of FDA device reports found that a quarter of reports were potentially AI-related while a third lacked enough information to tell (Handley et al., 2024), and reviews describe AI's potential for safety in general and in neurosurgery (De Micco et al., 2024; Sugiyama et al., 2025). Randomised trials of AI remain few and mostly single-centre (Han et al., 2024; Lam et al., 2022; Park et al., 2022); examples include AI-assisted colonoscopy (Xu et al., 2021) and deterioration analytics (Keim-Malpass et al., 2026). A systematic review of COVID-19 imaging models found

none fit for clinical use, mainly because of bias and poor validation (Roberts et al., 2021), a warning for any rapidly developed model.

2.6.9 Usability, acceptance and trust

Clinician acceptance follows technology-acceptance models: in a meta-analysis, intention to use AI decision support correlated with performance expectancy (r = 0.66), effort expectancy, social influence and facilitating conditions (Dingel et al., 2024); reviews of barriers and facilitators, national surveys and cross-disciplinary studies confirm these drivers (Scipion et al., 2025; Shi et al., 2026; Breitwieser et al., 2025; Fujimori et al., 2022), and human and technological factors interact (Knop et al., 2022). Usability methods for decision support include heuristic evaluation, live usability testing and dual methods (Cho et al., 2022; Richardson et al., 2019; Rizvi et al., 2025), with design lessons such as preferring brief, flexible, non-interruptive support (Blecker et al., 2019) and user-centred design (Chyzhyk et al., 2025; Blanes-Selva et al., 2023; Kunz et al., 2025); reviews summarise usability assessment and explainable decision support (Cubukcu & Canbaz, 2025; Abbas et al., 2025). Trust depends on clinicians' experience and perceived risk (Schulz et al., 2025), on explanations that can increase or decrease it (Rosenbacke et al., 2024), on workload and perceived risk (Shamszare & Choudhury, 2023), and on contextual explanation (Davis & Salwei, 2026); reviews of trust in clinical AI (Asan et al., 2020; Jones et al., 2023) and studies of trust calibration (Sakamoto et al., 2024; Lin-Martore et al., 2025; Cicin & Çetin Gürkan, 2026) show that the aim is calibrated trust. Prospective implementations show that early experiences shape sustained adoption (Naicker et al., 2026) and that user satisfaction can be good even when outcomes do not change significantly (Smith et al., 2025). Neurosurgical teams are broadly positive about AI but want it for imaging interpretation, planning and hazard alerts rather than autonomous surgery (Layard Horsfall et al., 2021; Boaro et al., 2025).

2.6.10 Pre-operative planning: tractography, extent of resection and visualisation

The surgical-planning agent builds on established neurosurgical tools. Tractography of the corticospinal tract and language pathways is accurate for subcortical resection (Ille et al., 2021), navigated tractography identifies high-risk language areas (Salvati et al., 2021), combining navigated transcranial magnetic stimulation with tractography helps preserve motor function, with greater distance to the tract associated with stable function (Eibl et al., 2024), and constrained spherical deconvolution and multi-level fibre tracking improve tract

reconstruction in glioma (Sheng et al., 2021; Zhylka et al., 2021). A preoperative tractography algorithm has been proposed for tumours in the motor pathway (Ordonez-Rubiano et al., 2023), and connectome-guided surgery is reviewed as an emerging paradigm (Patel et al., 2026). Automated extent-of-resection assessment on early post-operative MRI was reliable and prognostic (Luque et al., 2024), although another model determined extent of resection correctly in only 44.1% of cases (Zanier et al., 2023), and editing AI contours affected gross tumour volume accuracy (Hochreuter et al., 2024). Virtual and augmented reality support planning and awake craniotomy (Mofatteh et al., 2022; Gómez Amarillo et al., 2023; Zawy Alsofy et al., 2021), with open-source virtual-reality frameworks and markerless augmented reality under development (Fiederer et al., 2019; Ho et al., 2024; Hellum et al., 2022; Asadi et al., 2024), and intra-operative imaging reviews look towards AI integration (Bin-Alamer et al., 2023). Registration error estimation for MRI-ultrasound (Salari et al., 2023) and simulation for planning (Zhao et al., 2024) show how uncertainty can be made visible to surgeons. Reinforcement-learning path planning to the tumour has also been demonstrated (Dundar et al., 2022). None of these combines segmentation uncertainty with route planning, which is the specific feature the proof of concept will test.

2.6.11 Gap for O4

No reporting standard defines how to measure the quality of an escalation decision itself—whether the system deferred on the cases that needed a clinician and acted on the ones that did not—and evidence on how clinicians respond to a deferring system in brain-tumour care is absent. Evaluations in low-resource settings are rare. A multi-reader, multi-case evaluation that measures escalation appropriateness, missed escalations, workload, usability and trust in situ is the contribution of O4 and RQ4.

2.7 Cross-Cutting Themes

Reading across the four objectives, five themes recur and directly shape the design of this study.

1. **Evaluation is moving from outputs to processes.** Audits of multi-agent systems (Cemri et al., 2025; Zhu et al., 2025a), reliance frameworks (Guo et al., 2024b) and reporting guidelines (Vasey et al., 2022) all look at how a decision was reached, not only whether it was right. The audit trail of Objective 1 and the escalation-quality measures of Objective 4 follow this direction.

2. **Calibration is necessary but fragile.** Every objective depends on calibrated confidence, but calibration degrades under shift (Ovadia et al., 2019), after model updates and potentially after federated aggregation (Koutsoubis et al., 2024). Calibration is therefore treated as a quantity to monitor, not a property to assume.
3. **Simple baselines are hard to beat.** Maximum softmax response for failure detection (Jaeger et al., 2023; Bungert et al., 2023), nnU-Net for segmentation (Isensee et al., 2024) and FedAvg for federation (Manthe et al., 2024) repeatedly match more complex methods. Every experiment in this study includes these baselines.
4. **Human-AI teams can be worse than their parts.** Benefit depends on who decides which cases the AI handles (Dvijotham et al., 2023; Malone et al., 2024; Jabbour et al., 2025). This is the central argument for a principled escalation policy rather than AI assistance on every case.
5. **Local validation is the exception.** Few studies come from low- and middle-income settings (Ciecierski-Holmes et al., 2022), and none from Sri Lanka. A locally validated, federated and calibrated framework would add evidence where it is scarcest.

2.8 Synthesis

Four things are established. The perception tasks the framework requires are technically solved to a clinically useful standard on curated data, for MRI (Isensee et al., 2021; Pemberton et al., 2023) and for digitised slides (Lu et al., 2021; Wang et al., 2023). Tools for quantifying and calibrating uncertainty are mature (Guo et al., 2017; Lakshminarayanan et al., 2017; Angelopoulos & Bates, 2021), with formal mechanisms for acting on it (Geifman & El-Yaniv, 2017; Mozannar & Sontag, 2020). Federated learning is a proven route to multi-institutional training without data sharing (Pati et al., 2022; Lee et al., 2024a). And both ethical guidance and regulation require demonstrable human oversight of high-risk clinical AI (World Health Organization, 2021; European Union, 2024).

Four things remain open, and together they define this research (Table 2.5). The primitives for selective autonomy have not been assembled into a calibrated escalation policy for a dependent chain of clinical sub-decisions with asymmetric costs and finite expert capacity. Agentic architectures do not treat uncertainty as part of the contract between agents, so hand-offs are unqualified and accountability for composite decisions is undefined. Multi-source diagnosis is demonstrated on complete, high-quality cohorts rather than under the

incompleteness of real practice. And data-efficient, federated and calibrated learning have been studied separately, leaving unexamined how federated updates and domain adaptation change calibration and thereby silently change the safety behaviour of an escalation policy.

Table 2.5: Summary of open problems and how this research addresses them.

Open problem	Key evidence	Addressed by
Escalation policy for a dependent chain with asymmetric costs and finite clinician time	(Geifman & El-Yaniv, 2017; Mozannar & Sontag, 2020; Alves et al., 2024; Dvijotham et al., 2023)	O2 / RQ2 (WP3)
Uncertainty as part of the contract between agents; auditable hand-offs	(Cemri et al., 2025; Zhu et al., 2025a; Li et al., 2026; Liu et al., 2026c)	O1 / RQ1 (WP3, WP1)
Fusion and perception robust to missing, poor or local data	(Chen et al., 2022b; Saueressig et al., 2026; Adewole et al., 2026; Boyd et al., 2024)	O3 / RQ3 (WP2, WP4)
Effect of federation and adaptation on calibration and deferral	(Manthe et al., 2024; Ovadia et al., 2019; Koutsoubis et al., 2024)	O3 / RQ3 (WP5)
Measuring escalation quality, workload and trust in situ	(Vasey et al., 2022; Malone et al., 2024; Jabbour et al., 2025)	O4 / RQ4 (WP6, WP7)

2.9 Evidence Tables

Tables 2.6 to 2.9 summarise representative studies for each objective from the structured extraction records of the Research Paper Bank: the design and data, the main result, and the main limitation in relation to this study. Base papers are included in every table. The full records for all 559 papers, with links to the sources, are available in the Paper Bank on the progress website.

Table 2.6: Evidence table for multi-agent orchestration (O1): 30 representative studies.

Study	Aim and design	Data	Main result	Limitation for this study
Kim et al. (2024) (base)	To let LLM agents decide how much teamwork a medical question really needs, instead of fixing one setup for every case. A moderator agent first rates the case complexity. Easy cases go to one doctor agent, medium cases go to a multi-disciplinary team that discusses ...	Ten public benchmarks (50 random test samples each in main runs; 100 in Table 12 runs). Text: MedQA (USMLE, English test set 1,273 questions, 5 ...	Best accuracy on 7 of 10 medical benchmarks (MedQA, PMC-VQA, Path-VQA etc). Adding moderator review and external medical knowledge gave a further clear gain.	Only question-answer benchmarks. No real patient pathway, no calibrated confidence, and the human doctor is not part of the loop.
Ferber et al. (2025) (base)	To check if an LLM that can call specialised oncology tools can make better clinical decisions than the LLM alone. GPT-4 acts as the central agent and calls tools: vision transformers for mutation prediction from histology, MedSAM for radiology segmentation ...	(1) Knowledge base: about 6,800 medical documents and clinical scores from six sources: MDCalc (clinical scores), UpToDate and MEDITRON guidelines ...	Decision accuracy went from about 30% (GPT-4 alone) to about 87% with tools. Most tool calls were correct and most answers cited guidelines.	Only 20 cases, not prospective, no deferral rule. Errors in one tool can flow silently to the final answer.
Dvijotham et al. (2023) (base)	To build a system (CoDoC) that learns, for each case, whether to trust a locked diagnostic AI model or to defer to the clinician workflow. The goal is higher accuracy ... CoDoC is a small 'deferral AI' wrapped around a black-box diagnostic AI. It takes only the AI confidence score, and uses kernel density estimates of ...	(1) UK Mammography: CoDoC tuning data = 60,755 cases (1,198 biopsy-confirmed cancers) from the diagnostic AI's hyperparameter-tuning subset, split ...	UK mammography, deferral to double reading: sensitivity +3.4% (95% CI 1.3-5.4%, P<0.001) or specificity +0.8% (95% CI 0.6-1.0%, P<0.001) vs double reading, with 1.4 fewer human reads per case (published abstract: 25% fewer false ...	Deferral uses only the scalar AI confidence score; it does not co-train the diagnostic model and deferral model, does not use image features or case metadata, and was ...
Liu et al. (2025a) (base)	To give a clear conceptual framework for medical AI agents, review their clinical applications, and set out the technical, ethical and regulatory challenges to using ... A narrative (non-systematic) review. The authors searched PubMed, Google Scholar and arXiv for papers up to May 2025 and grouped the findings by ...	No dataset used. Literature sources: PubMed, Google Scholar and arXiv, articles published up to May 2025, including peer-reviewed articles, reviews ...	No new quantitative results. The review defines medical AI agents by four components: planning (LLM/VLM reasoning core), action (APIs, device interfaces, software libraries, robotic actuators), reflection (perception of images ...	No quantitative synthesis or systematic appraisal of evidence. Explainability has no standard metric. Long-term learning of agents is hard to evaluate. Regulation (FDA ...
Chen et al. (2025a)	To build a Multi-Agent Conversation (MAC) framework that copies a clinical Multi-Disciplinary Team (MDT) discussion, and to test whether it diagnoses rare diseases ... The authors built MAC on Microsoft AutoGen with one admin agent, several doctor agents and one supervisor agent that discuss a case until consensus ...	(1) Orphanet rare-disease database: pool of over 7000 rare diseases in 33 categories; 302 diseases sampled across all 33 categories (1 to 9 diseases ...	With GPT-4 and 4 doctor agents, MAC reached 34.11% most-likely-diagnosis accuracy, 48.12% possible-diagnosis accuracy and 78.26% helpful-test rate in primary consultation, versus 19.65%, 34.55% and 58.17% for GPT-4 alone. In ...	Uses text-only rare-disease case reports from the literature, not real prospective patients or images. There is no uncertainty estimate or deferral rule; consensus is ...
Wu et al. (2026b)	To build and test an LLM-based multi-agent clinical decision support system, with a central orchestrator and 7 red-flag specialist agents, that detects ... Seven red flags were derived from 10 clinical guidelines. A LangGraph orchestrator reads each vignette, outputs a JSON routing decision, and sends ...	Gold-standard set of 90 adult secondary-headache cases (free-text vignettes). Sources: (1) 70 cases from PMC-Patients (a dataset of 167,000 patient ...	Multi-agent with guideline prompting (GPrompt) gave the best F1 for every model: Qwen-30b 0.605 (vs 0.542 single-LLM QPrompt), GPT-OSS-20b 0.568 (vs 0.474), Qwen-14b 0.603 (vs 0.568 in table), Qwen-8b 0.594 (vs 0.557) ...	Best F1 of 0.605 is too low for clinical use. Tested only on curated published case reports, not real primary-care notes, and only 7 red flags. No calibrated confidence ...
Wang et al. (2025b)	To make multimodal medical diagnosis by vision-language models evidence-based and step by step, instead of one-hop question answering. The workflow uses retrieved ... A hierarchical agentic workflow with GPT-4o as the core VLM. At disease level, a RAG agent retrieves guideline text from a MedlinePlus knowledge base ...	(1) REFUGE2: retinal fundus images for glaucoma diagnosis (binary); test size Not reported. (2) MITEA: 3D echocardiography for heart disease ...	Glaucoma (REFUGE2): bAcc 90.4%, F1 76.4%, AUC 95.1% vs GPT-4o bAcc 56.4%, F1 21.1%. Heart disease (MITEA): bAcc 77.8%, F1 72.3% vs GPT-4o 56.8%, 28.1%. NEJM overall accuracy 81.7% vs MDAgent 73.8% and GPT-4o 70.9%. MIMIC chest ...	Depends on the availability of visual tools for each domain; qualitative steps still rely on a hallucination-prone VLM; the step reliability check is a VLM judgement ...
Yan et al. (2025)	To build a leakage-free, real-case, multi-department benchmark for end-to-end clinical diagnosis by LLMs, and to design an agent that routes each case to the best LLM ... ClinicalBench has 1,500 real de-identified cases (150 diseases x 10 cases, 24 departments) from Chinese Grade 3A hospitals and 8 tasks from ...	ClinicalBench: 1,500 real cases, 150 diseases x 10 cases each, 24 departments, average about 1,000 tokens per case, Chinese originals translated to ...	Best single LLM overall was InternLM2-20B-Chat (Avg 51.98, DWR 14.91); GPT-4 had the best CDR (33.27%) and Acceptability (11.37). ClinicalAgent variants raised CDR to 52.73 (Agent#3@1) and 54.00 (Agent#1@3), and Acceptability to ...	Imaging is given only as text reports, not images. No uncertainty, calibration or deferral to clinicians. No few-shot or RAG tested. Data only from mainland China and ...
Jiang et al. (2025)	To build a standard benchmark that tests whether LLM agents can carry out real clinical record tasks, such as reading labs and placing orders, inside a realistic ... Two internal-medicine	Patient profiles: 100 patients randomly sampled from a STARR cohort who had an inpatient sodium	Best overall success rate was 69.67% (Claude 3.5 Sonnet v2), then GPT-4o 64.00% and DeepSeek-V3 62.67%; Mistral v0.3 7B scored 4.00%. Most models did better on	No agent is reliable enough for unsupervised clinical action, especially for write actions and multi-step tasks. The benchmark covers

Study	Aim and design	Data	Main result	Limitation for this study
	physicians wrote 300 patient-specific tasks in 10 categories. The tasks run against a FHIR-compliant server holding ...	lab test ordered on the morning of 13 November 2023. ...	query tasks than on action tasks (Claude 85.33% vs 54.00%). Hard ...	only medical-record tasks, not ...
Fallahpour et al. (2025)	To build a training-free AI agent that combines many specialised chest X-ray (CXR) models with a large language model so it can answer complex, multi-step CXR questions. ... MedRAX uses GPT-4o as a reasoning engine in a ReAct loop (observe, think, act) built on LangChain and LangGraph. The LLM picks and calls specialised ...	ChestAgentBench (new): 675 chest cases with CXRs and full documentation from Eurorad (European Society of Radiology peer-reviewed case database), CC ...	MedRAX reached 63.1% overall accuracy on ChestAgentBench versus 57.9% (Llama-3.2-90B), 56.4% (GPT-4o), 39.5% (CheXagent) and 28.7% (LLaVA-Med). On CheXbench it scored 68.1% overall. On MIMIC-CXR report generation it had mF1-14 ...	No uncertainty quantification, so the agent cannot say when it is unsure or hand over to a clinician. It struggles when tools disagree. Benchmark questions are generated ...
Lou et al. (2026)	To build a chest X-ray (CXR) agent in which a central director LLM calls many CXR tools, checks each tool output against visual evidence, and then forms a virtual expert ... A multimodal LLM director (GPT-4o or Qwen-VL-Max) runs three stages: ReAct tool invocation with an Evidence-driven Validator (EDV) that lists ...	1) CheXbench (public): image perception and image-text reasoning tasks used: single disease identification (SDI) on MIMIC-CXR and CheXpert, multiple ...	CXRAgent (Qwen) reached 67.0% overall accuracy on CheXbench and 75.6% on Medical-CXR-VQA; CXRAgent (GPT-4o) reached 66.3% and 71.7%. On MIMIC-CXR report generation (GPT-4o director) it got RaTEScore 0.569, GREEN 0.359, RadGraph ...	Confidence from EDV is a qualitative LLM judgement (High/Medium/Low), not a calibrated probability, and there is no formal deferral rule to a clinician. Only chest ...
Nath et al. (2025)	To show that a medical vision-language model (VLM) becomes more accurate when it learns to call specialised medical expert models (segmentation and classification) on ... VILA-M3 is built on the open VILA framework (vision encoder + linear projector + LLM). It is fine-tuned on medical VQA, report, classification and ...	Training mixture (Supp. Table 7, original count -> balancing factor -> balanced count): USMLE medical text 10,178 x10 = 101,780; RadVQA 6,281 x16 = ...	Average score over 8 metrics: VILA-M3-40B 64.3, 13B 63.4, 8B 63.0, 3B 61.5 vs Med-Gemini 55.7 and task-specific SOTA 58.6 (about 9% over Med-Gemini, 6% over task SOTA). VQA-RAD accuracy 90.4 (40B), PathVQA 92.7, MIMIC-CXR BLEU-4 ...	Brain MRI is only used through a 2D input plus a 3D BraTS expert; no quantitative brain-tumour segmentation or diagnosis results are reported for the VLM. There is no ...
Ellis-Caleo et al. (2026)	To find which LLM system best writes short, tumour-board-ready patient summaries from the electronic health record, and then to deploy it and monitor it at the Stanford ... Five automated GPT-4.1 summarisers, from a single-note prompt to a high-autonomy multi-agent system, were compared with a physician reference and the ...	Randomly chosen 50 thoracic tumour-board cases from April to June 2025 (one academic quarter). Data: unstructured clinical notes in the Epic EHR ...	LLM judge agreed with physicians: exact agreement 89.3-89.4%, Cohen kappa 0.687-0.709, Gwet AC1 0.870-0.871; Fleiss kappa 0.731 (2 physicians) vs 0.708 (with LLM). Single-step, multi-step and low-autonomy multi-agent systems were ...	Single centre, single cancer type, text notes only (no imaging or genomics attachments). No uncertainty or deferral mechanism; no 'single agent with all tools' baseline. ...
Raza et al. (2026)	To review trust, risk and security management (TRiSM) for multi-agent systems built on large language models (AMAS). The review proposes a framework, a risk taxonomy and ... Structured literature review guided by three research questions. The authors searched six digital libraries (2022-2025) with a two-block Boolean ...	No patient data. Literature corpus: publications from January 2022 to December 2025 (plus earlier foundational works). 250 papers shortlisted, 180 ...	250 papers shortlisted after title/abstract screening; 180 primary studies kept after full-text screening, plus standards and reports. Journal articles made up 61.2% of the corpus. Outputs: a TRiSM framework for AMAS; a risk ...	No empirical validation of the framework or of CSS/TUE on a real system; weights in the composite scores are left to the user. Few clinical examples. No concrete method ...
Cemri et al. (2025)	To find out why LLM-based multi-agent systems (MAS) fail. The authors build a failure taxonomy (MAST) and a large annotated dataset of failed and successful MAS runs ... Six experts used Grounded Theory to study 150 execution traces from 5 MAS frameworks and derived 14 failure modes in 3 categories. Definitions were ...	MAST-Data: 1642 annotated execution traces from 7 MAS frameworks. Human-evaluated/human-annotated/LLM-annotated sets of 30 traces each for ChatDev ...	MAS failure rates were 41% to 86.7% on 7 open-source MAS. Failure categories across 1642 traces: system design issues 44.2%, inter-agent misalignment 32.3%, task verification 23.5%. Most common modes: step repetition 15.7% ...	MAST is not claimed to be exhaustive. Simple prompt or topology fixes give only modest gains and task success stays low. Inter-agent misalignment needs agents that model ...
Kirchhof et al. (2025)	To argue that the usual split of uncertainty into aleatoric and epistemic parts does not work for interactive, multi-turn LLM agents. It proposes new research directions ... A position paper built on a critical literature review and small worked examples. It shows with toy cases (a two-Dirac Bernoulli belief, a linear ...	Not applicable. No dataset is used by the authors. Datasets only mentioned through cited work: Natural Questions (56% ambiguous questions) ...	No new experiments. Key cited evidence: aleatoric and epistemic estimators are almost perfectly correlated, with rank correlations of 0.8 to 0.999 across methods (Mucsanyi et al., 2024); ambiguity appears in 56% of Natural ...	No benchmarks, metrics or methods yet exist for underspecification uncertainty, for deciding which and how many follow-up questions an agent should ask, or for faithful ...
Liu et al. (2025b)	To review uncertainty quantification (UQ) and confidence calibration methods for LLMs. It organises them in a new taxonomy by uncertainty dimension and computational ... A narrative survey. It defines four uncertainty dimensions (input, reasoning, parameter, prediction) and separates uncertainty U(x) from confidence ...	No datasets used by the authors. Benchmarks reviewed (Table 3): reading comprehension - TriviaQA, CoQA, RACE, CosmosQA, SQuAD, HotpotQA; reasoning ...	No new experiments. Main output is Table 2 (method taxonomy) and Table 3 (benchmarks). Cited numbers: over 30% of answers in medical QA contain factual errors; uncertainty-aware flagging could reduce diagnostic errors by up to ...	Few methods handle input (ambiguity) uncertainty. Most UQ works on a single output and cannot track uncertainty across multi-step reasoning chains or multi-agent plans. ...
Atf et al. (2025)	To review how uncertainty in large language models (LLMs)	Not applicable. No dataset was used.	No quantitative results. The main output is a conceptual	No empirical test of the proposed

Study	Aim and design	Data	Main result	Limitation for this study
	used in medicine can be measured, explained and managed. The paper also proposes a conceptual framework that ... A narrative (non-systematic) literature review combined with a philosophical discussion of uncertainty. The authors group the sources of uncertainty ...	The paper discusses general data types (EHR, clinical notes, imaging reports, genetic data, wearable data) as ...	framework with eight parts: probabilistic modelling and Bayesian inference; hybrid uncertainty reduction (deep ensembles, Monte Carlo dropout); linguistic confidence ...	framework. No benchmark, dataset or metric values are given. It is not shown how the uncertainty scores should be turned into concrete ...
Alves et al. (2025)	To provide a framework (OpenL2D) that creates realistic synthetic human experts with work-capacity limits, and a ready dataset (FiFAR) for benchmarking learning-to-defer ... OpenL2D simulates each expert's false-positive and false-negative error probabilities as sigmoid functions of pre-processed features and the model ...	(1) Bank Account Fraud (BAF) dataset, base variant version 1 (Kaggle, Jesus et al.): 1,000,000 bank account opening applications over 8 months ...	Synthetic experts showed inter-rater Cohen's kappa from 0.06 to 0.79 (mean 0.49) and pairwise agreement 0.32 to 0.95 (mean 0.75), close to values reported for real cardiologists. Classifier h: ROC AUC 0.71, ECE 4.9%. ...	Only binary classification and tabular data are supported. Experts are synthetic and cannot replace real human data. Expert-correctness models reach AUC of at most about ...
Vazhentsev et al. (2026)	To compare post-hoc uncertainty quantification (UQ) methods for selective prediction (abstention) in diagnosis tasks from medical text, and to propose a better hybrid ... Fine-tuned clinical Transformer encoders on mortality prediction, ICD-10 code prediction, outpatient diagnosis and mental-health detection, then ...	1) MIMIC-III mortality (binary, discharge summaries, 2001-2012): 34,692 patients, 260,195 records, 2 classes, record length quartiles (4.9k, 8.4k ...	Normalised RC-AUC (first 50%): MIMIC mortality best MC (PV) / MC (BALD) 0.49 vs SR 0.38, HUQ2-MD 0.47; OV dataset best HUQ-RDE 0.47 vs SR 0.40 and MC (SMP) 0.44; MIMIC-IV top-50 ICD-10 FR-AUC best HUQ2-MD 0.35 vs MD 0.26 and SR ...	Only encoder-only classifiers; no generative decoder LLMs; no method guarantees high uncertainty on errors; hybrid methods need a labelled validation set; individual ...
Ferber et al. (2026)	To test whether an autonomous LLM agent (MIRA) working inside a sandboxed, FHIR-based EHR can manage full emergency department cases (history, tests, diagnosis ... MIRA (GPT-4o physician agent with an o1-preview planning tool) talks to an LLM patient agent grounded in real MIMIC-IV histories and calls 11 ...	MIMIC-IV v2.2 (about 300,000 patients). Benchmark of 574 emergency department admissions across 8 diagnoses: appendicitis, cholecystitis ...	Diagnostic accuracy 88.9% vs MIMIC-IV labels (n = 574). On 311 matched cases, MIRA scored 87.8% vs 78.1% for board-certified physicians ($P < 0.001$) and 71.1% for the mixed cohort ($P < 0.001$). Procedure recall was 53.5% vs 38.3% ...	Retrospective sandbox simulation only; patient speech derived from structured HPI text; single public dataset with possible training-data contamination; no imaging ...
Liu et al. (2026b)	To build a multi-agent system that copies how a multidisciplinary tumour board (MDT) makes lesion-level cancer decisions, with traceable evidence and safety checks. To ... Five LLM agents (Diagnostic, Treatment, Safety, Monitoring, Coordinator) built on DeepSeek V3/R1 reason over a pan-cancer guideline knowledge base ...	Public QA benchmarks (filtered for oncology relevance): MedQA (general medical exam MCQ); MedQA-Cancer (cancer-only subset of MedQA); MedMCQA-Cancer ...	Benchmark accuracy 0.86 (MedQA), 0.89 (MedQA-Cancer), 0.75 (MedMCQA-Cancer) vs Llama3-70B 0.76/0.76/0.63 and Claude-3 0.79/0.75/0.69. BERTScore 0.66 (MedQARo), 0.63 (MedQuAD), 0.68 (ROND). On 405 real cases semantic similarity ...	Only four cancer types and retrospective Chinese data; no imaging or raw unstructured notes as direct input; no prospective deployment or patient outcome study; no ...
Sorka et al. (2025)	To test how well large language models (LLMs) handle neurology board-exam reasoning, and whether retrieval-augmented generation (RAG) or a five-agent framework improves ... The authors built a benchmark of 305 Israeli neurology board exam questions and graded each on three complexity dimensions. They tested 10 LLMs as ...	(1) Israeli Board Certification Exams in Neurology, June 2023 to September 2024, three exam periods: 305 four-option multiple-choice questions ...	Base accuracy ranged from 46.8% (DeepSeek-R1-8B, MedLLaMA3) to 90.9% (o1). RAG raised GPT-4o from 80.5% to 87.3% but cut Meditron-70B from 52.9% to 41.2%. The multi-agent system reached 94.6% with o1, 89.3% with GPT-4o and 89.2% ...	Text-only questions; no images, no real patient records, no uncertainty-based deferral to clinicians. The validator agent did not catch retrieval gaps and often ...
Loaiza-Bonilla et al. (2026)	To test prospectively whether a neuro-symbolic multi-agent AI platform with an oncology knowledge graph and human review can match cancer patients to clinical trials ... LLM agents extract and normalise patient data from EHR documents, and a curated oncology knowledge graph (OncoGraph) runs deterministic inclusion ...	Prospective cohort: 3804 consecutive patients with metastatic or progressive solid tumours, ECOG 0-2, balanced for cancer-type incidence; broad ...	F1 0.82 (95% CI about 0.81-0.83; sensitivity 0.84, specificity 0.84, precision 0.81) vs GPT-4 zero-shot F1 0.47, GPT-4 chain-of-thought 0.67 and GPT-4o 0.47. Screening time fell from about 120 min to about 30 min per patient. ...	Only identification efficiency measured, not consent or enrollment; single health system; commercial platform with author conflicts of interest; underlying models ...
Lu et al. (2026)	To review agentic AI systems in computational pathology, classify them with an operational taxonomy, and judge their evidence, costs, risks and the conditions for safe ... Narrative review that classifies systems on three operational dimensions: dynamic control flow (D), inference-time tool selection (T) and external ...	No primary data. Datasets and benchmarks reviewed: Camelyon16 (WSI metastasis detection), PathVQA (patch-level VQA), WSI-VQA, WSI-Caption, PathMMU ...	Agentic systems show technical feasibility but no proven clinical benefit; most are retrospective prototypes at autonomy Level 1 or Level 1-2, and none had regulatory clearance for autonomous diagnosis. Only one prospective study ...	No prospective pathology cohort studies, few matched ablations separating agentic organisation from backbone and data, poor reporting of latency and cost, no measure of ...
Sharma et al. (2026)	To build a multi-agent LLM system that supports Emergency Severity Index (ESI) triage in two stages (before and after vital signs) while the clinician keeps the final ... An orchestrator agent built with LangGraph routes work between	ESI Implementation Handbook v4 (AHRQ, 2012) case studies, text vignettes, N = 60: 30 practice cases and 30 competency cases. ESI	Phase 1 (symptoms only): exact ESI match 66.67% (95% CI 55.00-78.33%), within +/-1 level 95.00%, high/low priority accuracy 86.67%, high-priority sensitivity 87.18% (34/39), specificity 85.71%. Phase 2 (with	Only simulated vignettes were used. There is no prospective test in a real emergency department, no real patients, no demographic or equity analysis, no multilingual or ...

Study	Aim and design	Data	Main result	Limitation for this study
	four GPT-4.1 agents: intake interview, clinical reasoning with RAG over the ESI ...	distribution ...	vitals): exact match ...	
Presacan et al. (2026)	To review how LLMs abstain (refuse, defer or ask for clarification) in healthcare, and to propose a decision-theoretic model and an evaluation framework (MedSAFE) for ... A PRISMA-guided systematic review of 4 databases plus manual search (January 2021 to September 2025) screened 1,046 records and included 28 studies ...	Review corpus: searches in PubMed, ACM DL, IEEE Xplore, Semantic Scholar plus manual Google Scholar/Google search; last search 23 September 2025 ...	Review: 1,046 records screened, 92 full texts assessed, 28 included; nearly all studies from 2024-2025; uncertainty-driven abstention is the most common motivation; extrinsic methods dominate and fine-tuning is the only intrinsic ...	Few benchmarks test abstention in realistic, multi-turn clinical settings; most treat abstention as binary; no unified predictor joins epistemic and normative signals ...
Zhao et al. (2025)	To build an uncertainty-aware pathology AI (PICTURE) that separates glioblastoma from primary CNS lymphoma (PCNSL) on FFPE and frozen-section slides, and flags tumour ... Whole-slide images are tiled, features are extracted with nine pathology foundation models, and a one-hidden-layer MLP classifies glioblastoma vs ...	2141 GBM/PCNSL slides (1760 FFPE, 381 frozen) from five centres: Mayo 273 FFPE (one slide per patient; 166 GBM, 107 PCNSL; development set); UPenn ...	Held-out Mayo test AUROC 0.998 (95% CI 0.995-1.0). External FFPE AUROC: UPenn 0.996, BWH 0.987, Vienna 0.992, TVGH 0.992. Frozen sections: UPenn 0.958, BWH 0.987, Vienna 0.982, TVGH 0.924. OOD detection of 67 unseen CNS tumour ...	Mostly Caucasian patients; small PCNSL numbers per site; only the glioblastoma vs PCNSL pair; no multi-omics; no prospective clinical deployment or measurement of how ...
Kim et al. (2025b)	To test whether working together with a large language model (LLM) changes the accuracy and speed of brain MRI differential diagnosis by radiology residents. It also ... A retrospective, single-centre, randomised cross-over usability study. Six radiology residents read 40 difficult brain MRI cases, half with ...	Local imaging database, Klinikum rechts der Isar. 40 brain MRI exams acquired between 01/01/2016 and 12/31/2023, split at random into two sets of 20. ...	LLM-assisted reading gave higher accuracy: 70/114 (61.4%) vs 53/114 (46.5%) correct, p = 0.033 (chi-square); median numeric score 1 vs 0, p = 0.021; Cohen's h = 0.30. Mixed model beta = 0.150, SE = 0.065, p = 0.021. No change in ...	Only inexperienced residents and a controlled setting were tested. The LLM was text-only, so readers had to describe images by hand. No uncertainty signal or deferral ...
Lekadir et al. (2025)	To set an international consensus guideline for building and deploying trustworthy AI tools in healthcare. It covers the whole AI lifecycle, from design to deployment ... A modified Delphi consensus run over 24 months (8 rounds) by the FUTURE-AI Consortium, which grew to 117 experts from 50 countries. Six guiding ...	No patient data. Participants: round 1, six working groups of 3 experts each (18 experts); rounds 2-5, 72 experts (AI scientists, clinicians ...	The final guideline has 30 recommendations: fairness 3, universality 4, traceability 6, usability 5, robustness 3, explainability 2, general 7. All were approved with less than 5% disagreement in the final vote. Round 1 produced ...	Recommendations are generic and do not prescribe specific techniques. Liability and who audits remain open. Current regulations block post-release model updates. ...

Table 2.7: Evidence table for uncertainty, calibration and deferral (O2): 30 representative studies.

Study	Aim and design	Data	Main result	Limitation for this study
Angelopoulos & Bates (2021) (base)	To explain conformal prediction in a practical way. Tutorial with code: split conformal, adaptive sets, conformal risk control, examples on images and medicine.	Worked examples on ImageNet classification (prediction sets), regression with quantile regression, and other examples (tumour segmentation, time ...	Shows how to turn any model output into prediction sets with a guaranteed coverage.	Guarantee needs exchangeable data, which is not true across hospitals.
Verma & Nalisnick (2022) (base)	To build a learning-to-defer (L2D) system whose probability that the human expert is correct is properly calibrated, without losing accuracy. Shows that the softmax-based L2D loss of Mozannar and Sontag (2020) does not give valid or calibrated expert-correctness probabilities. Proposes a ...	CIFAR-10 with a simulated expert (correct with probability 0.70-0.75 on some classes, random on the others); HAM10000 dermoscopy images with a ...	On CIFAR-10 with a simulated expert, ECE of the expert-correctness estimate fell from 7.58% (softmax) to 3.01% (OvA). Accuracy was equal or better than the softmax method on hate speech, galaxy zoo, HAM10000 skin lesions and ...	Single expert; simulated experts on most datasets; does not handle a budget on expert time or different error costs.
Kwon & Kim (2026) (base)	To build a clinical triage system that gives set-valued predictions with coverage guarantees and defers low-confidence cases to clinicians when this lowers the expected ... Calibrated probabilistic model + split conformal prediction (plus group-conditional Mondrian for gender-stratified coverage and importance-weighted ...	Early sepsis prediction EHR data with temporally separated ID and OOD test splits (details in paper).	For early sepsis prediction, at 80% coverage selective deferral reduced error on retained cases by 49.6% (in-distribution) and 46.7% (out-of-distribution); low ECE on both splits; near-perfect NPV at 95% sensitivity; coverage ...	Tabular EHR sepsis task, not imaging; single deferral threshold; costs assumed.
Mehta et al. (2022)	To create a fair way to score uncertainty maps in brain tumour segmentation and compare methods. New score that rewards	BraTS multiparametric MRI (T1, T1-Gd, T2, FLAIR) of glioma patients;	Ranked many teams. Ensemble and test-time augmentation methods were common among top teams.	Voxel-level score only, not linked to clinical decisions.

Study	Aim and design	Data	Main result	Limitation for this study
	confidence on correct voxels and uncertainty on wrong ones, and penalises throwing away too many correct voxels. Used to rank ...	BraTS 2019: 335 training, 125 validation, 166 test patients; QU-BraTS 2020 test ...		
Charusaie et al. (2022)	To show, in theory and practice, when learning a classifier that complements the human expert beats training the classifier alone, and how to learn deferral with few ... Compares joint learning (classifier and rejector trained together) with staged learning (classifier first, rejector after). Gives generalisation ...	CIFAR-10 (32x32, 10 classes) with the simulated expert of Mozannar and Sontag (2020): perfect on the first 5 classes, random on the other 5. Settings ...	The accuracy gap between joint and staged learning shrinks as the classifier gets more complex, as the theory predicts. When only a small part of the data has expert labels, staged and semi-supervised joint learning can be more ...	Experiments use simulated experts on CIFAR-10 only; no medical data, no workload limit.
Tailor et al. (2024)	To build a learning-to-defer system that works with experts it has never seen before, without retraining for each new expert. L2D-Pop: meta-learning over a population of experts. A small context set of the new expert's past decisions is embedded and fed to the rejector, so ...	Synthetic 2D clusters; Traffic Signs (GTSRB, train set down-sampled to 10,000); CIFAR-10; HAM10000 skin lesion images; CIFAR-100 merged into 20 ...	Across Traffic Signs, CIFAR-10 and HAM10000, L2D-Pop raised the expert's accuracy on deferred cases and overall system accuracy compared with a single-expert L2D model; the gain grew as experts differed more. Cross-attention ...	Experts simulated from class-wise skills; real clinicians' changing behaviour not tested.
Mao et al. (2024a)	To design surrogate losses with strong theoretical guarantees for learning to defer when there are several experts, each with its own cost. Adds one output per expert to the label set and learns prediction and deferral together with a new family of surrogate losses (built from comp-sum ...	SVHN and CIFAR-10 image classification.	System accuracy rose as more experts were available: CIFAR-10 73.31% (1 expert), 77.12% (2), 78.71% (3); SVHN 92.08%, 93.18%, 93.46%.	Experts are other models, not humans; mainly theoretical; no workload limit.
Verma et al. (2023)	To extend learning to defer to many experts with consistent losses, calibrated estimates of each expert's correctness, and a principled way to ask a subset of experts. Derives two consistent multi-expert surrogates (softmax and one-vs-all), analyses how well each estimates P(expert j is correct | x), and proposes ...	Hate Speech (tweets with real annotator labels), Galaxy-Zoo (real volunteer annotations), HAM10000 dermoscopy images (7 diagnostic categories ...	OvA never fell below the single-classifier baseline as experts were added on HAM10000 and Galaxy-Zoo, while softmax and mixture-of-experts sometimes did. Softmax mis-calibration spreads across experts; OvA does not (with some ...	Most experts simulated; querying several experts costs more clinician time, which is not limited in the model.
Alves et al. (2024)	To make learning to defer practical by handling different costs for false positives and false negatives, needing only one human decision per training case, and ... DeCCaF: (1) an ML classifier; (2) a human expertise model that predicts each expert's probability of being correct, trained with supervised learning ...	Bank Account Fraud dataset (version 1): 1 million synthetic bank-account applications based on anonymised real data, 1.1% fraud prevalence, with ...	Across fraud-detection scenarios with teams of 9 synthetic analysts and individual capacity limits, DeCCaF significantly beat the baselines, with an average 8.4% reduction in misclassification cost.	Fraud data and synthetic experts, not clinical; static capacity per batch.
Jaeger et al. (2023)	To unify how misclassification detection, selective classification and out-of-distribution detection are evaluated, and to test methods under a realistic range of ... Proposes the FD-Shifts benchmark: confidence scoring functions (CSFs) are evaluated by the area under the risk-coverage curve (AURC) on i.i.d. data ...	CIFAR-10, CIFAR-100, SVHN, TinyImageNet, iWildCam (Wilds), BREEDS (ImageNet sub-population shift), CAMELYON-17-Wilds (histopathology, hospital ...	None of the evaluated literature methods beat the simple maximum softmax response (MSR) baseline across realistic failure sources; MSR or MC-dropout MSR was best or near-best on all i.i.d. and most shifted test sets except ...	Natural and some medical-domain images (CAMELYON pathology); no 3D MRI; no human deferral.
Bungert et al. (2023)	To test whether current confidence scoring functions can prevent silent failures of medical image classifiers under realistic distribution shifts, and to provide a tool ... Benchmark of 8 confidence scoring functions on four biomedical classification tasks with i.i.d. tests, image corruptions, acquisition shifts (new ...	Dermoscopy: ISIC 2020 + derm7pt + PH2 + HAM10000, benign vs malignant; acquisition shifts with MSKCC or Hospital Clinic Barcelona as target domain ...	No confidence scoring function reliably prevented silent failures; none beat maximum softmax response across shifts; MC-dropout MSR was best overall and DG-MCD-MSR best on i.i.d. data in 3 of 4 tasks.	2D classification only; no MRI brain tumour data; no human-in-the-loop evaluation.
Mukhoti et al. (2023)	To get reliable epistemic and aleatoric uncertainty from a single forward pass of one network, matching deep ensembles without their cost. Deep Deterministic Uncertainty (DDU): train a standard softmax network with a well-regularised feature space (residual connections + spectral ...	CIFAR-10 and CIFAR-100 (in-distribution) vs SVHN, CIFAR-100 and Tiny-ImageNet (OOD); ImageNet vs ImageNet-O; active learning on MNIST / Dirty-MNIST ...	DDU beat the single-pass methods DUQ and SNGP on OOD detection and matched 5-member deep ensembles on CIFAR-10/100 vs SVHN / Tiny-ImageNet, and on ImageNet vs ImageNet-O; it also worked well for active learning.	Classification only; OOD detection on natural images; not tested for medical segmentation or deferral.
Angelopoulos et al. (2024a)	To extend conformal prediction from controlling coverage to controlling the expected value of any monotone loss, such as	Gut polyp segmentation pooled from Kvasir, Hyper-Kvasir, CVC-	Worked examples controlled the target risk at alpha = 0.1: polyp segmentation false negative rate mean 0.0987	Guarantee is marginal (on average), not per patient; assumes exchangeable data.

Study	Aim and design	Data	Main result	Limitation for this study
	the false negative rate of a segmentation. Conformal risk control: on n calibration points compute the loss L_i(lambda) as a function of a threshold lambda; choose the smallest lambda such ...	ColonDB, CVC-ClinicDB and ETIS-Larib (n = 1,000 calibration, 781 evaluation); MS COCO ...	(SD 0.0114) over 1,000 random splits; MS COCO multilabel FNR 0.0996 (SD 0.0052); also hierarchical ImageNet graph ...	
Angelopoulos et al. (2025)	To calibrate any trained model so that its predictions meet explicit finite-sample statistical guarantees on one or several risks, without refitting the model. Learn then Test (LTT): turn the choice of model parameters (thresholds lambda) into multiple hypothesis testing. For each candidate lambda test ...	Synthetic autoregressive loss processes (V-shaped risk) for numerical comparison; MS COCO multi-label classification; MS COCO instance segmentation ...	Worked examples achieved the target guarantees: false discovery rate control in multi-label classification, intersection-over-union control in instance segmentation, simultaneous control of outlier-detection type-1 error and set ...	Marginal guarantees; requires exchangeable calibration data and enough calibration size.
Mehrtens et al. (2023a)	To show the limitations and assumptions of conformal prediction that practitioners must know before using it in medical image classification. Trains ResNet-34 models on dermatology and histopathology data and demonstrates, with controlled experiments, how conformal guarantees fail under ...	HAM10000 (7 skin-lesion classes) and CAMELYON17 (50 whole-slide images with annotated tumour regions from 5 clinics, cut into 256x256 non-overlapping ...	Conformal predictions were unreliable under distribution shift in inputs and labels; they should not be used to select confident predictions to improve accuracy; coverage was not guaranteed for subsets such as individual classes ...	Points out problems; does not provide a fix for shift.
Zhang et al. (2023b)	To make conformal prediction actually meet very small user-specified error rates (e.g. 0.5%) in medical image classification. Reliable-Region-Based Conformal Prediction (RR-CP): estimates on the calibration set a 'reliable region' of the score space where the set-valued ...	MedMNIST: OrganAMNIST (11 classes; 34,581 / 6,491 / 17,778 train/val/test), OrganCMNIST (11; 13,000 / 2,392 / 8,268), OrganSMNIST (11; 13,940 / 2,452 ...	With alpha = 0.5%, RR-CP met the target error rate on 4 of 5 MedMNIST datasets while classic CP, RAPS and ERAPS failed on most; e.g. BloodMNIST actual error 0.39% vs 0.85% (CP), 0.59% (RAPS), 0.74% (ERAPS).	Small 2D MedMNIST images; no distribution shift; larger set sizes.
Borges et al. (2026)	To give a single, well-founded confidence score for a whole segmentation, so the model can abstain on images whose segmentation is likely poor (image-level selective ... Derives the optimal image-level confidence estimator under the Dice metric when pixel posteriors are known (intractable for real images), then ...	Brain tumour: BraTS 2020 (3D nnU-Net); breast ultrasound: BUSI (UNeXt); skin lesions: ISIC 2018 (UNeXt); multiple sclerosis white-matter lesions ...	In simulations SDC's risk-coverage curve was almost identical to the optimal estimator. On six medical segmentation tasks (including BraTS 2020 brain tumour with 3D nnU-Net), SDC beat earlier image-level confidence measures ...	Binary or per-class Dice; image-level abstention only, not region-level escalation.
Sambyal et al. (2023)	To study how training regime, architecture and dataset affect the calibration (not just accuracy) of medical image classifiers. Empirical study of ResNet-18, ResNet-50 and WideResNet trained three ways - fully supervised from random initialisation, fully supervised with ...	Diabetic Retinopathy (EyePACS, 35K high-resolution fundus images, 5 severity classes, varied cameras); Histopathologic Cancer (modified ...	Rotation-based self-supervised pre-trained models were significantly better calibrated with comparable or better accuracy than fully supervised models; weight distributions and similarity of learned representations correlated ...	Classification only; one self-supervised method; small COVID dataset.
Lambert et al. (2024a)	To review uncertainty quantification (UQ) methods for deep learning in medical image classification and segmentation, how they are evaluated, and the open challenges. Systematic literature search (Google Scholar and PubMed, June 2022) for supervised DL studies with UQ in medical image classification or segmentation ...	Not applicable (review of published studies on MRI, CT, PET, X-ray and other imaging).	Groups methods into Bayesian (variational inference, MC dropout, SWAG/Laplace), ensemble, test-time augmentation, deterministic single-pass (evidential, distance-aware) and post-hoc approaches; catalogues evaluation metrics ...	Review up to mid-2022; no quantitative meta-analysis.
Koutsoubis et al. (2025)	To review federated learning (FL), privacy-preserving FL and uncertainty quantification within FL for medical imaging, and identify gaps. Narrative review of FL algorithms, privacy threats from shared gradients and defences (differential privacy, homomorphic encryption, secure ...	Not applicable (review).	FL lets institutions train jointly without sharing data, but gradients can leak information; privacy defences trade off against performance; UQ in FL is hard because client data differ, and standard calibration and UQ methods ...	Lack of studies combining privacy-preserving FL with uncertainty quantification on real multi-site medical data.
Bangalore Yogananda et al. (2025)	To predict glioma IDH mutation status non-invasively from MRI and give each prediction a calibrated confidence score that clinicians can act on. Multi-contrast 2D voxel-wise segmentation network (MC-net, nnU-Net pipeline) classifies each tumour voxel as IDH-mutant or wildtype; percentage of ...	2,481 pre-operative glioma cases with multi-contrast MRI and IDH status from 5 public sources (TCIA, EGD, UCSF, UPenn, IvyGAP) and ...	On 2,481 gliomas from eight institutions, MC-net + BLR reached 96.4% and 95.1% accuracy on diverse test groups, AUC 0.98; confidence scores were well calibrated (Brier score 0.0125).	IDH only (not full WHO 2021 integrated diagnosis); US/European cohorts; retrospective.
Elyassirad et al. (2025)	To apply conformal prediction to voxel-wise glioma segmentation and use the resulting uncertainty to flag cases with poor segmentation. U-Net segmentation trained on	UCSF glioma dataset (495 patients; WHO grade 2: 56, grade 3: 43, grade 4: 396) split 70/10/10/10 into	Base Dice 0.8628 (internal UCSF test) and 0.8257 (external UPenn). Conformal coverage 0.9982 internal and 0.9977 external. UR correlated strongly and	Two datasets; whole-tumour style segmentation; retrospective.

Study	Aim and design	Data	Main result	Limitation for this study
	UCSF gliomas; conformal threshold chosen from nonconformity scores on an internal or external calibration set; defines ...	train/validation/calibration/test; UPenn ...	negatively with Dice (r = -0.827, p < 0.001). Uncertain cases (20.4%, UR > ...	
Prince et al. (2023)	To add predictive uncertainty and an abstention mechanism to a deep learning classifier that diagnoses adamantinomatous craniopharyngioma (ACP) from preoperative MRI. Re-trains and calibrates the authors' earlier ACP classifier with a Bayesian deep learning approach (predictive distributions instead of point ...	Preoperative MRI from 86 paediatric suprasellar tumour patients across multiple institutions (same dataset as the authors' prior ACP study).	The calibrated model under-performed the earlier published results, showing the original model was over-fit; the mean of predictive distributions was not informative but the variance was. With abstention, accuracy rose from 80.8% ...	Small cohort (86 patients); paediatric suprasellar tumours only; high abstention rate.
Wang et al. (2025a)	To improve meningioma radiotherapy target (GTV) segmentation on MRI and provide pixel-wise uncertainty using a spherical-projection U-Net. SPU-Net: U-Net with spherical image projection that maps 2D slices onto a sphere from multiple projection centres, giving several segmentation ...	76 supratentorial meningioma patients treated with radiotherapy; GTVs contoured by one experienced radiation oncologist on high-resolution ...	SPU-Net vs standard U-Net: sensitivity 0.758 vs 0.746, Dice 0.760 vs 0.742, mean Hausdorff 0.612 vs 0.744 cm, HD95 2.682 vs 2.912 cm, plus an uncertainty map that U-Net lacks.	Single centre, single annotator, 76 patients; meningioma rather than glioma.
Alshamlan (2026)	To measure whether confidence- and entropy-based selective prediction reduces risk in a brain-MRI vision-language classifier, and at what coverage cost, within a ... LoRA-fine-tuned LLaVA-Med classifier with a disjunctive rejection rule (reject if softmax confidence low OR predictive entropy high); companion ...	BRISC2025 brain MRI classification test partition (1,000 images); BraTS 2020 for segmentation (74 held-out cases) and glioma grading (high vs low ...	Withholding 1.80% of predictions (95% CI 1.14-2.83) raised selective accuracy to 97.76% (96.63-98.52) and cut selective risk from 3.40% to 2.24% (34.1% relative reduction). Withheld cases were strongly enriched for errors (66.7% ...	Single operating point (continuous score AUROC not identifiable); 2D classification dataset; modules not evaluated together clinically.
Angelopoulos et al. (2024b)	To deploy a black-box medical imaging AI model safely in a new hospital by sorting patients into low-risk, high-risk and uncertain groups with statistical guarantees on ... Conformal triage: using a representative calibration set from the deployment site, choose two thresholds on the AI score so the high-risk group has ...	Head CT exams from Massachusetts General Hospital, scored by a head CT AI model developed previously by Do and colleagues (trained in a different ...	On a previously developed head CT model applied at Massachusetts General Hospital (a different setting from training), false positives fell from 233 (45%) to 8 (5%) while abstaining on only 14% of exams.	Requires a labelled representative sample from the new site; preprint; single head CT task.
Gade et al. (2024)	To estimate the uncertainty of a deep learning prostate segmentation with conformal prediction and test its effect on prostate volume calculation. Deep learning prostate segmentation on T2 MRI; pixel-wise conformal prediction at 85% confidence flags unreliable pixels; prostate volume ...	377 multi-centre 3T axial T2-weighted prostate MRI exams of biopsied men at risk of prostate cancer (age 66.64 +/- 7.47 years) with PI-RADS 2.1 ...	CP significantly reduced relative volume difference (RVD -2.81 +/- 8.85 vs -8.01 +/- 11.50 for DL alone). PV_CP agreed better with the reference: mean difference 1.27 mL (95% LoA -13.64 to 16.17) vs 6.07 mL (-14.29 to 26.42); ICC ...	Prostate, not brain; volume assessment only; retrospective.
Mehrtens et al. (2023b)	To evaluate common uncertainty and robustness methods for whole-slide image classification, focusing on selective classification under domain shift and label noise. Comparison of Deep Ensembles, Monte-Carlo Dropout, Stochastic Variational Inference, Test-Time Augmentation and ensembles of these at tile level ...	Tile level: CAMELYON17 lesion-level annotated set - 50 breast lymph-node WSIs from 5 clinics and 3 scanners (realistic domain shift). Slide level ...	Ensembles of methods generally gave the best uncertainty estimates and the most robustness to domain shift and label noise; other methods showed no systematic gain (unlike natural-image benchmarks); rejecting the most uncertain ...	Tasks limited to tumour detection in specific cancers; label-noise simulated.
Angelopoulos & Bates (2023)	To give a practical, self-contained introduction to conformal prediction and related distribution-free uncertainty quantification. Tutorial/monograph: split conformal prediction with any pre-trained model, score functions for classification and regression (e.g., adaptive ...	Illustrative examples (e.g., ImageNet classification, regression and medical-style examples).	Conformal methods produce prediction sets or intervals that contain the true answer with a user-chosen probability (e.g., 90%) without distributional or model assumptions, using only a held-out calibration set; set size naturally ...	Guarantees are marginal (on average), not per-patient; exchangeability assumption can break under site shift.
Yang et al. (2024)	To unify out-of-distribution (OOD) detection with related problems and review methods and benchmarks. Survey proposing 'generalised OOD detection' covering anomaly detection, novelty detection, open-set recognition, OOD detection and outlier ...	Not applicable (reviews CIFAR, ImageNet and other OOD benchmarks).	The five problems differ mainly in what counts as in-distribution and whether labels are used; the unified view clarifies settings and evaluation; open issues include near-OOD detection, evaluation realism and foundation-model ...	Mostly natural-image benchmarks; limited medical evaluation.

Table 2.8: Evidence table for data-efficient, multi-source and federated models (O3): 30 representative studies.

Study	Aim and design	Data	Main result	Limitation for this study
Isensee et al. (2021) (base)	To make segmentation work out-of-the-box on any new medical dataset without manual tuning. Rules that set preprocessing, network (2D/3D U-Net), training and post-processing automatically from dataset properties.	Medical Segmentation Decathlon phase 1: BrainTumour (BraTS multimodal MRI, 3 labels: oedema, non-enhancing, enhancing), Heart, Liver, Hippocampus ...	Beat most specialised methods on 23 public datasets and many challenges.	Needs strong GPU and no uncertainty output by default.
Baid et al. (2021) (base)	To describe the BraTS 2021 dataset and tasks. Large multi-institution multi-parametric MRI (T1, T1Gd, T2, FLAIR) with expert labels for enhancing tumour, tumour core and whole tumour. Also MGMT ...	About 2,000 multi-institutional glioma cases (up from 660 in earlier BraTS) with pre-operative mpMRI: native T1, post-contrast T1 (T1Gd), T2 ...	Around 2,000 glioma cases, the biggest public brain tumour MRI set at that time.	Mostly gliomas from Western centres, pre-processed and standardised.
Pati et al. (2022) (base)	To run a real worldwide federated study for glioblastoma boundary detection. 71 sites on 6 continents trained a segmentation model together with FL, data never left the site.	25,256 MRI scans (>5 TB) of 6,314 glioblastoma patients from 71 sites (T1, T1Gd, T2, FLAIR); public initial model trained on 231 cases from 16 sites ...	Big improvement over the model trained on public data for both surgically targetable tumour and whole tumour boundaries.	Large sites with good IT. Little on what happens with a handful of small sites.
Wang et al. (2023) (base)	To classify adult-type diffuse gliomas into WHO 2021 integrated types and grades directly from routine H&E whole-slide images. Patch clustering (K-means on ResNet-50 features, 9 clusters) to select informative patches, a patch-level CNN classifier and a patient-level ...	2,624 patients: training 1,362 and validation 340 (FAHZZU), internal test 289 (FAHZZU), external test 1 305 (HPPH), external test 2 328 (XHCMU) ...	AUCs above 0.90 for all major types: six-category task AUCs 0.904-0.995 across cohorts; three-type task 0.938-0.983; IDH-mutant A2-3 vs IDH-wildtype glioblastoma 0.935-0.984.	Chinese cohorts only; no uncertainty estimation or deferral to pathologists.
Ma et al. (2024)	To build one promptable segmentation model for many medical image types. Fine-tuned Segment Anything on over 1.5 million medical image-mask pairs across 10 modalities. User gives a bounding box.	1,570,263 medical image-mask pairs covering 10 modalities (CT, MRI, chest X-ray, dermoscopy, endoscopy, ultrasound, mammography, OCT, pathology, and ...	Better than SAM and comparable or better than specialist models on many internal and external tests.	2D slices, needs a box prompt, weaker on very small or unclear edges.
Chen et al. (2024c)	To pretrain one strong pathology feature model useful for many tasks. Self-supervised (DINOv2) training on over 100 million tissue patches from about 100,000 slides, tested on 34 tasks.	Pre-training: Mass-100K - over 100 million tissue patches from 100,426 diagnostic H&E WSIs across 20 major tissue types (MGH, Brigham and Women's ...	Better than earlier pathology encoders on most tasks, including rare cancer classification and few-shot learning.	Trained on US hospital data. Licence and compute limits.
Zenk et al. (2025)	To benchmark federated aggregation methods and the real-world robustness of brain tumour segmentation algorithms across many international sites, through the FeTS ... Task 1: simulated federated learning on a multicentric BraTS-derived dataset partitioned by institution; teams innovate on weight aggregation and ...	Multi-institutional glioma mpMRI derived from BraTS with institution labels for partitioning (Task 1); test data distributed across 32 institutions ...	Adaptive weight aggregation and selective client sampling improved both segmentation performance and efficiency (no trade-off). In Task 2, algorithms generalised well on average (median Dice mostly > 0.9 per institution) but ...	Glioma only; Task 1 FL is simulated; failure analysis per site is descriptive.
Linardos et al. (2025)	To evaluate new federated weight-aggregation methods for robust and communication-efficient glioma segmentation in a standardised FL setup. Six teams' aggregation methods were run in the same simulated FL setup on a multi-institutional BraTS-derived dataset and ranked by a cumulative ...	BraTS-derived multi-institutional glioma mpMRI: 1,251 training cases, 219 validation cases, 570 hidden test cases; labels ET, TC, WT; data ...	A PID-controller-based aggregation method won: mean Dice 0.733 (ET), 0.761 (TC), 0.751 (WT), HD95 33.922 / 33.623 / 32.309 mm, and best convergence score 0.764, surpassing top methods from earlier FeTS editions.	Simulated FL; absolute Dice lower than centralised models; glioma only.
Manthe et al. (2024)	To benchmark global, personalised and clustered federated learning algorithms for brain tumour segmentation on the FeTS 2022 dataset. Implements federated methods from three classes - global (one final model, e.g. FedAvg, FedProx, FedNova, FedAdam), personalised (one model per ...	FeTS 2022 (BraTS-derived) glioma mpMRI partitioned by institution (natural, highly unbalanced partition), plus artificial IID partition and a ...	Standard FedAvg already performed very well; some methods in each class gave slight improvements and could reduce bias of the final model towards the dominant data distribution in the federation. Behaviour differed under IID and ...	Simulated federation; one dataset; small gains over FedAvg.
Dai et al. (2024)	To do federated brain tumour segmentation when some hospitals only have a subset of MRI sequences, while giving	BraTS 2020 multimodal glioma MRI (FLAIR, T1c, T1, T2); clients	On BraTS 2020 simulating clients with only FLAIR, T1c, T1 or T2, FedMEMA outperformed recent	Simulated missing modalities; assumes one server with full-modal data.

Study	Aim and design	Data	Main result	Limitation for this study
	each hospital a personalised model. FedMEMA: one exclusive encoder per MRI modality shared across clients; a server with full-modal data trains a fusion decoder that combines all ...	simulated each holding a single modality, server holding full-modal data.	multimodal and personalised FL methods in mean Dice for clients and server; ablations confirmed the value of anchors and cross-attention ...	
Jiang et al. (2023)	To personalise federated segmentation models for each participating hospital (inside) and adapt them at test time for new hospitals outside the federation (outside). Inside personalisation: lightweight gradient-based local adaptation that accumulates global gradients (common knowledge) and local gradients ...	Prostate T2 MRI from 6 sources (12, 12, 30, 30, 13, 19 volumes) as clients A-F; retinal fundus optic disc/cup from 4 institutions (101, 159, 400, 400 ...	Significant Dice improvements over state-of-the-art personalised FL (pFedMe, FedRep, etc.) for inside clients and over test-time adaptation methods (TTT, Tent) for outside clients on prostate MRI and fundus segmentation.	2D segmentation; small datasets; no brain MRI.
Wang et al. (2024a)	To personalise federated segmentation models at both feature and prediction level for transformer-based networks. FedDP: (1) Local Query (LQ) - the query embedding layer of self-attention is kept private per site to adapt to local feature distributions; (2) ...	EndoPolyp (polyp segmentation from endoscopy, 4 sites A-D); RIF (retinal fundus optic disc/cup, 4 sites).	Outperformed state-of-the-art personalised FL methods on polyp segmentation (EndoPolyp) and optic disc/cup segmentation (RIF), e.g. RIF average Dice above FedAvg 91.06 and FineTune 91.52; EndoPolyp average above FedAvg 82.20.	2D endoscopy and fundus only; no MRI.
Lin et al. (2024a)	To do federated medical image segmentation when hospitals use different kinds of cheap (weak) annotations - points, scribbles, boxes, blocks - and have different data ... FedLPPA: a learnable universal knowledge prompt plus personalised prompts for each site's data distribution and annotation sparsity, fused with image ...	Fundus: Drishti-GS1 (50:51 train:test, scribble), RIM-ONE-r3 (99:60, scribble style 2), REFUGE-train (320:80, bounding box), REFUGE-val (320:80 ...	Outperformed state-of-the-art federated and weakly-supervised methods across fundus, OCTA, endoscopy and prostate MRI segmentation with heterogeneous weak labels, approaching fully supervised performance.	2D tasks; no brain tumour; prompts add complexity.
Chen et al. (2024a)	To choose which local images to annotate in federated active learning when hospitals have different data distributions. Federated Evidential Active Learning (FEAL): local and global models output Dirichlet distributions (evidential learning) to capture aleatoric and ...	Fed-ISIC skin lesions (4 sources: 12,413 / 3,954 / 3,363 / 2,259 images); Fed-Camelyon breast histology (5 centres: 59,436 / 34,904 / 85,054 / ...	FEAL beat other federated active learning methods on five multi-centre datasets (2 classification, 3 segmentation), reaching near full-annotation performance with a fraction of labels.	No brain MRI; annotation budgets simulated.
Lin et al. (2023)	To segment brain tumours by fusing MRI sequences the way radiologists read them, combining CNN and Transformer strengths. Groups input MRI sequences into two branches by imaging principle (e.g. T1 / T1ce vs T2 / FLAIR); dual-branch hybrid encoder with a ...	BraTS 2021 multimodal MRI (T1, T1ce, T2, FLAIR) training data with labels, split for training and testing.	Outperformed CNN-based (VNet, ResUNet, UNet++, AttentionUNet, DynUNet - BraTS21 winner) and transformer-based (TransBTS, UNETR, SwinUNETR, etc.) models on BraTS 2021 in Dice and HD95.	Single dataset (BraTS 2021); grouping fixed to the four BraTS sequences.
Liu et al. (2023)	To learn brain tumour segmentation that is robust to missing MRI sequences using self-supervised pre-training. Two stages. Stage 1: multimodal masked autoencoder (M3AE) - randomly drop whole modalities and mask random patches of the rest, and reconstruct them ...	BraTS 2018 and BraTS 2020 multimodal MRI (T1, T1c, T2, FLAIR).	On BraTS 2018 and 2020, outperformed U-HVED, ACN, SMU-Net and RFNet on average over missing-modality combinations, as a single 'catch-all' model.	Simulated missing modalities; BraTS only.
Cox et al. (2024)	To build a 3D brain MRI foundation model that needs few labels for tumour and lesion segmentation. Two-stage self-supervised pre-training of a Swin UNETR-style transformer (41,400 healthy UK Biobank brains, then disease data), followed by ...	Stage 1 pre-training: UK Biobank brain MRI of 41,400 generally healthy participants (T1w, T2-FLAIR; 82,800 images). Stage 2 and fine-tuning: BraTS ...	Beat previous challenge-winning fully supervised solutions on BraTS and ATLAS v2.0; larger models and more pre-training data improved results.	Pre-training data mainly from one high-income cohort (UK Biobank); heavy compute.
Ma et al. (2025)	To create a promptable 3D and video medical segmentation model and speed up large-scale annotation. Fine-tuned SAM2 on 455,000+ 3D image-mask pairs and 76,000 video frames; box or point prompts propagate across slices; human-in-the-loop annotation ...	Over 455,000 3D image-mask pairs (363,161 CT, 14,818 PET, 77,154 MRI) and 76,000 annotated video frames (19,232 ultrasound, 56,462 endoscopy). User ...	Outperformed previous models across organs, lesions and modalities; cut manual annotation cost by more than 85% in studies covering 5,000 CT lesions, 3,984 liver MRI lesions and 251,550 echo frames.	Still needs a human prompt; not designed to give calibrated confidence.
Lu et al. (2024)	To build a visual-language foundation model for pathology that works across many tasks with little or no task-specific training data. CONCH: iBOT-pretrained ViT-B image encoder plus a text encoder and multimodal decoder (CoCa-style), trained with contrastive and captioning losses on ...	Pre-training: 1.79 million cleaned image-text pairs from PubMed Central open access and in-house educational material, filtered to 1.17 million ...	State of the art on 13 benchmarks; zero-shot accuracy 90.0% for NSCLC subtyping, 89.3% for RCC subtyping and 84.0% for BRCA subtyping, where PLIP and BiomedCLIP were near chance (50.7-55.3%).	Captions mainly from English-language education and PubMed sources; rare tumours and brain tumours less covered.

Study	Aim and design	Data	Main result	Limitation for this study
Saeed et al. (2023)	To test whether deep learning can really predict glioblastoma MGMT promoter methylation from MRI. Extensive evaluation of ResNets, DenseNets, EfficientNets, ViT and Swin with ImageNet and SimCLR pre-training, modality fusion and ROI crops, using ...	RSNA-MICCAI BraTS 2021 classification task: 585 glioblastoma patients with T1w, T1wCE, T2w, FLAIR and MGMT labels; also BraTS Task 1 ...	All models stayed near chance (AUC about 0.49-0.63; e.g. ViT 0.549, Swin 0.570, EfficientNet-b1 0.631); loss stayed near 0.69 (random), while the same pipeline reached AUC 0.856 on NoduleMNIST3D.	Shows MRI-only MGMT prediction is unreliable; calls for external cohort validation.
Lee et al. (2024a)	To build and test a federated learning platform for paediatric posterior fossa tumour classification and segmentation across many international hospitals. FL-PedBrain: joint classification and segmentation network trained by federated learning across 19 sites (16 in-network, 3 external) on T2-weighted ...	1,468 children (mean age 7.6 years; 48% female) with posterior fossa tumours from 19 institutions: 596 medulloblastoma, 210 ependymoma, 335 pilocytic ...	Less than 1.5% drop in classification and 3% drop in segmentation versus centralised training; F1 0.856 (FL) vs 0.877 (CDS), above a prior pooled pilot (0.800); FL raised segmentation by 20-30% at three out-of-network sites.	T2 only; four tumour types; logistical overhead of FL deployment.
Albalawi et al. (2024)	To classify brain tumours on MRI with federated learning combined with transfer learning while keeping data private. Modified ImageNet-pretrained VGG16 (adjusted filters, extra dropout) trained with federated averaging across simulated clients on a 4-class MRI ...	7,023 brain MRI images in 4 classes (glioma, meningioma, no tumour, pituitary) combined from figshare, SARTAJ and Br35H; SARTAJ glioma images ...	Overall accuracy 98%; precision 0.99 glioma, 0.95 meningioma, 1.00 no tumour, 0.98 pituitary.	Simulated clients split from one public 2D dataset; no real multi-hospital heterogeneity; no external test.
Boyd et al. (2024)	To build expert-level paediatric low-grade glioma segmentation on T2 MRI when only small paediatric datasets exist. In-domain stepwise transfer learning with nnU-Net: train on adult BraTS 2021, fine-tune on paediatric data, then fine-tune again with the encoder or ...	Children's Brain Tumor Network (CBTN) consortium (n=184) and a paediatric cancer centre (n=100) T2-weighted MRI of pLGG; external test n=60; clinical ...	Best model (Transfer-Encoder) median Dice 0.877 vs 0.812 baseline ($p<0.05$); external test Dice 0.834 vs inter-expert 0.861 ($p=0.13$); AI segmentations acceptable in 80.2% vs 65.4% for experts; experts identified AI segmentations ...	T2-only; low-grade glioma only; small external set (60).
Luo et al. (2024)	To develop a brain metastasis segmentation system and test, in a randomized crossover multi-reader study, whether it improves radiologists' contours and speed. VB-Net-based segmentation system (BMSS) trained on 488 patients; prospective 5-centre test set; 5 residents and 5 attending radiologists contoured ...	Development: 488 patients with 10,338 brain metastases (Sun Yat-sen University Cancer Center, Aug 2019 - Nov 2020, from 624 consecutive patients). ...	BMSS alone median Dice 0.91 (95% CI 0.90-0.92), similar in internal and external sets ($p=0.67$); with AI readers' Dice rose from 0.87 to 0.92 ($p<0.001$) and contouring time fell by 42% per lesion; residents improved more (+0.05 vs ...	Metastases only; single sequence; Chinese centres.
Farahani et al. (2026)	To pool the diagnostic accuracy of deep learning MRI models for predicting glioma IDH mutation and 1p/19q co-deletion and find what drives differences between studies. PRISMA systematic review of 5 databases to March 2025; quality by Radiomics Quality Score and QUADAS-2; bivariate random-effects meta-analysis and ...	1,517 unique publications screened from PubMed, Scopus, Embase, Web of Science and Google Scholar; 104 included, 72 in meta-analysis (many using ...	104 studies reviewed, 72 meta-analysed. Test-cohort IDH: sensitivity 0.80 (0.77-0.83), specificity 0.85 (0.81-0.87). 1p/19q: sensitivity 0.75 (0.65-0.82), specificity 0.82 (0.75-0.88). Segmentation method and degree of DL ...	Few studies with true external or prospective validation; heterogeneous MRI protocols.
Tampu et al. (2026)	To evaluate state-of-the-art weakly supervised pathology methods and foundation-model features for classifying paediatric brain tumours on a national multicentre cohort. Patch features from ResNet50, UNI and CONCH aggregated with ABMIL or CLAM for three hierarchical tasks (category, family, type); generalisation ...	1,449 H&E WSIs from 540 children and young adults (age 8.5 +/- 4.9 years; 284 male, 253 female) from six Swedish university hospitals (2013-2023) ...	Best: UNI + ABMIL with MCC 0.76 +/- 0.04 (category), 0.63 +/- 0.04 (family), 0.60 +/- 0.05 (type); UNI and CONCH beat ResNet50 out-of-site, but the in-site to out-of-site drop was similar for all extractors.	Rare tumour types have few cases; scanner variability remains a source of error.
Saueressig et al. (2026)	To classify WHO 2021 glioma subtypes by fusing histopathology and MRI foundation-model features, even without patient-paired training data. Frozen pathology and MRI foundation models as unimodal experts producing patch tokens; three fusion classifiers (late fusion, early fusion, mixture ...	WSI: EBRAINS (497 glioblastoma, 132 astrocytoma, 157 oligodendroglioma). MRI: UCSF-PDGM (398 / 84 / 15) and EGD (249 / 62 / 58). Paired test: TCGA ...	Mixture-of-experts model best: AUC 0.98 on validation and 0.94 on an independent 171-patient matched test set; multimodal models beat unimodal ones and still matched unimodal baselines when only one modality was available.	Only three WHO subtypes; TCGA is the only paired test cohort.
Piffer et al. (2026)	To test deep learning segmentation across diverse paediatric brain tumour types and MRI sequence combinations using a single-centre clinical dataset. 3D nnU-Net trained on 121 and tested on 53 paediatric patients for four subregions (whole tumour, T2-hyperintensity, enhancing tumour, cystic ...	174 paediatric patients (0-18 years) at University Children's Hospital Zurich, 2014-2022, pre-treatment MRI (T1, T1-C, T2, FLAIR); HGG, LGG ...	Mean Dice 0.85 for whole tumour and T2-hyperintensity (human variability 0.86), 0.75 enhancing tumour, 0.26 cystic component; T1 + T1-C + T2 nearly matched the full protocol; manual contouring time reduced by up to 83%.	Cystic components poorly segmented; single centre (Zurich).
LaBella et al. (2026)	To provide a large public multi-institutional dataset of meningioma radiotherapy-planning MRI with expert gross	750 RT planning MRI exams from 747 patients at Duke, UCSF, SUNY	750 RT planning MRIs from 747 patients (500 training with GTV labels, plus validation and test sets); 570 scans	Single sequence (T1c); GTV only, no clinical target volume or organs at risk.

Study	Aim and design	Data	Main result	Limitation for this study
	tumour volume labels for automated segmentation research. Data descriptor: retrospective collection from 7 US/UK centres; native-resolution T1 post-contrast MRI, AFNI defacing (no skull stripping), nnU-Net ...	Upstate, University of Washington, University of Missouri, King's College London and ...	public with 500 annotated; intact and postoperative meningiomas; used for the BraTS 2024 MEN-RT challenge.	
Wu et al. (2025)	To adapt the Segment Anything Model (SAM) to medical images efficiently, including 3D scans, without full fine-tuning. Medical SAM Adapter (Med-SA): small adapter modules inserted into SAM; Space-Depth Transpose (SD-Trans) to process 3D volumes with the 2D model, and ...	17 medical segmentation tasks across CT, MRI, ultrasound, fundus and dermoscopy (e.g., BTCV multi-organ CT, BraTS brain tumour, REFUGE optic ...	Med-SA outperformed state-of-the-art medical segmentation methods (e.g., TransUNet, UNETR, Swin-UNETR, diffusion-based models) on the 17 tasks, including brain tumour, abdominal organs (BTCV), optic disc/cup, thyroid nodule and ...	Still needs prompts and labelled fine-tuning data; compute of SAM backbone.

Table 2.9: Evidence table for clinical evaluation of human-AI teams (O4): 30 representative studies.

Study	Aim and design	Data	Main result	Limitation for this study
Vasey et al. (2022) (base)	To give a reporting guideline for the first live clinical studies of AI decision support. International Delphi consensus with experts and stakeholders.	Expert panel from 20 pre-defined stakeholder categories: 123 experts in Delphi round 1, 138 in round 2, 16 at the virtual consensus meeting and 16 in ...	Checklist of 17 AI-specific items and 10 generic items, covering human factors, safety, and user interaction.	Guideline only, it does not say which metrics to use for deferral.
Chen et al. (2024b) (base)	To quantify how much human-AI collaboration reduces clinicians' image-interpretation workload and how it affects diagnostic performance. Systematic review and random-effects meta-analysis of 36 studies (4 databases) comparing reading time or reading quantity before and after AI ...	36 included studies of image-based disease detection (e.g., mammography, chest radiography, other modalities) reporting reading time or reading ...	Concurrent AI assistance cut reading time by 27.20% (95% CI 18.22-36.18%); reading quantity fell by 44.47% (40.68-48.26%) with AI as second reader and by 61.72% (47.92-75.52%) with AI pre-screening; relative sensitivity 1.12 ...	High heterogeneity and uneven study quality; few studies define reading time clearly; most are retrospective.
Liu et al. (2020)	To extend CONSORT for randomised trials that test AI interventions. Delphi process and consensus meeting.	29 candidate items assessed by 103 international stakeholders in a two-stage Delphi survey; 31 stakeholders at a two-day consensus meeting; checklist ...	14 new items, for example describing the AI version, input data, how humans interact with the output, and error analysis.	For RCTs, which come after early-stage studies.
Collins et al. (2024)	To update TRIPOD for prediction models built with machine learning. Delphi survey and consensus meeting.	Delphi round 1: 170 of 292 invited completed (22 countries; mainly UK 52, US 31, Netherlands 23, Canada 20); round 2: 200 of 395 invited completed ...	27-item checklist covering fairness, data, model development, evaluation and open science.	Covers model development and validation, not live human use.
Bansal et al. (2021)	To test if AI explanations help humans and AI together do better than either alone. User studies on text tasks with AI recommendations, with and without explanations and confidence.	Three tasks: sentiment of beer reviews (Beer) and Amazon book reviews (Amzbook), and LSAT logical reasoning questions; 50 test examples per domain ...	Explanations made people accept AI advice more, even when the AI was wrong. Team performance was not better than just showing confidence.	Non-medical tasks with crowd workers.
Mozannar et al. (2023b)	To teach people when to rely on, ignore or collaborate with an AI by learning data-grounded rules described in natural language and teaching them in onboarding. IntegrAI: collect human decisions with AI advice, discover regions in an embedding space (CLIP or text embeddings) where human reliance should ...	BDD100K driving images (is there a traffic light? noisy images) and MMLU multiple-choice questions; participants recruited on Prolific ...	Onboarding with learned rules improved human-AI team accuracy by 5.2% over no onboarding on traffic-light detection (BDD); on MMLU question answering onboarding had no effect, and showing only AI-integration recommendations hurt ...	Lay crowd workers, not clinicians; gains task-dependent.
Stutz et al. (2023)	To make conformal prediction guarantees valid when ground-truth labels are ambiguous because experts disagree. Aggregate expert annotations into a non-degenerate label distribution (plausibilities) and apply Monte Carlo conformal	Dermatology (skin condition) dataset with multiple dermatologist annotations per case (Roy et al. 2022; classifier with 72.6% top-3 accuracy)	In dermatology classification with strong expert disagreement, standard CP on voted labels calibrated for 72% coverage under-covered expert annotations by about 10% on average (aggregated coverage 0.62 vs voted	Needs multiple expert annotations per case; synthetic and dermatology data only.

Study	Aim and design	Data	Main result	Limitation for this study
	prediction that samples ...	...	0.73); Monte Carlo ...	
Kofler et al. (2023)	To find out how well standard segmentation metrics match expert radiologists' judgement of segmentation quality, and to design losses closer to that judgement. Psychophysical rating experiments: experts rated glioma MRI and ultrasound segmentations on a 6-star Likert scale; correlations with many metrics ...	Multimodal 3D glioma MRI (BraTS-style; segmentations of whole tumour, tumour core, enhancing tumour, necrosis, oedema) from reference and CNN ...	Standard metrics and losses correlated only moderately with expert ratings, worst for the clinically important enhancing tumour; experts consistently rated CNN segmentations better than the human reference labels; new compound ...	Rating experiments on limited case sets; loss design did not succeed.
Maier-Hein et al. (2024)	To give researchers a structured, problem-aware way to choose validation metrics for biomedical image analysis. Multi-stage Delphi consensus by an international consortium of 73 experts (156 contributors overall, plus crowdsourced pitfalls) producing the ...	Expert consortium of 73 members (image analysis, ML, statistics, epidemiology, biology, medicine); 156 researchers contributed (84 acknowledged) ...	Metrics Reloaded framework and online tool recommending metric pools (e.g., Dice plus a boundary metric such as NSD for segmentation; calibration metrics when probabilities are used) with documented pitfalls, demonstrated on many ...	Covers classification-type problems only (not regression or reconstruction); needs updates as methods change.
McDuff et al. (2025)	To test whether an LLM optimised for diagnostic reasoning (AMIE) produces better differential diagnoses alone and as an aid to clinicians on difficult cases. 20 clinicians assessed 302 challenging NEJM clinicopathological case reports; each case read by two clinicians who first gave an unassisted ...	302 real-world challenging case reports (New England Journal of Medicine clinicopathological conferences); 20 generalist clinicians; each case read ...	AMIE alone beat unassisted clinicians (top-10 accuracy 59.1% vs 33.6%, p=0.04). Clinicians assisted by AMIE reached 51.7% top-10 accuracy vs 36.1% unassisted (McNemar 45.7, p<0.01) and 44.4% with search only (McNemar 4.75 ...	Text case reports only (no images or real-time history-taking); case reports are unusually difficult and may not reflect routine practice.
Bucinca et al. (2021)	To test whether cognitive forcing functions reduce people's overreliance on incorrect AI suggestions compared with simple explainable AI. Randomized online experiment (199 Mechanical Turk participants) on a nutrition task; three cognitive forcing designs (on demand, update, wait) versus ...	199 Amazon Mechanical Turk participants; meal images where participants replaced the highest-carbohydrate ingredient with a low-carb alternative of ...	Cognitive forcing functions significantly reduced overreliance on wrong AI suggestions compared with simple explainable AI, but participants rated them as less usable and liked them least - a trade-off between performance and ...	Lay participants and a non-medical task; short sessions.
Wang et al. (2021a)	To understand clinicians' real post-adoption experience with an AI clinical decision support system in rural clinics of a developing country. Ethnographic field study: in-situ observations, semi-structured interviews and contextual inquiries with 22 clinicians in 6 rural clinics in China ...	22 clinicians (9 female) from 6 rural clinics in China using the Brilliant Doctor AI-CDSS embedded in the EHR; clinics recruited via presidents who ...	Tensions found: misalignment with local context and workflow (short visits, high load), technical limitations and usability barriers (poor EHR interoperability, low confidence scores such as 30%), and lack of transparency and ...	Single commercial system in one country; no quantitative outcomes.
Reverberi et al. (2022)	To study experimentally how clinicians integrate AI advice and whether human-AI teams outperform either alone in medical diagnosis. Multicentre 2x2 mixed-design experiment: 21 endoscopists (10 experts, 11 non-experts) optically diagnosed 504 real colonoscopy lesion videos without ...	504 short video clips of colorectal lesions prospectively acquired in real colonoscopies with AI device output; 21 endoscopists from Austria, Israel ...	Endoscopists were influenced by AI (OR 3.05) but sensibly: they followed it more when it was correct (OR 3.48) than incorrect (OR 1.85), weighting their own and the AI's reliability case by case (Bayesian-like), so the human-AI ...	Optical diagnosis of colorectal polyps only; offline video reading rather than live procedures.
Yu et al. (2024)	To measure how the effect of AI assistance varies between individual radiologists and what predicts who benefits or is harmed. Large reader study: 140 radiologists, 324 Stanford chest X-ray cases and 15 pathologies, with and without CheXpert AI probabilities (two designs ...	324 retrospective Stanford patient cases (chest X-ray plus clinical history: indication, vitals, labs) with 15 pathologies; 140 radiologists: 107 in ...	AI effects varied widely between radiologists; experience, subspecialty and familiarity with AI tools did not reliably predict benefit; lower performers did not consistently gain more; incorrect AI predictions harmed radiologist ...	Chest X-ray only; AI not tuned for teamwork; no selective deferral.
Gaube et al. (2021)	To test how physicians respond to diagnostic advice depending on whether it is labelled as coming from AI or a human expert, and on whether it is accurate. Online experiment: physicians read chest X-ray cases with diagnostic advice (all written by human experts, some deliberately inaccurate) labelled as ...	Chest X-ray cases with expert-written advice; 265 physicians: 138 radiologists (high task expertise) and 127 internal or emergency medicine ...	Radiologists rated advice as lower quality when labelled as AI; internal/emergency medicine physicians did not. Diagnostic accuracy was significantly worse with inaccurate advice regardless of the claimed source - showing ...	Advice was not from a real AI; single modality; no confidence information shown.
Nicolson et al. (2025)	To evaluate with real clinicians how a prototype-based explainable AI model affects trust, reliance and performance, including a new behaviour-based measure of ... Three-stage reader study with 10 sonographers estimating gestational age from fetal ultrasound images: (1) alone, (2) with model prediction, (3) with ...	Fetal ultrasound images for gestational age estimation; 10 sonographers.	Model predictions reduced clinician mean absolute error from 23.5 to 15.7 days; explanations gave a further non-significant reduction to 14.3 days; effects of explanations varied, with some participants worse; explanations ...	Only 10 participants; single task (gestational age); immediate effects only.
Brändl et al. (2025)	To classify brain tumours from sparse nanopore methylation	Heidelberg brain tumour classifier	Accuracy 94.52% for 91 CNS classes (97.72% for	Prospective evidence still limited (10

Study	Aim and design	Data	Main result	Limitation for this study
	data live during surgery, within one hour of biopsy. MethyLYZR: Bernoulli naive Bayes classifier over about 428,000 CpG sites with per-class methylation probabilities and CpG weights, updated live as ...	v11b4 reference methylation set (91 CNS classes) for training; over 200 brain tumour samples nanopore-sequenced ...	clinically grouped classes) at about 7,500 CpGs, stable with up to 10% methylation-calling errors (93.73-94.92%); highly accurate results within 15 minutes of sequencing and within ...	intraoperative cases); needs sequencing infrastructure.
Djirackor et al. (2021)	To show that intraoperative nanopore DNA methylation classification of brain tumours is accurate, fast and changes neurosurgical strategy. Ultra-low coverage whole-genome nanopore sequencing of tumour DNA, copy number analysis and an ad hoc random forest trained on the CNS methylation ...	105 patients (50 paediatric, 55 adult) across 6 independent cohorts; 20 evaluated intraoperatively.	Concordant with final neuropathology in 93 of 105 cases (89%); correct in 6/6 cases where frozen section was inconclusive; median turnaround 97 minutes (91-161) to the operating room; precise diagnosis would have changed surgical ...	Tumour classes limited to reference set; moderate cohort size; turnaround still above 1 hour in many cases.
Knop et al. (2022)	To summarise the technological characteristics and human factors that influence effective collaboration between medical professionals and AI-enabled clinical decision ... Literature review of 3 meta-databases: over 1,000 articles screened, 101 assessed in full text, 7 included and synthesised; comparison with factors ...	Over 1,000 articles from 3 meta-databases; 101 full-text assessed; 7 (6.9%) included.	Key factors: training data quality, performance, explainability, adaptability (technology); medical expertise, technological expertise, personality, cognitive biases and trust (human). Less experienced physicians tend to benefit ...	Very few empirical studies (7); little on patient outcomes or theory.
Dingel et al. (2024)	To identify predictors of health care practitioners' intention to use AI clinical decision support systems using the UTAUT model. Meta-analysis of 17 studies (from 7,731 records): random-effects correlations, relative weight analysis, and meta-analytic moderation and mediation ...	7,731 records from databases, forward searches, conference programmes and personal contacts (searches Oct 2022 - Nov 2023); 17 studies (0.22%) ...	Intention to use correlated with performance expectancy (r=0.66), effort expectancy (0.55), social influence (0.66), facilitating conditions (0.66), positive attitude (0.63), trust (0.73, strongest), innovativeness (0.54) ...	Intention rather than actual use; few studies per moderator; mostly high-income settings.
Feng et al. (2022)	To propose how hospitals should continuously monitor and update clinical AI algorithms after deployment to keep them safe and effective. Perspective with a worked example (acute hypotensive episode early-warning model): adapt quality-assurance tools such as statistical process control ...	Not applicable (illustrative example: acute hypotensive episode prediction model in ICU/operating room).	Recommends hospital 'AI-QI' units; monitoring of performance (AUC, MSE) and of marginal and conditional distribution shifts; distinguishing population change from changes in disease relationships; careful, regulated procedures ...	Conceptual; updating methods still immature; regulatory pathways for adaptive AI evolving.
Han et al. (2024)	To map randomised controlled trials of AI in clinical practice: where, in which specialties, with what endpoints and with what reporting quality. Scoping review of RCTs evaluating AI interventions in clinical practice; extraction of country, specialty, AI type, endpoints, demographics, centres ...	86 RCTs of AI in clinical practice; participant median age 57.3 years (IQR 49.9-62.0); median 48.9% male; 63 trials published since 2021.	86 RCTs, led by the USA and China, mostly deep learning for imaging (gastroenterology, radiology); 70 of 86 (81%) reported positive primary endpoints, mostly diagnostic yield/performance (54%); mostly single-centre ...	Few multicentre trials, few patient-relevant outcomes, poor demographic reporting, likely publication bias.
Ciecierski-Holmes et al. (2022)	To evaluate the scope and nature of real-world AI applications for strengthening health systems in low- and middle-income countries. Systematic scoping review of Scopus, EMBASE, MEDLINE, Global Health, APA PsycInfo and Google Scholar grey literature (1 Jan 2009 - 30 Sep 2021) ...	Records 2009-2021 from 5 databases plus grey literature; 10 included studies from LMICs (e.g., China rural CDSS, South Africa radiotherapy planning ...	Only 10 studies met criteria (CDSS, treatment planning, triage assistants, chatbots); only half reported algorithms and training data; challenges included reliability, mixed workflow impact, poor user-friendliness, lack of fit to ...	Very few real-world evaluations in LMICs; none in brain tumour care; little cost evidence.
Mathenge et al. (2022)	To test whether immediate AI-supported diabetic retinopathy screening feedback increases referral uptake in a low-resource African setting. RAIDERS: investigator-masked, parallel-group RCT at 4 Rwandan facilities (March-July 2021); patients with AI-detected referable DR randomised 1:1 to ...	823 adults (>=18 years) with diabetes screened at 4 clinics in Rwanda; 275 with referable DR randomised (mean age 50.7 years; 58.2% women); outcome ...	Of 823 eligible patients, 275 (33.4%) had referable DR and were randomised (136 intervention, 139 control); referral adherence 51.5% vs 39.6% (p=0.048), a 30.1% relative increase; intervention OR 1.74 (1.05-2.88), with older age ...	AI accuracy not separately validated in this population; 60% urban participants; short follow-up.
Kibudde et al. (2024)	To assess the time savings and contour quality of AI auto-segmentation of organs at risk for radiotherapy in two low- and middle-income countries. 10 randomly selected patients (5 head and neck, 5 prostate) had planning CT contoured by an FDA-approved AI tool (CARINA-AI) and manually by ...	10 planning CT scans (5 head and neck, 5 prostate cancer); LMIC1 Uganda Cancer Institute, LMIC2 National Cancer Center of Mongolia; reference ...	AI took about 2 minutes per patient vs 57-84 minutes manually (about 10 minutes total human time including review); AI pelvic contours agreed better with the reference than LMIC manual contours (mean DSC 0.834 vs 0.807 and 0.844 ...	Only 10 patients; small structures still poor; requires internet/cloud access.
Layard Horsfall et al. (2021)	To evaluate the attitudes of neurosurgeons and the wider surgical team toward AI in neurosurgery. Two-stage cross-sectional survey: stage 1 open-question qualitative survey (33 respondents) with thematic analysis; stage 2 quantitative 5-point ...	Stage 1: 33 respondents; Stage 2: 100 respondents (neurosurgeons, anaesthetists, nurses, operating room practitioners).	Six themes: imaging interpretation and preoperative diagnosis, team coordination, operative planning, real-time hazard alerts, autonomous surgery, postoperative follow-up. Agreement: imaging interpretation 62%, operative planning ...	Convenience sample via social media; mainly high-income respondents; attitudes not behaviour.

Study	Aim and design	Data	Main result	Limitation for this study
Yevudza et al. (2024)	To identify barriers and opportunities for access to neuro-oncology care in Sub-Saharan Africa. PRISMA-guided literature review of PubMed and Google Scholar (1 Jan 2003 - 20 Jun 2023), excluding systematic reviews and meta-analyses, with ...	Studies 2003-2023 on neuro-oncology capacity and challenges in Sub-Saharan Africa from PubMed and Google Scholar (e.g., Nigeria, Ghana, Kenya studies ...	Barriers: constrained resources (scanners, neurosurgeons, pathology, radiotherapy), insufficient training, cultural beliefs, low awareness, delayed diagnosis and treatment, and poor incidence data; opportunities: research ...	Heterogeneous, mostly small single-centre studies; lack of registry data.
Roberts et al. (2021)	To systematically review machine learning models for COVID-19 diagnosis or prognosis from chest X-ray and CT and assess whether any are fit for clinical use. Systematic review of bioRxiv, medRxiv, arXiv, EMBASE and MEDLINE (2020); quality screening with CLAIM, RQS and PROBAST-type checks; analysis of ...	2,212 records from 5 databases (1 Jan - 3 Oct 2020); 62 studies included.	2,212 studies found, 2,150 after de-duplication, 415 relevant, 320 full-text screened, 62 included after quality review (37 deep learning). None of the models was judged of potential clinical use, mainly because of biased or ...	Specific to COVID-19 imaging, but pitfalls generalise.
Sounderajah et al. (2025)	To develop a reporting guideline for diagnostic accuracy studies that evaluate AI (STARD-AI). Literature review (55 candidate items), international scoping survey, patient and public involvement, two-round modified Delphi (209 then 143 ...	Delphi round 1: 209 participants (45 items); round 2: 143 participants (26 items); over 240 international stakeholders overall.	40-item checklist: STARD 2015 items plus 18 new or modified items (4 modified, 14 new) covering dataset eligibility at dataset and participant level, data source and collection, annotation, capture devices and software versions ...	Reporting standard only; does not judge study quality.
Malone et al. (2024)	To find out when combinations of humans and AI perform better than humans or AI alone. Preregistered systematic review and meta-analysis of experimental studies (ACM DL, Web of Science, AIS eLibrary; 1 Jan 2020 - 30 Jun 2023) that ...	106 experimental studies, 370 effect sizes (2020-2023).	106 studies with 370 effect sizes. On average, human-AI combinations performed worse than the best of humans or AI alone (Hedges' g = -0.23, 95% CI -0.39 to -0.07), but better than humans alone. Losses occurred in decision-making ...	Most studies used lay participants and short tasks; few clinical deployments.

References

Abadi, M., Chu, A., Goodfellow, I., McMahan, H. B., Mironov, I., Talwar, K., & Zhang, L. (2016). Deep learning with differential privacy. ACM SIGSAC Conference on Computer and Communications Security. https://doi.org/10.1145/2976749.2978318

Abbas Q, Jeong W, Lee SW (2025). Explainable AI in Clinical Decision Support Systems: A Meta-Analysis of Methods, Applications, and Usability Challenges. Healthcare (Basel, Switzerland). https://doi.org/10.3390/healthcare13172154

Adewole M, Rudie JD, Gbadamosi A, et al. (2026). Brain tumor segmentation in Sub-Saharan Africa patient population: The BraTS-Africa challenge. Neuro-Oncology Advances. https://doi.org/10.1093/noajnl/vdag082

Adnan M, Kalra S, Cresswell JC, Taylor GW, Tizhoosh HR (2022). Federated learning and differential privacy for medical image analysis. Scientific reports. https://doi.org/10.1038/s41598-022-05539-7

Al-Anezi FM (2026). Generative Artificial Intelligence in Healthcare: Automation Bias, Deskilling, and Cognitive Implications - A Systematic Review. Journal of healthcare leadership. https://doi.org/10.2147/JHL.S590498

Al-Bazzaz H, Janicijevic M, Strand F (2024). Reader bias in breast cancer screening related to cancer prevalence and artificial intelligence decision support-a reader study. European radiology. https://doi.org/10.1007/s00330-023-10514-5

Albalawi E, T R M, Thakur A, Kumar VV, Gupta M, et al. (2024). Integrated approach of federated learning with transfer learning for classification and diagnosis of brain tumor. BMC Medical Imaging. https://doi.org/10.1186/s12880-024-01261-0

Ali Y, Taleb A, Höhne MM, Lippert C (2021). Self-Supervised Learning for 3D Medical Image Analysis using 3D SimCLR and Monte Carlo Dropout. arXiv preprint. https://arxiv.org/abs/2109.14288

Alijani S, Meibodi FA, Najjaran H (2026). Quantifying Volumetric Risk: Class-Aware Asymmetric Weighted Conformal Prediction for 3D Medical Image Segmentation. arXiv preprint. https://arxiv.org/abs/2610.09392

Alshamlan H (2026). Uncertainty-Aware Selective Prediction in Brain-Tumour MRI Classification: Selective Risk Reduction and Component-Wise Reliability Findings. Bioengineering. https://doi.org/10.3390/bioengineering13101184

Alves JV, Leitao D, Jesus S, Sampaio MOP, Liebana J, Saleiro P, et al. (2024). Cost-Sensitive Learning to Defer to Multiple Experts with Workload Constraints. Transactions on Machine Learning Research (arXiv preprint). https://doi.org/10.48550/arxiv.2403.06906

Alves JV, Leitao D, Jesus S, Sampaio MOP, Liebana J, Saleiro P, et al. (2025). A benchmarking framework and dataset for learning to defer in human-AI decision-making. Scientific Data, 12:506. https://doi.org/10.1038/s41597-025-04664-y

Alzyood M (2026). A responsible AI framework for infection surveillance in low-resource settings: ethics, opportunities and threats for LMICs (EOT-LMICs). International health. https://doi.org/10.1093/inthealth/ihag033

Angelopoulos AN, Bates S (2021). A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification. arXiv 2107.07511 (tutorial). https://arxiv.org/abs/2107.07511

Angelopoulos AN, Bates S (2023). Conformal Prediction: A Gentle Introduction. Foundations and Trends® in Machine Learning. https://doi.org/10.1561/2200000101

Angelopoulos AN, Bates S, Candes EJ, Jordan MI, Lei L (2025). Learn then Test: Calibrating Predictive Algorithms to Achieve Risk Control. Annals of Applied Statistics. https://arxiv.org/abs/2110.01052

Angelopoulos AN, Bates S, Fisch A, Lei L, Schuster T (2024a). Conformal Risk Control. ICLR 2024. https://doi.org/10.48550/arXiv.2208.02814

Angelopoulos AN, Pomerantz S, Do S, Bates S, et al. (2024b). Conformal Triage for Medical Imaging AI Deployment. medRxiv. https://doi.org/10.1101/2024.02.09.24302543

Ankomah CT, Ayivor LE, Nyame I, et al. (2025). How We Won BraTS-SSA 2025: Brain Tumor Segmentation in the Sub-Saharan African Population Using Segmentation-Aware Data Augmentation and Model Ensembling. arXiv (MICCAI BrainLes 2025, LNCS). https://arxiv.org/abs/2510.03568

Asadi Z, Castillo JP, Asadi M, Sinclair DS, Kersten-Oertel M (2024). iSurgARy: A mobile augmented reality solution for ventriculostomy in resource-limited settings. arXiv preprint. https://arxiv.org/abs/2410.00001

Asan O, Bayrak AE, Choudhury A (2020). Artificial Intelligence and Human Trust in Healthcare: Focus on Clinicians. Journal of medical Internet research. https://doi.org/10.2196/15154

Atf Z, Safavi-Naini SAA, Lewis PR, Mahjoubfar A, Naderi N, Savage TR, et al. (2025). The challenge of uncertainty quantification of large language models in medicine. arXiv (cs.AI), arXiv:2504.05278v1. https://arxiv.org/abs/2504.05278

Awuah WA, Adebusoye FT, Wellington J, David L, Salam A, Weng Yee AL, et al. (2024). Recent Outcomes and Challenges of Artificial Intelligence, Machine Learning, and Deep Learning in Neurosurgery. World neurosurgery: X. https://doi.org/10.1016/j.wnsx.2024.100301

Babendererde N, Lemke N, Stieber J, Fuchs M, Weng Z, Eich ML, et al. (2026). Nationwide federated learning for histopathology: secure deployment across Germany behind firewalls. NPJ digital medicine. https://doi.org/10.1038/s41746-026-02958-y

Babic B, Umihanic S, Osmanovic H, Selak N, Sehic-Kozica E, Moranjkic L, et al. (2026). Clinical evaluation of large language model recommendations in melanoma: comparison with multidisciplinary tumor board decisions in a real-world cohort. Frontiers in oncology. https://doi.org/10.3389/fonc.2026.1856390

Bachmann R, Gunes G, Hangaard S, Nexmann A, Lisouski P, Boesen M, et al. (2024). Improving traumatic fracture detection on radiographs with artificial intelligence support: a multi-reader study. BJR open. https://doi.org/10.1093/bjro/tzae011

Baid U, Ghodasara S, Mohan S, Bilello M, Calabrese E, Bakas S, et al. (2021). The RSNA-ASNR-MICCAI BraTS 2021 Benchmark on Brain Tumor Segmentation and Radiogenomic Classification. arXiv 2107.02314. https://arxiv.org/abs/2107.02314

Baker CR, Pease M, Sexton DP, Abumoussa A, Chambless LB (2024). Artificial intelligence innovations in neurosurgical oncology: a narrative review. Journal of neuro-oncology. https://doi.org/10.1007/s11060-024-04757-5

Band N, Rudner TGJ, Feng Q, Filos A, Nado Z, Dusenberry MW, et al. (2022). Benchmarking Bayesian Deep Learning on Diabetic Retinopathy Detection Tasks. arXiv preprint. https://arxiv.org/abs/2211.12717

Bangalore Yogananda CG, Truong NCD, Wagner BC, et al. (2025). Bridging the clinical gap: Confidence informed IDH prediction in brain gliomas using MRI and deep learning. Neuro-Oncology Advances. https://doi.org/10.1093/noajnl/vdaf142

Bansal G, Wu T, Zhou J, Fok R, Nushi B, Kamar E, et al. (2021). Does the Whole Exceed its Parts? The Effect of AI Explanations on Complementary Team Performance. ACM CHI 2021. https://arxiv.org/abs/2006.14779

Bas A, Ozturk-Isik E (2026). IDH Mutation Assessment in Gliomas from Anatomical MRI Using Deep Learning: A Comparative Analysis of Centralized and Federated Learning Frameworks. Diagnostics (Basel, Switzerland). https://doi.org/10.3390/diagnostics16040623

Bedi S, Mlauzi I, Shin D, Koyejo S, Shah NH (2025). The Optimization Paradox in Clinical AI Multi-Agent Systems. arXiv preprint. https://arxiv.org/abs/2506.06574

Bernhardt M, Ribeiro FDS, Glocker B (2022). Failure Detection in Medical Image Classification: A Reality Check and Benchmarking Testbed. Transactions on Machine Learning Research. https://arxiv.org/abs/2205.14094

Bertram CA, Aubreville M, Donovan TA, Bartel A, Wilm F, Marzahl C, et al. (2022). Computer-assisted mitotic count using a deep learning-based algorithm improves interobserver reproducibility and accuracy. Veterinary pathology. https://doi.org/10.1177/03009858211067478

Bin-Alamer O, Abou-Al-Shaar H, Gersey ZC, Huq S, Kallos JA, McCarthy DJ, et al. (2023). Intraoperative Imaging and Optical Visualization Techniques for Brain Tumor Resection: A Narrative Review. Cancers. https://doi.org/10.3390/cancers15194890

Bittencourt MS (2025). Will You Tell Them When They Are Wrong? Automation Bias and Artificial Intelligence in Medicine. Arquivos brasileiros de cardiologia. https://doi.org/10.36660/abc.20250300

Blanes-Selva V, Asensio-Cuesta S, Doñate-Martínez A, Pereira Mesquita F, García-Gómez JM (2023). User-centred design of a clinical decision support system for palliative care: Insights from healthcare professionals. Digital health. https://doi.org/10.1177/20552076221150735

Blecker S, Pandya R, Stork S, Mann D, Kuperman G, Shelley D, et al. (2019). Interruptive Versus Noninterruptive Clinical Decision Support: Usability Study. JMIR human factors. https://doi.org/10.2196/12469

Blondeel M, Codella N, Preston S, Qiu H, Schettini L, Tuan F, et al. (2025). Demo: Healthcare Agent Orchestrator (HAO) for Patient Summarization in Molecular Tumor Boards. arXiv preprint. https://arxiv.org/abs/2509.06602

Boaro A, Mezzalira E, Siddi F, Bagattini C, Gabrovsky N, Marchesini N, et al. (2025). Knowledge, interest and perspectives on Artificial Intelligence in Neurosurgery. A global survey. Brain & spine. https://doi.org/10.1016/j.bas.2024.104156

Borges BLC, Pacheco BM, Silva D (2026). Soft Dice Confidence: A Near-Optimal Confidence Estimator for Selective Prediction in Semantic Segmentation. Machine Learning (Springer). https://doi.org/10.1007/s10994-026-07096-w

Boyd A, Ye Z, Prabhu SP, Tjong MC, Zha Y, et al. (2024). Stepwise Transfer Learning for Expert-level Pediatric Brain Tumor MRI Segmentation in a Limited Data Scenario. Radiology: Artificial Intelligence. https://doi.org/10.1148/ryai.230254

Breitwieser M, Zirknitzer S, Poslusny K, Freude T, Scholsching J, Bodenschatz K, et al. (2025). AI in Fracture Detection: A Cross-Disciplinary Analysis of Physician Acceptance Using the UTAUT Model. Diagnostics (Basel, Switzerland). https://doi.org/10.3390/diagnostics15162117

Brooke, J. (1996). SUS: a quick and dirty usability scale. In Usability evaluation in industry (pp. 189-194). Taylor and Francis.

Brändl B, Steiger M, Kubelt C, et al. (2025). Rapid brain tumor classification from sparse epigenomic data. Nature Medicine. https://doi.org/10.1038/s41591-024-03435-3

Bucinca Z, Malaya MB, Gajos KZ (2021). To Trust or to Think: Cognitive Forcing Functions Can Reduce Overreliance on AI in AI-assisted Decision-making. Proceedings of the ACM on Human-Computer Interaction (CSCW). https://doi.org/10.1145/3449287

Bulten W, Balkenhol M, Belinga JA, Brilhante A, Çakır A, Egevad L, et al. (2021). Artificial intelligence assistance significantly improves Gleason grading of prostate biopsies by pathologists. Modern pathology : an official journal of the United States and Canadian Academy of Pathology, Inc. https://doi.org/10.1038/s41379-020-0640-y

Bungert TJ, Kobelke L, Jaeger PF (2023). Understanding Silent Failures in Medical Image Classification. MICCAI 2023. https://doi.org/10.48550/arXiv.2307.14729

Buçinca Z, Lin P, Gajos KZ, Glassman EL (2020). Proxy Tasks and Subjective Measures Can Be Misleading in Evaluating Explainable AI Systems. arXiv preprint. https://arxiv.org/abs/2001.08298

Campanella G, Chen S, Singh M, Verma R, Muehlstedt S, Zeng J, et al. (2025). A clinical benchmark of public self-supervised pathology foundation models. Nature communications. https://doi.org/10.1038/s41467-025-58796-1

Cao S, Liu A, Huang C (2024). Designing for Appropriate Reliance: The Roles of AI Uncertainty Presentation, Initial User Decision, and User Demographics in AI-Assisted Decision-Making. arXiv preprint. https://arxiv.org/abs/2401.05612

Cao Y, Mozannar H, Feng L, Wei H, An B (2023). In Defense of Softmax Parametrization for Calibrated and Consistent Learning to Defer. arXiv preprint. https://arxiv.org/abs/2311.01106

Cemri M, Pan MZ, Yang S, Agrawal LA, Chopra B, Tiwari R, et al. (2025). Why Do Multi-Agent LLM Systems Fail?. NeurIPS 2025 Datasets and Benchmarks Track (arXiv:2503.13657). https://arxiv.org/abs/2503.13657

Chan B, Kim B, Schonfeld E, Nageeb G, Pant A, Sjoholm A, et al. (2025). Publicly Available Datasets for Artificial Intelligence in Neurosurgery: A Systematic Review. Journal of clinical medicine. https://doi.org/10.3390/jcm14165674

Charusaie M, Mozannar H, Sontag D, Samadi S (2022). Sample Efficient Learning of Predictors that Complement Humans. ICML 2022. https://arxiv.org/abs/2207.09584

Charusaie M, Samadi S (2024). A Unifying Post-Processing Framework for Multi-Objective Learn-to-Defer Problems. arXiv preprint. https://arxiv.org/abs/2407.12710

Chen J, Ma B, Cui H, Xia Y (2024a). Think Twice Before Selection: Federated Evidential Active Learning for Medical Image Analysis with Domain Shifts. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). https://doi.org/10.1109/CVPR52733.2024.01087

Chen M, Wang Y, Wang Q, et al. (2024b). Impact of human and artificial intelligence collaboration on workload reduction in medical image interpretation. npj Digital Medicine. https://doi.org/10.1038/s41746-024-01328-w

Chen RJ, Chen C, Li Y, et al. (2022a). Scaling Vision Transformers to Gigapixel Images via Hierarchical Self-Supervised Learning. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). https://doi.org/10.1109/CVPR52688.2022.01567

Chen RJ, Ding T, Lu MY, Williamson DFK, Jaume G, Mahmood F, et al. (2024c). Towards a general-purpose foundation model for computational pathology (UNI). Nature Medicine. https://arxiv.org/abs/2308.15474

Chen RJ, Krishnan RG (2022). Self-Supervised Vision Transformers Learn Visual Concepts in Histopathology. arXiv preprint. https://arxiv.org/abs/2203.00585

Chen RJ, Lu MY, Wang J, Williamson DFK, Rodig SJ, Lindeman NI, et al. (2022b). Pathomic Fusion: An Integrated Framework for Fusing Histopathology and Genomic Features for Cancer Diagnosis and Prognosis. IEEE Transactions on Medical Imaging. https://arxiv.org/abs/1912.08937

Chen X, Yi H, You M, Liu W, Wang L, Li H, et al. (2025a). Enhancing diagnostic capability with multi-agents conversational large language models. npj Digital Medicine (2025) 8:159. https://doi.org/10.1038/s41746-025-01550-0

Chen Z, Luo Y, Sra M (2025b). Engaging with AI: How Interface Design Shapes Human-AI Collaboration in High-Stakes Decision-Making. arXiv preprint. https://arxiv.org/abs/2501.16627

Cheung MY, Veeraraghavan A, Balakrishnan G (2025). COMPASS: Robust Feature Conformal Prediction for Medical Segmentation Metrics. arXiv preprint. https://arxiv.org/abs/2509.22240

Chin SC, Zhang X, Khang LY, Yang W (2024). CONSULT: Contrastive Self-Supervised Learning for Few-shot Tumor Detection. arXiv preprint. https://arxiv.org/abs/2410.11307

Cho H, Keenan G, Madandola OO, Dos Santos FC, Macieira TGR, Bjarnadottir RI, et al. (2022). Assessing the Usability of a Clinical Decision Support System: Heuristic Evaluation. JMIR human factors. https://doi.org/10.2196/31758

Chopra M, Sparrenberg L, Terheyden JH, Sifa R (2026). Knowing When Not to Predict: Self Supervised Learning and Abstention for Safer DR Screening. arXiv preprint. https://arxiv.org/abs/2605.19133

Choudhury A, Asan O (2020). Role of Artificial Intelligence in Patient Safety Outcomes: Systematic Literature Review. JMIR medical informatics. https://doi.org/10.2196/18599

Chuang WY, Yu WH, Liu YJ, Huang CC, Lee KF, Hwang CC, et al. (2026). Assessing the influence of two deep-learning assistance modes on pathologists in cancer identification. Diagnostic pathology. https://doi.org/10.1186/s13000-026-01747-1

Chunduru P, Phillips JJ, Molinaro AM (2022). Prognostic risk stratification of gliomas using deep learning in digital pathology images. Neuro-oncology advances. https://doi.org/10.1093/noajnl/vdac111

Chyzhyk D, Arregi M, Errazquin M, Ariceta A, Sevilla M, Álvarez R, et al. (2025). Exploring Usability of a Clinical Decision Support System for Cancer Care: A User-Centered Study. Cancer reports (Hoboken, N.J.). https://doi.org/10.1002/cnr2.70173

Cicin FN, Çetin Gürkan G (2026). How physicians embrace AI: insights from technology acceptance and trust theories. Frontiers in digital health. https://doi.org/10.3389/fdgth.2026.1722087

Ciecierski-Holmes T, Singh R, Axt M, et al. (2022). Artificial intelligence for strengthening healthcare systems in low- and middle-income countries: a systematic scoping review. npj Digital Medicine. https://doi.org/10.1038/s41746-022-00700-y

Collins GS, Moons KGM, Dhiman P, Riley RD, Beam AL, Logullo P, et al. (2024). TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. BMJ. https://doi.org/10.1136/bmj-2023-078378

Cox J, Liu P, Stolte SE, et al. (2024). BrainSegFounder: Towards 3D Foundation Models for Neuroimage Segmentation. Medical Image Analysis. https://doi.org/10.1016/j.media.2024.103301

Cubukcu HC, Canbaz H (2025). Usability assessments in clinical decision support systems. Northern clinics of Istanbul. https://doi.org/10.14744/nci.2024.22316

d'Assier MAB, Vorontsov E, Kadoury S (2022). M-GenSeg: Domain Adaptation For Target Modality Tumor Segmentation With Annotation-Efficient Supervision. arXiv preprint. https://arxiv.org/abs/2212.07276

Dai Q, Wei D, Liu H, et al. (2024). Federated Modality-Specific Encoders and Multimodal Anchors for Personalized Brain Tumor Segmentation. AAAI Conference on Artificial Intelligence. https://doi.org/10.1609/aaai.v38i2.27909

Das BC, Amini MH, Wu Y (2024). In-depth Analysis of Privacy Threats in Federated Learning for Medical Data. arXiv preprint. https://arxiv.org/abs/2409.18907

Davis SE, Salwei ME (2026). Explainability in context: calibrating appropriate trust and reliance in artificial intelligence. Journal of the American Medical Informatics Association : JAMIA. https://doi.org/10.1093/jamia/ocag082

De Micco F, Di Palma G, Ferorelli D, De Benedictis A, Tomassini L, Tambone V, et al. (2024). Artificial intelligence in healthcare: transforming patient safety with intelligent systems-A systematic review. Frontiers in medicine. https://doi.org/10.3389/fmed.2024.1522554

Dehdab R, Afat S, Mankertz F, Brendel JM, Maalouf N, Werner S, et al. (2026). When AI joins the table: evaluating large language model performance in soft tissue sarcoma tumor board decisions. Journal of cancer research and clinical oncology. https://doi.org/10.1007/s00432-026-06432-w

Despotovic V, Kim S, Hau A, Kakoichankava A, Klamminger GG, Borgmann FBK, et al. (2023). Glioma subtype classification from histopathological images using in-domain and out-of-domain transfer learning: An experimental study. arXiv preprint. https://arxiv.org/abs/2309.17223

Dingel J, Kleine AK, Cecil J, et al. (2024). Predictors of Health Care Practitioners' Intention to Use AI-Enabled Clinical Decision Support Systems: Meta-Analysis Based on the Unified Theory of Acceptance and Use of Technology. Journal of Medical Internet Research. https://doi.org/10.2196/57224

Dissez G, Tay N, Dyer T, Tam M, Dittrich R, Doyne D, et al. (2022). Enhancing Early Lung Cancer Detection on Chest Radiographs with AI-assistance: A Multi-Reader Study. arXiv preprint. https://arxiv.org/abs/2208.14742

Djirackor L, Halldorsson S, Niehusmann P, et al. (2021). Intraoperative DNA methylation classification of brain tumors impacts neurosurgical strategy. Neuro-Oncology Advances. https://doi.org/10.1093/noajnl/vdab149

Dolezal JM, Srisuwananukorn A, Karpeyev D, Ramesh S, et al. (2022). Uncertainty-informed deep learning models enable high-confidence predictions for digital histopathology. Nature Communications. https://doi.org/10.1038/s41467-022-34025-x

Du Y, Li S, Torralba A, Tenenbaum JB, Mordatch I (2023). Improving Factuality and Reasoning in Language Models through Multiagent Debate. International Conference on Machine Learning (ICML 2024). https://arxiv.org/abs/2305.14325

Dundar TT, Yurtsever I, Pehlivanoglu MK, Yildiz U, Eker A, Demir MA, et al. (2022). Machine Learning-Based Surgical Planning for Neurosurgery: Artificial Intelligent Approaches to the Cranium. Frontiers in surgery. https://doi.org/10.3389/fsurg.2022.863633

Dvijotham K, Winkens J, Barsbey M, Ghaisas S, Stanforth R, Pawlowski N, et al. (2023). Enhancing the reliability and accuracy of AI-enabled diagnosis via complementarity-driven deferral to clinicians. Nature Medicine 29(7):1814-1820. https://doi.org/10.1038/s41591-023-02437-x

Eibl T, Schrey M, Liebert A, Ritter L, Lange R, Steiner HH, et al. (2024). Significance of navigated transcranial magnetic stimulation and tractography to preserve motor function in patients undergoing surgery for motor eloquent gliomas. Heliyon. https://doi.org/10.1016/j.heliyon.2024.e28115

Ellis-Caleo T, Keyes T, Ambers N, Bekheet F, Yim W, Kotecha N, et al. (2026). Development, Evaluation, and Deployment of a Multi-Agent System for Thoracic Tumor Board. arXiv (arXiv:2604.12161). https://arxiv.org/abs/2604.12161

Eloy C, Marques A, Pinto J, Pinheiro J, Campelos S, Curado M, et al. (2023). Artificial intelligence-assisted cancer diagnosis improves the efficiency of pathologists in prostatic biopsies. Virchows Archiv : an international journal of pathology. https://doi.org/10.1007/s00428-023-03518-5

Elyassirad D, Gheiji B, Vatanparast M, Ahmadzadeh AM, et al. (2025). CONSeg: Voxelwise Uncertainty Quantification for Glioma Segmentation Using Conformal Prediction. American Journal of Neuroradiology. https://doi.org/10.3174/ajnr.A8914

Erdur AC, Scholz D, Pan J, Wiestler B, Rueckert D, Peeken JC (2026). Agentic Large Language Models for Training-Free Neuro-Radiological Image Analysis. arXiv preprint. https://arxiv.org/abs/2604.16729

European Union. (2024). Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act), Article 14: Human oversight. Official Journal of the European Union. https://eur-lex.europa.eu/eli/reg/2024/1689/oj

Fallahpour A, Ma J, Munim A, Lyu H, Wang B (2025). MedRAX: Medical Reasoning Agent for Chest X-ray. Proceedings of the 42nd International Conference on Machine Learning (ICML 2025), PMLR 267 (arXiv:2502.02673v2). https://arxiv.org/abs/2502.02673

Fan Z, Tang J, Chen W, Wang S, Wei Z, Xie J, et al. (2024). AI Hospital: Benchmarking Large Language Models in a Multi-agent Medical Interaction Simulator. arXiv (preprint, arXiv:2402.09742v3). https://arxiv.org/abs/2402.09742

Farahani S, Hejazi M, Ieva AD, Fatemizadeh E, Liu S (2025a). Towards a Multimodal MRI-Based Foundation Model for Multi-Level Feature Exploration in Segmentation, Molecular Subtyping, and Grading of Glioma. arXiv preprint. https://arxiv.org/abs/2503.06828

Farahani S, Hejazi M, Ieva AD, Liu S (2025b). FoundBioNet: A Foundation-Based Model for IDH Genotyping of Glioma from Multi-Parametric MRI. arXiv preprint. https://arxiv.org/abs/2508.06756

Farahani S, Hejazi M, Tabassum M, Di Ieva A, et al. (2026). Diagnostic performance of deep learning for predicting glioma isocitrate dehydrogenase and 1p/19q co-deletion in MRI: a systematic review and meta-analysis. European Radiology. https://doi.org/10.1007/s00330-025-11898-2

Farquhar S, Kossen J, Kuhn L, Gal Y (2024). Detecting hallucinations in large language models using semantic entropy. Nature, Vol 630, pp. 625-630 (20 June 2024). https://doi.org/10.1038/s41586-024-07421-0

Fayjie AR, Borah J, Carbone F, Tack J, Vandewalle P (2024). Predictive uncertainty estimation in deep learning for lung carcinoma classification in digital pathology under real dataset shifts. arXiv preprint. https://arxiv.org/abs/2408.08432

Feng J, Phillips RV, Malenica I, et al. (2022). Clinical artificial intelligence quality improvement: towards continual monitoring and updating of AI algorithms in healthcare. npj Digital Medicine. https://doi.org/10.1038/s41746-022-00611-y

Ferber D, El Nahhas OSM, Wolflein G, Wiest IC, Clusmann J, Kather JN, et al. (2025). Development and validation of an autonomous artificial intelligence agent for clinical decision-making in oncology. Nature Cancer 2025 (preprint arXiv 2404.04667). https://arxiv.org/abs/2404.04667

Ferber D, Hilgers L, Höper C, Kinny-Köster B, Eckardt JN, Egger-Heidrich K, et al. (2026). Towards autonomous medical artificial intelligence agents. Nature, 655:1282. https://doi.org/10.1038/s41586-026-10675-5

Fiederer LDJ, Alwanni H, Völker M, Schnell O, Beck J, Ball T (2019). A Research Framework for Virtual Reality Neurosurgery Based on Open-Source Tools. arXiv preprint. https://arxiv.org/abs/1908.05188

Filos A, Farquhar S, Gomez AN, Rudner TGJ, Kenton Z, Smith L, et al. (2019). A Systematic Comparison of Bayesian Deep Learning Robustness in Diabetic Retinopathy Tasks. arXiv preprint. https://arxiv.org/abs/1912.10481

Fisch A, Jaakkola T, Barzilay R (2022). Calibrated Selective Classification. Transactions on Machine Learning Research. https://doi.org/10.48550/arXiv.2208.12084

Fok R, Weld DS (2023). In Search of Verifiability: Explanations Rarely Enable Complementary Performance in AI-Advised Decision Making. arXiv preprint. https://arxiv.org/abs/2305.07722

Fourkioti O, Vries MD, Jin C, Alexander DC, Bakal C (2023). CAMIL: Context-Aware Multiple Instance Learning for Cancer Detection and Subtyping in Whole Slide Images. The Twelfth International Conference on Learning Representations, ICLR 2024. https://arxiv.org/abs/2305.05314

Frazer HML, Peña-Solorzano CA, Kwok CF, Elliott MS, Chen Y, Wang C, et al. (2024). Comparison of AI-integrated pathways with human-AI interaction in population mammographic screening for breast cancer. Nature communications. https://doi.org/10.1038/s41467-024-51725-8

Fujimori R, Liu K, Soeno S, Naraba H, Ogura K, Hara K, et al. (2022). Acceptance, Barriers, and Facilitators to Implementing Artificial Intelligence-Based Decision Support Systems in Emergency Departments: Quantitative and Qualitative Evaluation. JMIR formative research. https://doi.org/10.2196/36501

Futami F, Fujisawa M (2024). Information-theoretic Generalization Analysis for Expected Calibration Error. arXiv preprint. https://arxiv.org/abs/2405.15709

Gade M, Nguyen KM, Gedde S, Fernandez-Quilez A (2024). Impact of uncertainty quantification through conformal prediction on volume assessment from deep learning-based MRI prostate segmentation. Insights into Imaging. https://doi.org/10.1186/s13244-024-01863-w

Gadermayr M, Tschuchnig M (2022). Multiple Instance Learning for Digital Pathology: A Review on the State-of-the-Art, Limitations & Future Potential. arXiv preprint. https://arxiv.org/abs/2206.04425

Gaillochet M, Desrosiers C, Lombaert H (2026). Anatomically-aware conformal prediction for medical image segmentation with random walks. arXiv preprint. https://arxiv.org/abs/2601.18997

Gal Y, Ghahramani Z (2016). Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning. ICML 2016. https://arxiv.org/abs/1506.02142

Galada S, Halder T, Deo K, Krish RP, Jadhav K (2024). PRISM: Privacy-preserving Inter-Site MRI Harmonization via Disentangled Representation Learning. arXiv preprint. https://arxiv.org/abs/2411.06513

Gandouz M, Holzmann H, Heider D (2021). Machine learning with asymmetric abstention for biomedical decision-making. BMC Medical Informatics and Decision Making. https://doi.org/10.1186/s12911-021-01655-y

Gaube S, Suresh H, Raue M, et al. (2021). Do as AI say: susceptibility in deployment of clinical decision-aids. npj Digital Medicine. https://doi.org/10.1038/s41746-021-00385-9

Geifman Y, El-Yaniv R (2017). Selective Classification for Deep Neural Networks. NeurIPS 2017. https://arxiv.org/abs/1705.08500

Graham S, Jahanifar M, Vu QD, Maskoliunaite V, Petroska D, Valkiuniene RB, et al. (2026). AI-assisted mitotic counting improves reproducibility and efficiency across multiple tumour types. arXiv preprint. https://arxiv.org/abs/2610.01813

Grande HZ, Alves JH, Laroca R, Hochuli AG (2026a). Generalizable Brain Tumor Segmentation with Self-Training and Tumor-Aware Deformations. arXiv preprint. https://arxiv.org/abs/2609.02600

Grande HZ, Pitol JG, Schuck LB, Serenato RV, Laroca R, Hochuli AG (2026b). On the Role of MRI Sequences in Cross-Dataset Generalization for Brain Tumor Segmentation. arXiv preprint. https://arxiv.org/abs/2608.29944

Grovik E, Yi D, Iv M, Tong E, Nilsen LB, et al. (2021). Handling missing MRI sequences in deep learning segmentation of brain metastases: a multicenter study. npj Digital Medicine. https://doi.org/10.1038/s41746-021-00398-4

Gu H, Yang C, Magaki S, Zarrin-Khameh N, Lakis NS, Cobos I, et al. (2024). Majority Voting of Doctors Improves Appropriateness of AI Reliance in Pathology. arXiv preprint. https://arxiv.org/abs/2404.04485

Guan H, Yap P, Bozoki A, Liu M (2023). Federated Learning for Medical Image Analysis: A Survey. Pattern Recognition, volume 151, 2024. https://arxiv.org/abs/2306.05980

Guarino VE, Winklmayr C, Franzen J, Rumberger JL, Pfeuffer M, Greven S, et al. (2026). Better than Average: Spatially-Aware Aggregation of Segmentation Uncertainty Improves Downstream Performance. arXiv preprint. https://arxiv.org/abs/2603.29941

Gunesli GN, Bilal M, Raza SEA, Rajpoot NM (2021). FedDropoutAvg: Generalizable federated learning for histopathology image classification. arXiv preprint. https://arxiv.org/abs/2111.13230

Guo C, Pleiss G, Sun Y, Weinberger KQ (2017). On Calibration of Modern Neural Networks. ICML 2017. https://arxiv.org/abs/1706.04599

Guo T, Chen X, Wang Y, Chang R, Pei S, Chawla NV, et al. (2024a). Large Language Model Based Multi-agents: A Survey of Progress and Challenges. Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence (IJCAI-24), Survey Track, pp. 8048-8057. https://doi.org/10.24963/ijcai.2024/890

Guo Z, Wu Y, Hartline J, Hullman J (2024b). A Decision Theoretic Framework for Measuring AI Reliance. arXiv preprint. https://arxiv.org/abs/2401.15356

Gutbrod M, Rauber D, Nunes DW, Palm C (2025). OpenMIBOOD: Open Medical Imaging Benchmarks for Out-Of-Distribution Detection. arXiv preprint. https://arxiv.org/abs/2503.16247

Gómez Amarillo DF, Ordóñez-Rubiano EG, Ramírez-Sanabria AD, Figueredo LF, Vargas-Osorio MP, Ramon JF, et al. (2023). Augmented reality for intracranial meningioma resection: a mini-review. Frontiers in neurology. https://doi.org/10.3389/fneur.2023.1269014

Hager P, Jungmann F, al. RH, et al. (2024). Evaluation and mitigation of the limitations of large language models in clinical decision-making. Nature Medicine. https://doi.org/10.1038/s41591-024-03097-1

Hamidinekoo A, Pieciak T, Afzali M, Akanyeti O, Yuan Y (2020). Glioma Classification Using Multimodal Radiology and Histology Data. arXiv preprint. https://arxiv.org/abs/2011.05410

Han R, Acosta JN, Shakeri Z, et al. (2024). Randomised controlled trials evaluating artificial intelligence in clinical practice: a scoping review. The Lancet Digital Health. https://doi.org/10.1016/S2589-7500(24)00047-5

Handley JL, Krevat SA, Fong A, Ratwani RM (2024). Artificial intelligence related safety issues associated with FDA medical device reports. NPJ digital medicine. https://doi.org/10.1038/s41746-024-01357-5

Hao R, Namdar K, Liu L, Khalvati F (2021). A Transfer Learning-Based Active Learning Framework for Brain Tumor Classification. Frontiers in Artificial Intelligence. https://doi.org/10.3389/frai.2021.635766

Hart, S. G., & Staveland, L. E. (1988). Development of NASA-TLX (Task Load Index): results of empirical and theoretical research. Advances in Psychology, 52, 139-183. https://doi.org/10.1016/S0166-4115(08)62386-9

Hatamizadeh A, Nath V, Tang Y, Yang D, Roth HR, Xu D (2022). Swin UNETR: Swin Transformers for Semantic Segmentation of Brain Tumors in MRI Images. MICCAI BrainLes Workshop 2021. https://arxiv.org/abs/2201.01266

Hauke F, Wienholt P, Kuhl C, Ferber D, Kather JN, Nebelung S, et al. (2026). Bayesian uncertainty estimation improves clinical decision making in medical AI agents. arXiv preprint. https://arxiv.org/abs/2607.20582

He G, Kuiper L, Gadiraju U (2023). Knowing About Knowing: An Illusion of Human Competence Can Hinder Appropriate Reliance on AI Systems. arXiv preprint. https://arxiv.org/abs/2301.11333

He K, Ji W, Zhou T, Li Z, Huo J, Zhang X, et al. (2021). Cross-Modality Brain Tumor Segmentation via Bidirectional Global-to-Local Unsupervised Domain Adaptation. arXiv preprint. https://arxiv.org/abs/2105.07715

Hekler A, Kuhn L, Buettner F (2025). Beyond Overconfidence: Foundation Models Redefine Calibration in Deep Neural Networks. arXiv preprint. https://arxiv.org/abs/2506.09593

Hellum O, Kersten-Oertel M, Xiao Y (2022). Assessment of user-interaction strategies for neurosurgical data navigation and annotation in virtual reality. arXiv preprint. https://arxiv.org/abs/2212.08688

Hemmer P, Thede L, Vossing M, Jakubik J, Kuhl N (2023). Learning to Defer with Limited Expert Predictions. Proceedings of the AAAI Conference on Artificial Intelligence, 37(5) (AAAI 2023); arXiv:2304.07306. https://doi.org/10.1609/aaai.v37i5.25742

Hewitt KJ, Wiest IC, Carrero ZI, Bejan L, Millner TO, Brandner S, et al. (2024). Large language models as a diagnostic support tool in neuropathology. The journal of pathology. Clinical research. https://doi.org/10.1002/2056-4538.70009

Hindawi S, Szubstarski B, Boernert E, Tackenberg B, Wuerfel J (2025). Federated learning for lesion segmentation in multiple sclerosis: a real-world multi-center feasibility study. Frontiers in neurology. https://doi.org/10.3389/fneur.2025.1620469

Ho HL, Wang Y, Wang A, Bai L, Ren H (2024). Web-based Augmented Reality with Auto-Scaling and Real-Time Head Tracking towards Markerless Neurointerventional Preoperative Planning and Training of Head-mounted Robotic Needle Insertion. arXiv preprint. https://arxiv.org/abs/2411.02410

Hochreuter KM, Ren J, Nijkamp J, Korreman SS, Lukacova S, Kallehauge JF, et al. (2024). The effect of editing clinical contours on deep-learning segmentation accuracy of the gross tumor volume in glioblastoma. Physics and imaging in radiation oncology. https://doi.org/10.1016/j.phro.2024.100620

Hollon T, Jiang C, Chowdury A, Nasir-Moin M, et al. (2023). Artificial-intelligence-based molecular classification of diffuse gliomas using rapid, label-free optical imaging. Nature Medicine. https://doi.org/10.1038/s41591-023-02252-4

Holtzman Gazit M, Faran R, Stepovoy K, Peles O, Shamir RR (2022). Post-operative glioblastoma multiforme segmentation with uncertainty estimation. Frontiers in Human Neuroscience. https://doi.org/10.3389/fnhum.2022.932441

Hong S, Zhuge M, Chen J, Zheng X, Cheng Y, Zhang C, et al. (2023). MetaGPT: Meta Programming for A Multi-Agent Collaborative Framework. International Conference on Learning Representations (ICLR 2024). https://arxiv.org/abs/2308.00352

Hong Z, Yue Y, Chen Y, Cong L, Lin H, Luo Y, et al. (2024). Out-of-distribution Detection in Medical Image Analysis: A survey. arXiv preprint. https://arxiv.org/abs/2404.18279

Hou X, Jiang C, Kondepudi A, Lyu Y, Chowdury A, Lee H, et al. (2024). A self-supervised framework for learning whole slide representations. arXiv preprint. https://arxiv.org/abs/2402.06188

Hsu WW, Guo JM, Pei L, Chiang LA, Li YF, Hsiao JC, et al. (2022). A weakly supervised deep learning-based method for glioma subtype classification using WSI and mpMRIs. Scientific reports. https://doi.org/10.1038/s41598-022-09985-1

Hu LS, Wang L, Hawkins-Daarud A, Eschbacher JM, et al. (2021a). Uncertainty quantification in the radiogenomics modeling of EGFR amplification in glioblastoma. Scientific Reports. https://doi.org/10.1038/s41598-021-83141-z

Hu X, Zeng D, Xu X, Shi Y (2021b). Semi-supervised Contrastive Learning for Label-efficient Medical Image Segmentation. arXiv preprint. https://arxiv.org/abs/2109.07407

Hüllermeier E, Waegeman W (2021). Aleatoric and epistemic uncertainty in machine learning: an introduction to concepts and methods. Machine Learning. https://doi.org/10.1007/s10994-021-05946-3

Ille S, Schwendner M, Zhang W, Schroeder A, Meyer B, Krieg SM (2021). Tractography for Subcortical Resection of Gliomas Is Highly Accurate for Motor and Language Function: ioMRI-Based Elastic Fusion Disproves the Severity of Brain Shift. Cancers. https://doi.org/10.3390/cancers13081787

Ilse M, Tomczak JM, Welling M (2018). Attention-based Deep Multiple Instance Learning. ICML 2018. https://arxiv.org/abs/1802.04712

Im S, Hyeon J, Rha E, Lee J, Choi HJ, Jung Y, et al. (2021). Classification of Diffuse Glioma Subtype from Clinical-Grade Pathological Images Using Deep Transfer Learning. Sensors (Basel, Switzerland). https://doi.org/10.3390/s21103500

International Agency for Research on Cancer. (2024). GLOBOCAN 2024: Sri Lanka fact sheet. Global Cancer Observatory, IARC/WHO. https://gco.iarc.who.int/media/globocan/factsheets/populations/144-sri-lanka-fact-sheet.pdf

Isensee F, Jaeger PF, Kohl SAA, Petersen J, Maier-Hein KH (2021). nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation. Nature Methods. https://arxiv.org/abs/1809.10486

Isensee F, Wald T, Ulrich C, et al. (2024). nnU-Net Revisited: A Call for Rigorous Validation in 3D Medical Image Segmentation. MICCAI 2024 (LNCS). https://doi.org/10.1007/978-3-031-72114-4_47

Islam M, Reza MT, Kaosar M, Parvez MZ (2022). Effectiveness of Federated Learning and CNN Ensemble Architectures for Identifying Brain Tumors Using MRI Images. Neural Processing Letters. https://doi.org/10.1007/s11063-022-11014-1

Islam MR (2026). An Explainable Agentic AI Framework for Uncertainty-Aware and Abstention-Enabled Acute Ischemic Stroke Imaging Decisions. arXiv preprint. https://arxiv.org/abs/2601.01008

Jabal MS, Zhang J, LaBella D, Houk JL, Zhang D, Rudie JD, et al. (2026). Agentic Automation of BT-RADS Scoring: End-to-End Multi-Agent System for Standardized Brain Tumor Follow-up Assessment. arXiv preprint. https://arxiv.org/abs/2603.21494

Jabbour S, Fouhey D, Banovic N, Shepard SD, Kazerooni E, Sjoding MW, et al. (2025). Selective Prediction Reduces the Negative Effects of Automation Bias Overall but Increases False Negatives. arXiv preprint. https://arxiv.org/abs/2508.07617

Jaeger PF, Luth CT, Klein L, Bungert TJ (2023). A Call to Reflect on Evaluation Practices for Failure Detection in Image Classification. ICLR 2023. https://arxiv.org/abs/2211.15259

Jeong J, Kim S, Pan L, Hwang D, Kim D, Choi J, et al. (2025). Reducing the workload of medical diagnosis through artificial intelligence: A narrative review. Medicine. https://doi.org/10.1097/MD.0000000000041470

Jiang M, Wang Z, Dou Q (2022). HarmoFL: Harmonizing Local and Global Drifts in Federated Learning on Heterogeneous Medical Images. AAAI Conference on Artificial Intelligence. https://doi.org/10.1609/aaai.v36i1.19993

Jiang M, Yang H, Cheng C, Dou Q (2023). IOP-FL: Inside-Outside Personalization for Federated Medical Image Segmentation. IEEE Transactions on Medical Imaging. https://doi.org/10.1109/TMI.2023.3263072

Jiang Y, Black KC, Geng G, Park D, Zou J, Ng AY, et al. (2025). MedAgentBench: A Virtual EHR Environment to Benchmark Medical LLM Agents. NEJM AI (preprint version read: arXiv:2501.14654v2). https://doi.org/10.1056/aidbp2500144

Jin L, Sun T, Liu X, Cao Z, Liu Y, et al. (2023). A multi-center performance assessment for automated histopathological classification and grading of glioma using whole slide images. iScience. https://doi.org/10.1016/j.isci.2023.108041

Jin W, Fatehi M, Guo R, Hamarneh G (2024). Evaluating the clinical utility of artificial intelligence assistance and its explanation on the glioma grading task. Artificial Intelligence in Medicine, 148:102751. https://doi.org/10.1016/j.artmed.2023.102751

Jones C, Thornton J, Wyatt JC (2023). Artificial intelligence and clinical decision support: clinicians' perspectives on trust, trustworthiness, and liability. Medical law review. https://doi.org/10.1093/medlaw/fwad013

Joshi S, Parbhoo S, Doshi-Velez F (2022). Learning-to-defer for sequential medical decision-making under uncertainty. Transactions on Machine Learning Research. https://arxiv.org/abs/2109.06312

Jungo A, Reyes M (2019). Assessing Reliability and Challenges of Uncertainty Estimations for Medical Image Segmentation. arXiv preprint. https://arxiv.org/abs/1907.03338

Kang DY, DeYoung PN, Tantiongloc J, Coleman TP, Owens RL (2021). Statistical uncertainty quantification to augment clinical decision support: a first implementation in sleep medicine. npj Digital Medicine. https://doi.org/10.1038/s41746-021-00515-3

Karagoz MA, Nalbantoglu OU, Fox GC (2024). Residual Vision Transformer (ResViT) Based Self-Supervised Learning Model for Brain Tumor Classification. arXiv preprint. https://arxiv.org/abs/2411.12874

Karami M, Kebriaei H, Ghassemi F, Azadegan H (2026). Byzantine robust federated learning for heterogeneous brain MRI using multisignal gradient fingerprinting and adaptive trust aggregation. Scientific reports. https://doi.org/10.1038/s41598-026-55855-5

Karargyris A, Umeton R, Sheller MJ, Aristizabal A, et al. (2023). Federated benchmarking of medical artificial intelligence with MedPerf. Nature Machine Intelligence. https://doi.org/10.1038/s42256-023-00652-2

Karimireddy, S. P., Kale, S., Mohri, M., Reddi, S., Stich, S., & Suresh, A. T. (2020). SCAFFOLD: stochastic controlled averaging for federated learning. Proceedings of the 37th International Conference on Machine Learning, PMLR 119, 5132-5143.

Kazerooni AF, Khalili N, Liu X, Haldar D, Jiang Z, Zapaishchykova A, et al. (2024). BraTS-PEDs: Results of the Multi-Consortium International Pediatric Brain Tumor Segmentation Challenge 2023. Machine.Learning.for.Biomedical.Imaging. 3 (2025). https://arxiv.org/abs/2407.08855

Keim-Malpass J, Ratcliffe SJ, Clark MT, Krahn KN, Monfredi OJ, Hamil S, et al. (2026). A randomized controlled trial of artificial intelligence-based analytics for clinical deterioration. Scientific reports. https://doi.org/10.1038/s41598-026-39051-z

Kerrigan G, Smyth P, Steyvers M (2021). Combining Human Predictions with Model Probabilities via Confusion Matrices and Calibration. NeurIPS 2021. https://arxiv.org/abs/2109.14591

Keswani V, Lease M, Kenthapadi K (2021). Towards Unbiased and Accurate Deferral to Multiple Experts. AAAI/ACM Conference on AI, Ethics, and Society (AIES 2021). https://doi.org/10.1145/3461702.3462516

Kiani A, Uyumazturk B, Rajpurkar P, Wang A, Gao R, Jones E, et al. (2020). Impact of a deep learning assistant on the histopathologic classification of liver cancer. NPJ digital medicine. https://doi.org/10.1038/s41746-020-0232-8

Kibudde S, Kavuma A, Hao Y, et al. (2024). Impact of Artificial Intelligence-Based Autosegmentation of Organs at Risk in Low- and Middle-Income Countries. Advances in Radiation Oncology. https://doi.org/10.1016/j.adro.2024.101638

Kim SH, Schramm S, Riedel EO, Schmitzer L, Rosenkranz E, Kertels O, et al. (2025a). Automation bias in AI-assisted detection of cerebral aneurysms on time-of-flight MR angiography. La Radiologia medica. https://doi.org/10.1007/s11547-025-01964-6

Kim SH, Wihl J, Schramm S, et al. (2025b). Human-AI collaboration in large language model-assisted brain MRI differential diagnosis: a usability study. European Radiology. https://doi.org/10.1007/s00330-025-11484-6

Kim Y, Park C, Jeong H, Chan YS, Xu X, McDuff D, et al. (2024). MDAgents: An Adaptive Collaboration of LLMs for Medical Decision-Making. NeurIPS 2024. https://arxiv.org/abs/2404.15155

Kirchhof M, Kasneci G, Kasneci E (2025). Position: Uncertainty Quantification Needs Reassessment for Large-language Model Agents. Proceedings of the 42nd International Conference on Machine Learning (ICML 2025), Position Paper Track, PMLR 267, Vancouver, Canada. https://arxiv.org/abs/2505.22655

Klang E, Omar M, Raut G, Agbareia R, Timsina P, Freeman R, et al. (2025). Orchestrated multi agents sustain accuracy under clinical-scale workloads compared to a single agent. medRxiv : the preprint server for health sciences. https://doi.org/10.1101/2025.08.22.25334049

Knop M, Weber S, Mueller M, Niehaves B (2022). Human Factors and Technological Characteristics Influencing the Interaction of Medical Professionals With Artificial Intelligence-Enabled Clinical Decision Support Systems: Literature Review. JMIR Human Factors. https://doi.org/10.2196/28639

Kofler F, Ezhov I, Isensee F, et al. (2023). Are we using appropriate segmentation metrics? Identifying correlates of human expert perception for CNN training beyond rolling the DICE coefficient. Journal of Machine Learning for Biomedical Imaging (MELBA). https://doi.org/10.59275/j.melba.2023-dg1f

Koutsoubis N, Waqas A, Yilmaz Y, Ramachandran RP, et al. (2025). Privacy-preserving Federated Learning and Uncertainty Quantification in Medical Imaging. Radiology: Artificial Intelligence. https://doi.org/10.1148/ryai.240637

Koutsoubis N, Waqas A, Yilmaz Y, Ramachandran RP, Schabath M, Rasool G (2024). Future-Proofing Medical Imaging with Privacy-Preserving Federated Learning and Uncertainty Quantification: A Review. arXiv preprint. https://arxiv.org/abs/2409.16340

Kuhn L, Gal Y, Farquhar S (2023). Semantic Uncertainty: Linguistic Invariances for Uncertainty Estimation in Natural Language Generation. ICLR 2023 (International Conference on Learning Representations). https://arxiv.org/abs/2302.09664

Kunz LM, Metzger M, Schaefer C, Pohlmeier R, Petrovic Vorkapic J, Nosch M (2025). User experience study to evaluate a clinical decision support system prototype supporting continuous kidney replacement therapy in a simulated ICU environment. BMC medical informatics and decision making. https://doi.org/10.1186/s12911-025-03165-7

Kurc T, Bakas S, Ren X, Bagari A, Momeni A, Huang Y, et al. (2020). Segmentation and Classification in Digital Pathology for Glioma Research: Challenges and Deep Learning Approaches. Frontiers in neuroscience. https://doi.org/10.3389/fnins.2020.00027

Kwon H, Kim DJ (2026). Conformal selective prediction with cost aware deferral for safe clinical triage under distribution shift. Scientific Reports. https://doi.org/10.1038/s41598-026-40637-w

Kwon H, Kim SM, Oh SH, Jang M, Kim MG, Bae HM, et al. (2026). Artificial intelligence-assisted breast ultrasound: modest AUROC improvement and shorter interpretation time without significant change in diagnostic accuracy. Frontiers in radiology. https://doi.org/10.3389/fradi.2026.1747783

Kömen J, Marienwald H, Dippel J, Hense J (2024). Do Histopathological Foundation Models Eliminate Batch Effects? A Comparative Study. arXiv preprint. https://arxiv.org/abs/2411.05489

LaBella D, Schumacher K, Mix M, Leu K, et al. (2026). The 2024 Brain Tumor Segmentation Challenge Meningioma Radiotherapy (BraTS-MEN-RT) dataset. Scientific Data. https://doi.org/10.1038/s41597-026-06649-x

Lakshminarayanan B, Pritzel A, Blundell C (2017). Simple and Scalable Predictive Uncertainty Estimation using Deep Ensembles. NeurIPS 2017. https://arxiv.org/abs/1612.01474

Lam TYT, Cheung MFK, Munro YL, Lim KM, Shung D, Sung JJY (2022). Randomized Controlled Trials of Artificial Intelligence in Clinical Practice: Systematic Review. Journal of medical Internet research. https://doi.org/10.2196/37188

Lambert B, Forbes F, Doyle S, Dehaene H, Dojat M (2024a). Trustworthy clinical AI solutions: A unified review of uncertainty quantification in Deep Learning models for medical image analysis. Artificial Intelligence in Medicine. https://doi.org/10.1016/j.artmed.2024.102830

Lambert B, Forbes F, Doyle S, Dojat M (2024b). Robust Conformal Volume Estimation in 3D Medical Images. arXiv preprint. https://arxiv.org/abs/2407.19938

Lang K, Dustler M, Dahlblom V, Akesson A, Andersson I, Zackrisson S (2021). Identifying normal mammograms in a large screening population using artificial intelligence. European Radiology. https://doi.org/10.1007/s00330-020-07165-1

Lawson McLean A, Wu Y, Lawson McLean AC, Hristidis V (2024). Large language models as decision aids in neuro-oncology: a review of shared decision-making applications. Journal of cancer research and clinical oncology. https://doi.org/10.1007/s00432-024-05673-x

Layard Horsfall H, Palmisciano P, Khan DZ, et al. (2021). Attitudes of the Surgical Team Toward Artificial Intelligence in Neurosurgery: International 2-Stage Cross-Sectional Survey. World Neurosurgery. https://doi.org/10.1016/j.wneu.2020.10.171

Le NLD, Khoi TD, Le NPH (2026). Counterfactual Context Auditing for Glioma Segmentation Failure Detection Across MRI Datasets. arXiv preprint. https://arxiv.org/abs/2607.07038

Lee EH, Han M, Wright J, Kuwabara M, Mevorach J, et al. (2024a). An international study presenting a federated learning AI platform for pediatric brain tumors. Nature Communications. https://doi.org/10.1038/s41467-024-51172-5

Lee J, Lim J, Byeon K, Kwak JT (2024b). Benchmarking Pathology Foundation Models: Adaptation Strategies and Scenarios. arXiv preprint. https://arxiv.org/abs/2410.16038

Lee J, Shin D, Oh SH, Kim H (2022). Method to Minimize the Errors of AI: Quantifying and Exploiting Uncertainty of Deep Learning in Brain Tumor Segmentation. Sensors. https://doi.org/10.3390/s22062406

Lekadir K, Frangi AF, Porras AR, et al. (2025). FUTURE-AI: international consensus guideline for trustworthy and deployable artificial intelligence in healthcare. BMJ. https://doi.org/10.1136/bmj-2024-081554

Li B, Yan T, Pan Y, Luo J, Ji R, Ding J, et al. (2024a). MMedAgent: Learning to Use Medical Tools with Multi-modal Agent. Findings of EMNLP 2024. https://arxiv.org/abs/2407.02483

Li G, HAAK H, Itani H, Khizbullin D, Ghanem B (2023). CAMEL: Communicative Agents for "Mind" Exploration of Large Language Model Society. Advances in Neural Information Processing Systems 36 (NeurIPS 2023). https://arxiv.org/abs/2303.17760

Li H, Nan Y, Ser JD, Yang G (2022). Region-based evidential deep learning to quantify uncertainty and improve robustness of brain tumor segmentation. Neural Computing and Applications. https://doi.org/10.1007/s00521-022-08016-4

Li H, Xue F, Chaitanya K, Luo S, Ezhov I, Wiestler B, et al. (2021a). Imbalance-Aware Self-Supervised Learning for 3D Radiomic Representations. arXiv preprint. https://arxiv.org/abs/2103.04167

Li J, Wang S, Zhang M, Li W, Lai Y, Kang X, et al. (2024b). Agent Hospital: A Simulacrum of Hospital with Evolvable Medical Agents. arXiv preprint (2405.02957). https://arxiv.org/abs/2405.02957

Li S, Fang X, Jin Y, Deng Y, Hu W, Wu B, et al. (2025a). Improving diagnostic accuracy in preoperative glioma classification: performance of knowledge-enhanced large language models compared with radiologists. NPJ precision oncology. https://doi.org/10.1038/s41698-025-01171-6

Li T, Sahu AK, Zaheer M, Sanjabi M, Talwalkar A, Smith V (2020). Federated Optimization in Heterogeneous Networks (FedProx). MLSys 2020. https://arxiv.org/abs/1812.06127

Li X, Jiang M, Xu G, Xia J, Chiu C, Jia M, et al. (2026). CRC-Router: Risk-Constrained Routing for Medical Agentic AI Systems. arXiv preprint. https://arxiv.org/abs/2609.30714

Li X, Jiang M, Xu G, Xia J, Jia M, Chen D, et al. (2025b). AT-CXR: Uncertainty-Aware Agentic Triage for Chest X-rays. arXiv preprint. https://arxiv.org/abs/2508.19322

Li X, Jiang M, Zhang X, Kamp M, Dou Q (2021b). FedBN: Federated Learning on Non-IID Features via Local Batch Normalization. ICLR 2021. https://arxiv.org/abs/2102.07623

Lin J, Lin J, Lu C, et al. (2023). CKD-TransBTS: Clinical Knowledge-Driven Hybrid Transformer With Modality-Correlated Cross-Attention for Brain Tumor Segmentation. IEEE Transactions on Medical Imaging. https://doi.org/10.1109/TMI.2023.3250474

Lin L, Liu Y, Wu J, et al. (2024a). FedLPPA: Learning Personalized Prompt and Aggregation for Federated Weakly-Supervised Medical Image Segmentation. IEEE Transactions on Medical Imaging. https://doi.org/10.1109/TMI.2024.3483221

Lin Z, Trivedi S, Sun J (2024b). Generating with Confidence: Uncertainty Quantification for Black-box Large Language Models. Transactions on Machine Learning Research (TMLR), May 2024; arXiv:2305.19187 (first posted 2023). https://arxiv.org/abs/2305.19187

Lin-Martore M, Kornblith A, Firnberg M, Haque A, O'Brien B (2025). Trust of Artificial Intelligence-Augmented Point-of-Care Ultrasound Among Pediatric Emergency Physicians. Journal of the American College of Emergency Physicians open. https://doi.org/10.1016/j.acepjo.2025.100173

Linardos A, Pati S, Baid U, et al. (2025). The MICCAI Federated Tumor Segmentation (FeTS) Challenge 2024: Efficient and Robust Aggregation Methods. Machine Learning for Biomedical Imaging (MELBA). https://doi.org/10.59275/j.melba.2025-5242

Liu F, Niu Y, Zhang Q, Wang K, Dong Z, Wong IN, et al. (2025a). A foundational architecture for AI agents in healthcare. Cell Reports Medicine. https://doi.org/10.1016/j.xcrm.2025.102374

Liu H, Wei D, Lu D, et al. (2023). M3AE: Multimodal Representation Learning for Brain Tumor Segmentation with Missing Modalities. AAAI Conference on Artificial Intelligence. https://doi.org/10.1609/aaai.v37i2.25253

Liu J, Ding C, Zhang D (2026a). Addressing data annotation scarcity in Brain Tumor Segmentation on 3D MRI scan Using a Semi-Supervised Teacher-Student Framework. arXiv preprint. https://arxiv.org/abs/2602.08797

Liu J, Gallego B, Barbieri S (2022). Incorporating uncertainty in learning to defer algorithms for safe computer-aided diagnosis. Scientific Reports. https://doi.org/10.1038/s41598-022-05725-7

Liu Q, Chen C, Qin J, Dou Q, Heng PA (2021). FedDG: Federated Domain Generalization on Medical Image Segmentation via Episodic Learning in Continuous Frequency Space. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). https://doi.org/10.1109/CVPR46437.2021.00107

Liu Q, Hu Z, Huang T, Niu Y, Zhang X, Ma S, et al. (2026b). EvoMDT: a self-evolving multi-agent system for structured clinical decision-making in multi-cancer. npj Digital Medicine (vol 9, article 124). https://doi.org/10.1038/s41746-025-02304-8

Liu T, Tan Z, Chen M, Yang X, Jiang H, Huang K (2024a). MedMAP: Promoting Incomplete Multi-modal Brain Tumor Segmentation with Alignment. arXiv preprint. https://arxiv.org/abs/2408.09465

Liu X, Chen T, Da L, Chen C, Lin Z, Wei H (2025b). Uncertainty Quantification and Confidence Calibration in Large Language Models: A Survey. Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '25), Toronto, Canada. https://doi.org/10.1145/3711896.3736569

Liu X, Cruz Rivera S, Moher D, Calvert MJ, SPIRIT-AI DA, Group CW (2020). Reporting guidelines for clinical trial reports for interventions involving artificial intelligence: the CONSORT-AI extension. Nature Medicine. https://doi.org/10.1038/s41591-020-1034-x

Liu Y, Luo G, Zhu Y (2024b). FedFMS: Exploring Federated Foundation Models for Medical Image Segmentation. arXiv preprint. https://arxiv.org/abs/2403.05408

Liu Y, Zhang Z, Liu R, Xitang M, Shin S, Lee H (2026c). TumorBoard: Evidence-Grounded Multi-Agent Decision Support for Longitudinal Neuro-Oncology. arXiv preprint. https://arxiv.org/abs/2608.03190

Loaiza-Bonilla A, Yost C, Kurnaz S, Tuysuz E, Thaker NG, Giritlioglu D, et al. (2026). Transforming oncology clinical trial matching through neuro-symbolic, multi-agent AI and an oncology-specific knowledge graph: a prospective evaluation in 3804 patients. ESMO Real World Data and Digital Oncology (vol 12, article 100706). https://doi.org/10.1016/j.esmorw.2026.100706

Loftus TJ, Shickel B, Ruppert MM, Balch JA, et al. (2022). Uncertainty-aware deep learning in healthcare: A scoping review. PLOS Digital Health. https://doi.org/10.1371/journal.pdig.0000085

Lou J, Yang Y, Yu Z, Fu Z, Han W, Huang Q, et al. (2026). CXRAgent: Director-Orchestrated Multi-Stage Reasoning for Chest X-Ray Interpretation. IEEE Transactions on Medical Imaging (accepted version read: arXiv:2510.21324v2). https://doi.org/10.1109/tmi.2026.3713585

Louis, D. N., Perry, A., Wesseling, P., et al. (2021). The 2021 WHO Classification of Tumors of the Central Nervous System: a summary. Neuro-Oncology, 23(8), 1231-1251. https://doi.org/10.1093/neuonc/noab106

Lu C, Lemay A, Chang K, Hobel K, Kalpathy-Cramer J (2022). Fair Conformal Predictors for Applications in Medical Imaging. AAAI 2022. https://doi.org/10.1609/aaai.v36i11.21459

Lu C, Yu Y, Karimireddy SP, Jordan MI, Raskar R (2023). Federated Conformal Predictors for Distributed Uncertainty Quantification. arXiv preprint. https://arxiv.org/abs/2305.17564

Lu MY, Chen B, Williamson DFK, et al. (2024). A visual-language foundation model for computational pathology. Nature Medicine. https://doi.org/10.1038/s41591-024-02856-4

Lu MY, Kong D, Lipkova J, Chen RJ, Singh R, Williamson DFK, et al. (2020). Federated Learning for Computational Pathology on Gigapixel Whole Slide Images. arXiv preprint. https://arxiv.org/abs/2009.10190

Lu MY, Williamson DFK, Chen TY, Chen RJ, Barbieri M, Mahmood F (2021). Data-efficient and weakly supervised computational pathology on whole-slide images (CLAM). Nature Biomedical Engineering. https://arxiv.org/abs/2004.09666

Lu X, Li Q, Gao Y, Dong W, Lyu M, Ma S, et al. (2026). Agentic systems in computational pathology: architectures, evidence, and translational challenges. Journal of Translational Medicine (vol 24, article 1111). https://doi.org/10.1186/s12967-026-08829-0

Luo C, Yang J, Liu Z, Jing D (2023). Predicting the recurrence and overall survival of patients with glioma based on histopathological images using deep learning. Frontiers in neurology. https://doi.org/10.3389/fneur.2023.1100933

Luo X, Yang Y, Yin S, Li H, Shao Y, et al. (2024). Automated segmentation of brain metastases with deep learning: A multi-center, randomized crossover, multi-reader evaluation study. Neuro-Oncology. https://doi.org/10.1093/neuonc/noae113

Luque L, Skogen K, MacIntosh BJ, Emblem KE, Larsson C, Bouget D, et al. (2024). Standardized evaluation of the extent of resection in glioblastoma with automated early post-operative segmentation. Frontiers in radiology. https://doi.org/10.3389/fradi.2024.1357341

Lyu X, Liang Y, Chen W, Ding M, Yang J, Huang G, et al. (2025). WSI-Agents: A Collaborative Multi-Agent System for Multi-Modal Whole Slide Image Analysis. arXiv preprint. https://arxiv.org/abs/2507.14680

Ma C, Wang H, Zhang Y, Shamout FE (2026). MedGuards: Multi-Agent System for Reliable Medical Error Detection and Correction. arXiv preprint. https://arxiv.org/abs/2606.25651

Ma J, He Y, Li F, Han L, You C, Wang B (2024). Segment Anything in Medical Images (MedSAM). Nature Communications. https://arxiv.org/abs/2304.12306

Ma J, Yang Z, Kim S, et al. (2025). MedSAM2: Segment Anything in 3D Medical Images and Videos. arXiv. https://arxiv.org/abs/2504.03600

Madras D, Pitassi T, Zemel R (2018). Predict Responsibly: Improving Fairness and Accuracy by Learning to Defer. NeurIPS 2018. https://arxiv.org/abs/1711.06664

Maier-Hein L, Reinke A, Godau P, et al. (2024). Metrics reloaded: recommendations for image analysis validation. Nature Methods. https://doi.org/10.1038/s41592-023-02151-z

Malone TW, Vaccaro MA, al. AC, et al. (2024). When combinations of humans and AI are useful: A systematic review and meta-analysis. Nature Human Behaviour. https://doi.org/10.1038/s41562-024-02024-1

Manthe M, Duffner S, Lartizien C (2024). Federated brain tumor segmentation: An extensive benchmark. Medical Image Analysis. https://doi.org/10.1016/j.media.2024.103270

Mao A, Mohri M, Zhong Y (2024a). Principled Approaches for Learning to Defer with Multiple Experts. ISAIM 2024 (Lecture Notes in Computer Science, Springer). https://doi.org/10.1007/978-3-031-63735-3_7

Mao A, Mohri M, Zhong Y (2024b). Realizable H-Consistent and Bayes-Consistent Loss Functions for Learning to Defer. arXiv preprint. https://arxiv.org/abs/2407.13732

Mathenge W, Whitestone N, Nkurikiye J, et al. (2022). Impact of Artificial Intelligence Assessment of Diabetic Retinopathy on Referral Service Uptake in a Low-Resource Setting: The RAIDERS Randomized Trial. Ophthalmology Science. https://doi.org/10.1016/j.xops.2022.100168

Mazurowski MA, Dong H, al. HG, et al. (2023). Segment anything model for medical image analysis: An experimental study. Medical Image Analysis. https://doi.org/10.1016/j.media.2023.102918

McDuff D, Schaekermann M, Tu T, et al. (2025). Towards accurate differential diagnosis with large language models. Nature. https://doi.org/10.1038/s41586-025-08869-4

McMahan HB, Moore E, Ramage D, Hampson S, Aguera y Arcas B (2017). Communication-Efficient Learning of Deep Networks from Decentralized Data (FedAvg). AISTATS 2017. https://arxiv.org/abs/1602.05629

Mehrtash A, Wells WM, Tempany CM, Abolmaesumi P, Kapur T (2020). Confidence Calibration and Predictive Uncertainty Estimation for Deep Medical Image Segmentation. IEEE Transactions on Medical Imaging. https://arxiv.org/abs/1911.13273

Mehrtens HA, Bucher T, Brinker TJ (2023a). Pitfalls of Conformal Predictions for Medical Image Classification. UNSURE Workshop at MICCAI 2023 (Lecture Notes in Computer Science). https://doi.org/10.1007/978-3-031-44336-7_20

Mehrtens HA, Kurz A, Bucher TC, Brinker TJ (2023b). Benchmarking common uncertainty estimation methods with histopathological images under domain shift and label noise. Medical Image Analysis. https://doi.org/10.1016/j.media.2023.102914

Mehta R, Filos A, Baid U, Sako C, McKinley R, Rebsamen M, et al. (2022). QU-BraTS: MICCAI BraTS 2020 Challenge on Quantifying Uncertainty in Brain Tumor Segmentation. Machine Learning for Biomedical Imaging (MELBA). https://arxiv.org/abs/2112.10074

Minderer M, Djolonga J, Romijnders R, Hubis F, Zhai X, Houlsby N, et al. (2021). Revisiting the Calibration of Modern Neural Networks. NeurIPS 2021. https://doi.org/10.48550/arXiv.2106.07998

Mitchell M, Wu S, Zaldivar A, Barnes P, Vasserman L, Hutchinson B, et al. (2019). Model Cards for Model Reporting. ACM FAT* 2019. https://arxiv.org/abs/1810.03993

Moehring A, Kutwal M, Huang R, Banerjee O, Jacobi A, Eber C, et al. (2025). A Dataset for Understanding Radiologist-Artificial Intelligence Collaboration. Scientific data. https://doi.org/10.1038/s41597-025-05054-0

Mofatteh M, Mashayekhi MS, Arfaie S, Chen Y, Mirza AB, Fares J, et al. (2022). Augmented and virtual reality usage in awake craniotomy: a systematic review. Neurosurgical review. https://doi.org/10.1007/s10143-022-01929-7

Mohri C, Hashimoto T (2024). Language Models with Conformal Factuality Guarantees. ICML 2024. https://arxiv.org/abs/2402.10978

Molinaro, A. M., Hervey-Jumper, S., Morshed, R. A., et al. (2020). Association of maximal extent of resection of contrast-enhanced and non-contrast-enhanced tumor with survival within molecular subgroups of patients with newly diagnosed glioblastoma. JAMA Oncology, 6(4), 495-503. https://doi.org/10.1001/jamaoncol.2019.6143

Montreuil Y, Carlier A, Ng LX, Ooi WT (2025). Why Ask One When You Can Ask k? Learning-to-Defer to the Top-k Experts. arXiv preprint. https://arxiv.org/abs/2504.12988

Mossina L, Dalmau J, andéol L (2024). Conformal Semantic Image Segmentation: Post-hoc Quantification of Predictive Uncertainty. arXiv preprint. https://arxiv.org/abs/2405.05145

Mossina L, Friedrich C (2025). Controlling False Positives in Image Segmentation via Conformal Prediction. arXiv preprint. https://arxiv.org/abs/2511.15406

Mozannar H, Lang H, Wei D, Sattigeri P, Das S, Sontag D (2023a). Who Should Predict? Exact Algorithms For Learning to Defer to Humans. International Conference on Artificial Intelligence and Statistics (AISTATS 2023), PMLR 206; arXiv:2301.06197. https://arxiv.org/abs/2301.06197

Mozannar H, Lee J, Wei D, Sattigeri P, Das S, Sontag D (2023b). Effective Human-AI Teams via Learned Natural Language Rules and Onboarding. NeurIPS 2023. https://doi.org/10.52202/075280-1328

Mozannar H, Sontag D (2020). Consistent Estimators for Learning to Defer to an Expert. ICML 2020. https://arxiv.org/abs/2006.01862

Mukherjee S, Gamble P, Ausin MS, Kant N, Aggarwal K, Manjunath N, et al. (2024). Polaris: A Safety-focused LLM Constellation Architecture for Healthcare. arXiv preprint. https://arxiv.org/abs/2403.13313

Mukhoti J, Kirsch A, Amersfoort J, Torr PHS, Gal Y (2023). Deep Deterministic Uncertainty: A New Simple Baseline. CVPR 2023. https://doi.org/10.1109/CVPR52729.2023.02336

Naicker S, Schmidt P, Shar B, Tariq A, Earnshaw A, McPhail S (2026). Implementing an Artificial Intelligence Decision Support System in Radiology: Prospective Qualitative Evaluation Study Using the Nonadoption Abandonment Scale-Up, Spread, and Sustainability (NASSS) Framework. Journal of medical Internet research. https://doi.org/10.2196/80342

Nampalle KB, Singh P, Narayan UV, Raman B (2023). Vision Through the Veil: Differential Privacy in Federated Learning for Medical Image Classification. arXiv preprint. https://arxiv.org/abs/2306.17794

Nath V, Li W, Yang D, Myronenko A, Zheng M, Lu Y, et al. (2025). VILA-M3: Enhancing Vision-Language Models with Medical Expert Knowledge. CVPR 2025 (IEEE/CVF Conference on Computer Vision and Pattern Recognition); arXiv:2411.12915v2. https://doi.org/10.1109/cvpr52734.2025.01378

National Cancer Control Programme. (2024). National cancer incidence and mortality data, Sri Lanka. Ministry of Health, Sri Lanka. https://www.nccp.health.gov.lk/

Nicolson A, Bradburn E, Gal Y, et al. (2025). The human factor in explainable artificial intelligence: clinician variability in trust, reliance, and performance. npj Digital Medicine. https://doi.org/10.1038/s41746-025-02023-0

Niehues JM, Quirke P, West NP, Grabsch HI, van Treeck M, Schirris Y, et al. (2023). Generalizable biomarker prediction from cancer pathology slides with self-supervised deep learning: A retrospective multi-centric study. Cell reports. Medicine. https://doi.org/10.1016/j.xcrm.2023.100980

Novak A, Ather S, Gill A, Aylward P, Maskell G, Cowell GW, et al. (2024). Evaluation of the impact of artificial intelligence-assisted image interpretation on the diagnostic performance of clinicians

in identifying pneumothoraces on plain chest X-ray: a multi-case multi-reader study. Emergency medicine journal : EMJ. https://doi.org/10.1136/emermed-2023-213620

Okati N, De A, Gomez-Rodriguez M (2021). Differentiable Learning Under Triage. NeurIPS 2021. https://arxiv.org/abs/2103.08902

Olsson H, Kartasalo K, Mulliqi N, Capuccini M, et al. (2022). Estimating diagnostic uncertainty in artificial intelligence assisted pathology using conformal prediction. Nature Communications. https://doi.org/10.1038/s41467-022-34945-8

Onyeze N, Sartawi S, Nayyer Z (2025). Effectiveness of AI-Based Tools in Detecting Diabetic Retinopathy in Low- and Middle-Income Countries: A Systematic Review of Diagnostic Performance and Implementation Feasibility. Cureus. https://doi.org/10.7759/cureus.96554

Ordonez-Rubiano EG, Johnson JM, Abdalá-Vargas N, Zorro OF, Marin-Munoz JH, Álvarez-Tobián R, et al. (2023). Preoperative tractography algorithm for safe resection of tumors located in the descending motor pathways zone. Surgical neurology international. https://doi.org/10.25259/SNI_230_2023

Ottesen JA, Yi D, Tong E, Iv M, Latysheva A, et al. (2022). 2.5D and 3D segmentation of brain metastases with deep learning on multinational MRI data. Frontiers in Neuroinformatics. https://doi.org/10.3389/fninf.2022.1056068

Ouyang C, Chen C, Li S, Li Z, Qin C, Bai W, et al. (2021). Causality-inspired Single-source Domain Generalization for Medical Image Segmentation. arXiv preprint. https://arxiv.org/abs/2111.12525

Ovadia Y, Fertig E, Ren J, Nado Z, Sculley D, Nowozin S, et al. (2019). Can You Trust Your Model's Uncertainty? Evaluating Predictive Uncertainty Under Dataset Shift. NeurIPS 2019. https://arxiv.org/abs/1906.02530

Pahud de Mortanges A, Luo H, Shu SZ, Kamath A, Suter Y, Shelan M, et al. (2024). Orchestrating explainable artificial intelligence for multimodal and longitudinal data in medical imaging. npj Digital Medicine. https://doi.org/10.1038/s41746-024-01190-w

Park SH, Choi JI, Fournier L, Vasey B (2022). Randomized Clinical Trials of Artificial Intelligence in Medicine: Why, When, and How?. Korean journal of radiology. https://doi.org/10.3348/kjr.2022.0834

Patel T, Yousuf Ibrahim H, Abbas M, Amir Ali M, Sohrab S, Nawaz K, et al. (2026). Emerging paradigms in connectome-guided neurosurgery for central nervous system tumors: a narrative review of structural and functional connectomics. Annals of medicine and surgery (2012). https://doi.org/10.1097/MS9.0000000000004915

Pati S, Baid U, Edwards B, Sheller M, Wang SH, Bakas S, et al. (2022). Federated learning enables big data for rare cancer boundary detection. Nature Communications. https://arxiv.org/abs/2204.10836

Pei L, Jones KA, Shboul ZA, Chen JY, Iftekharuddin KM (2021). Deep Neural Network Analysis of Pathology Images With Integrated Molecular Data for Enhanced Glioma Classification and Grading. Frontiers in oncology. https://doi.org/10.3389/fonc.2021.668694

Pemberton HG, Wu J, Kommers I, Muller DMJ, Hu Y, et al. (2023). Multi-class glioma segmentation on real-world data with missing MRI sequences: comparison of three deep learning algorithms. Scientific Reports. https://doi.org/10.1038/s41598-023-44794-0

Piffer A, Buchner JA, Gennari AG, Grehten P, et al. (2026). Enhancing efficiency in pediatric brain tumor segmentation using a pathologically diverse single-center clinical dataset. Neuro-Oncology Advances. https://doi.org/10.1093/noajnl/vdag024

Poceviciute M, Eilertsen G, Jarkman S, Lundstrom C (2022). Generalisation effects of predictive uncertainty estimation in deep learning for digital pathology. Scientific Reports. https://doi.org/10.1038/s41598-022-11826-0

Presacan O, Nik A, Ojha J, Thambawita V, Ionescu B, Riegler MA (2026). When silence is safer: a review and decision-theoretic framework for LLM abstention in healthcare. npj Digital Medicine. https://doi.org/10.1038/s41746-026-02882-1

Price, M., Ostrom, Q. T., et al. (2024). CBTRUS statistical report: primary brain and other central nervous system tumors diagnosed in the United States in 2017-2021. Neuro-Oncology, 26(Suppl. 6), vi1-vi85. https://doi.org/10.1093/neuonc/noae145

Prince EW, Ghosh D, Gorg C, Hankinson TC (2023). Uncertainty-Aware Deep Learning Classification of Adamantinomatous Craniopharyngioma from Preoperative MRI. Diagnostics. https://doi.org/10.3390/diagnostics13061132

Raees M, Papangelis K (2026). From Trust to Appropriate Reliance: Measurement Constructs in Human-AI Decision-Making. arXiv preprint. https://arxiv.org/abs/2604.23896

Raghu M, Blumer K, Corrado G, Kleinberg J, Obermeyer Z, Mullainathan S (2019). The Algorithmic Automation Problem: Prediction, Triage, and Human Effort. arXiv 1903.12220. https://arxiv.org/abs/1903.12220

Raji ID, Smart A, White RN, Mitchell M, Gebru T, Hutchinson B, et al. (2020). Closing the AI Accountability Gap: Defining an End-to-End Framework for Internal Algorithmic Auditing. ACM FAT* 2020. https://arxiv.org/abs/2001.00973

Raman N, Yee M (2021). Improving Learning-to-Defer Algorithms Through Fine-Tuning. arXiv preprint. https://arxiv.org/abs/2112.10768

Raza S, Sapkota R, Karkee M, Emmanouilidis C (2026). TRiSM for Agentic AI: A review of Trust, Risk, and Security Management in LLM-based Agentic Multi-Agent Systems. AI Open (also arXiv:2506.04133v5). https://doi.org/10.1016/j.aiopen.2026.02.006

Reddy DD, Saadat N, Holcomb JM, Wagner BC, Truong NC, Bowerman J, et al. (2026). The University of Texas Southwestern Glioma Dataset - MRI, Molecular Markers and Segmentations. Scientific data. https://doi.org/10.1038/s41597-026-07274-4

Ren AZ, Dixit A, Bodrova A, et al. (2023). Robots That Ask For Help: Uncertainty Alignment for Large Language Model Planners. CoRL 2023. https://arxiv.org/abs/2307.01928

Ren R, Zhu A, Li Y, Liu H, Huang G, Gu J, et al. (2025). Federated radiomics analysis of preoperative MRI across institutions: toward integrated glioma segmentation and molecular subtyping. Frontiers in radiology. https://doi.org/10.3389/fradi.2025.1648145

Reverberi C, Rigon T, Solari A, et al. (2022). Experimental evidence of effective human-AI collaboration in medical decision-making. Scientific Reports. https://doi.org/10.1038/s41598-022-18751-2

Riccardi MT, Nurchis MC, Savoia C, Gigli S, Marotta D, Di Lullo F, et al. (2026). Artificial Intelligence as triage in breast screening: simulated impact on workload and detection. The European Journal of Public Health. https://doi.org/10.1093/eurpub/ckag169.832

Richardson S, Feldstein D, McGinn T, Park LS, Khan S, Hess R, et al. (2019). Live Usability Testing of Two Complex Clinical Decision Support Tools: Observational Study. JMIR human factors. https://doi.org/10.2196/12471

Rivera JEH, Low DK, Xiong X, Ruzevick JJ, Child DD, Yim W, et al. (2026). CoRe-BT: A Multimodal Radiology-Pathology-Text Benchmark for Robust Brain Tumor Typing. arXiv preprint. https://arxiv.org/abs/2603.03618

Rizvi RF, Faisal S, Sussman M, Mendlick P, Brown S, Lindemann E, et al. (2025). Leveraging Dual Usability Methods to Evaluate Clinical Decision Support Among Patients With Traumatic Brain Injury: Mixed Methods Study. JMIR human factors. https://doi.org/10.2196/60268

Roberts M, Driggs D, al. MT, et al. (2021). Common pitfalls and recommendations for using machine learning to detect and prognosticate for COVID-19 using chest radiographs and CT scans. Nature Machine Intelligence. https://doi.org/10.1038/s42256-021-00307-0

Rodriguez-Ruiz A, Lång K, Gubern-Merida A, Teuwen J, Broeders M, Gennaro G, et al. (2019). Can we reduce the workload of mammographic screening by automatic identification of normal exams with artificial intelligence? A feasibility study. European radiology. https://doi.org/10.1007/s00330-019-06186-9

Rojas GEM, Voort SR, Pirkl CM, Kaushik S, Smits M, Klein S (2026). Towards Trustworthy AI for Glioma Diagnosis: A Task-Aware Evaluation of Uncertainty Quantification. Machine.Learning.for.Biomedical.Imaging. 2026 (2026). https://arxiv.org/abs/2609.39429

Rosbach E, Ammeling J, Ganz J, Bertram CA, Conrad T, Riener A, et al. (2026). Stuck on Suggestions: Automation Bias, the Anchoring Effect, and the Factors That Shape Them in Computational Pathology. Machine.Learning.for.Biomedical.Imaging. 2026 (2026). https://arxiv.org/abs/2603.11821

Rosbach E, Ganz J, Ammeling J, Riener A, Aubreville M (2024). Automation Bias in AI-Assisted Medical Decision-Making under Time Pressure in Computational Pathology. arXiv preprint. https://arxiv.org/abs/2411.00998

Rosenbacke R, Å M, McKee M, Stuckler D (2024). How Explainable Artificial Intelligence Can Increase or Decrease Clinicians' Trust in AI Applications in Health Care: Systematic Review. JMIR AI. https://doi.org/10.2196/53207

Roy S, Koehler G, Ulrich C, et al. (2023). MedNeXt: Transformer-Driven Scaling of ConvNets for Medical Image Segmentation. MICCAI 2023 (LNCS). https://doi.org/10.1007/978-3-031-43901-8_39

Saeed N, Ridzuan M, Alasmawi H, Sobirov I, Yaqub M (2023). MGMT promoter methylation status prediction using MRI scans? An extensive experimental evaluation of deep learning models. Medical Image Analysis. https://doi.org/10.1016/j.media.2023.102989

Sakamoto T, Harada Y, Shimizu T (2024). Facilitating Trust Calibration in Artificial Intelligence-Driven Diagnostic Decision Support Systems for Determining Physicians' Diagnostic Accuracy: Quasi-Experimental Study. JMIR formative research. https://doi.org/10.2196/58666

Salari S, Rasoulian A, Rivaz H, Xiao Y (2023). Dense Error Map Estimation for MRI-Ultrasound Registration in Brain Tumor Surgery Using Swin UNETR. arXiv preprint. https://arxiv.org/abs/2308.10784

Saldanha OL, Loeffler CML, Niehues JM, van Treeck M, Seraphin TP, Hewitt KJ, et al. (2023). Self-supervised attention-based deep learning for pan-cancer mutation prediction from histopathology. NPJ precision oncology. https://doi.org/10.1038/s41698-023-00365-0

Saldanha OL, Quirke P, West NP, James JA, Loughrey MB, et al. (2022). Swarm learning for decentralized artificial intelligence in cancer histopathology. Nature Medicine. https://doi.org/10.1038/s41591-022-01768-5

Salvati LF, De Marco R, Palmieri G, Minardi M, Massara A, Pesaresi A, et al. (2021). The Relevant Role of Navigated Tractography in Speech Eloquent Area Glioma Surgery: Single Center Experience. Brain sciences. https://doi.org/10.3390/brainsci11111436

Sambyal AS, Niyaz U, Krishnan NC, Bathula DR (2023). Understanding calibration of deep neural networks for medical image classification. Computer Methods and Programs in Biomedicine. https://doi.org/10.1016/j.cmpb.2023.107816

Sanai, N., & Berger, M. S. (2008). Glioma extent of resection and its impact on patient outcome. Neurosurgery, 62(4), 753-766. https://doi.org/10.1227/01.neu.0000318159.21731.cf

Saueressig C, Scholz D, Raffler P, Delbridge C, et al. (2026). Multimodal fusion of pathology and radiology foundation models for WHO 2021 glioma subtyping. npj Precision Oncology. https://doi.org/10.1038/s41698-026-01366-5

Schemmer M, Bartos A, Spitzer P, Hemmer P, Kühl N, Liebschner J, et al. (2023). Towards Effective Human-AI Decision-Making: The Role of Human Learning in Appropriate Reliance on AI Advice. International Conference on Information Systems (ICIS 2023). https://arxiv.org/abs/2310.02108

Schemmer M, Kühl N, Benz C, Satzger G (2022). On the Influence of Explainable AI on Automation Bias. arXiv preprint. https://arxiv.org/abs/2204.08859

Schirris Y, Gavves E, Nederlof I, Horlings HM, Teuwen J (2021). DeepSMILE: Contrastive self-supervised pre-training benefits MSI and HRD classification directly from H&E whole-slide images in colorectal and breast cancer. Medical Image Analysis Volume 79, July 2022, 102464. https://arxiv.org/abs/2107.09405

Schmidgall S, Ziaei R, Harris C, Reis E, Jopling J, Moor M (2024). AgentClinic: a multimodal agent benchmark to evaluate AI in simulated clinical environments. arXiv preprint (2405.07960). https://arxiv.org/abs/2405.07960

Schmutz M, Sommer S, Sander J, Graumann D, Raffler J, Soto-Rey I, et al. (2025). Large language model processing capabilities of ChatGPT 4.0 to generate molecular tumor board recommendations-a critical evaluation on real world data. The oncologist. https://doi.org/10.1093/oncolo/oyaf293

Schulz PJ, Kee KM, Lwin MO, Goh WW, Chia KY, Cheung MFK, et al. (2025). Clinical experience and perception of risk affect the acceptance and trust of using AI in medicine. Frontiers in digital health. https://doi.org/10.3389/fdgth.2025.1620127

Scipion CEA, Manchester MA, Federman A, Wang Y, Arias JJ (2025). Barriers to and facilitators of clinician acceptance and use of artificial intelligence in healthcare settings: a scoping review. BMJ open. https://doi.org/10.1136/bmjopen-2024-092624

Seker ME, Koyluoglu YO, Ozaydin AN, Gurdal SO, Ozcinar B, Cabioglu N, et al. (2024). Diagnostic capabilities of artificial intelligence as an additional reader in a breast cancer screening program. European radiology. https://doi.org/10.1007/s00330-024-10661-3

Shafique A, Dy A, Qin X, Alshamlan N, Androutsos D, Done SJ, et al. (2025). Lightweight self supervised learning framework for domain generalization in histopathology. Scientific reports. https://doi.org/10.1038/s41598-025-20411-0

Shamszare H, Choudhury A (2023). Clinicians' Perceptions of Artificial Intelligence: Focus on Workload, Risk, Trust, Clinical Decision Making, and Clinical Integration. Healthcare (Basel, Switzerland). https://doi.org/10.3390/healthcare11162308

Shao Z, Bian H, Chen Y, et al. (2021). TransMIL: Transformer based Correlated Multiple Instance Learning for Whole Slide Image Classification. Advances in Neural Information Processing Systems (NeurIPS). https://arxiv.org/abs/2106.00908

Sharma K, Sivadas H, Edwards C, Gunja N, Reddy S (2026). ED-triage-agent: a multi-agent framework for human-in-the-loop emergency triage. International Journal of Medical Informatics. https://doi.org/10.1016/j.ijmedinf.2026.106652

Sheller MJ, Edwards B, Reina GA, Martin J, Pati S, Kotrotsou A, et al. (2020). Federated learning in medicine: facilitating multi-institutional collaborations without sharing patient data. Scientific Reports. https://doi.org/10.1038/s41598-020-69250-1

Sheller MJ, Reina GA, Edwards B, Martin J, Bakas S (2018). Multi-Institutional Deep Learning Modeling Without Sharing Patient Data: A Feasibility Study on Brain Tumor Segmentation. arXiv preprint. https://arxiv.org/abs/1810.04304

Shen Y, Luo Y, Shen D, Ke J (2022). RandStainNA: Learning Stain-Agnostic Features from Histology Slides by Bridging Stain Augmentation and Normalization. arXiv preprint. https://arxiv.org/abs/2206.12694

Sheng Z, Yu J, Chen Z, Sun Y, Bu X, Wang M, et al. (2021). Constrained-Spherical Deconvolution Tractography in the Evaluation of the Corticospinal Tract in Glioma Surgery. Frontiers in surgery. https://doi.org/10.3389/fsurg.2021.646465

Shi X, Tian Z, Guo Q, Qiao B, Wang X (2026). Acceptance of Artificial Intelligence in Clinical Practice Among Chinese Physicians: Nationwide Cross-Sectional Survey Using Extended Unified Theory of Acceptance and Use of Technology and Explainable Machine Learning. JMIR medical informatics. https://doi.org/10.2196/85270

Shi Y, Wang G (2025). Few-Shot Generation of Brain Tumors for Secure and Fair Data Sharing. arXiv preprint. https://arxiv.org/abs/2504.00150

Singh S, Sikka H, Kotti S, Trask A (2020). Benchmarking Differentially Private Residual Networks for Medical Imagery. arXiv preprint. https://arxiv.org/abs/2005.13099

Smith BH, Wolfe JG, Karam A, Demarkschalk BM, Hrdlicka CM, Nasr DM, et al. (2025). Prospective Evaluation of Artificial Intelligence Imaging Support Software for Acute Ischemic Stroke in the Mayo Clinic Telestroke Network. Mayo Clinic proceedings. Innovations, quality & outcomes. https://doi.org/10.1016/j.mayocpiqo.2025.100631

Song CW, Baek BH, Kim SK, Yoon W, Lee YY, Park I, et al. (2026). Impact of Large Language Model Assistance on Radiologists' Diagnostic Performance for Brain Tumors by Experience Level. Journal of clinical medicine. https://doi.org/10.3390/jcm15041673

Song Y, Jeong M, Sung M (2025). Trustworthy Agents for Electronic Health Records through Confidence Estimation. arXiv preprint. https://arxiv.org/abs/2508.19096

Sorin V, Klang E, Sklair-Levy M, Cohen I, Zippel DB, Balint Lahat N, et al. (2023). Large language model (ChatGPT) as a support tool for breast tumor board. NPJ breast cancer. https://doi.org/10.1038/s41523-023-00557-8

Sorka M, Gorenshtein A, Aran D, Shelly S (2025). A multi-agent approach to neurological clinical reasoning. PLOS Digital Health 4(12): e0001106. https://doi.org/10.1371/journal.pdig.0001106

Sounderajah V, Guni A, al. XL, et al. (2025). The STARD-AI reporting guideline for diagnostic accuracy studies using artificial intelligence. Nature Medicine. https://doi.org/10.1038/s41591-025-03953-8

Srinivasan T, Thomason J (2025). Adjust for Trust: Mitigating Trust-Induced Inappropriate Reliance on AI Assistance. arXiv preprint. https://arxiv.org/abs/2502.13321

Stutz D, Roy AG, Matejovicova T, Strachan P, Cemgil AT, Doucet A (2023). Conformal prediction under ambiguous ground truth. Transactions on Machine Learning Research. https://doi.org/10.48550/arxiv.2307.09302

Sugiyama T, Sugimori H, Tang M, Fujimura M (2025). Artificial Intelligence for Patient Safety and Surgical Education in Neurosurgery. JMA journal. https://doi.org/10.31662/jmaj.2024-0141

Swaminathan A, Lopez I, Wang W, Srivastava U, Tran E, Bhargava-Shah A, et al. (2023). Selective prediction for extracting unstructured clinical data. Journal of the American Medical Informatics Association, 31(1):188-197. https://doi.org/10.1093/jamia/ocad182

Swaroop S, Buçinca Z, Gajos KZ, Doshi-Velez F (2023). Accuracy-Time Tradeoffs in AI-Assisted Decision Making under Time Pressure. arXiv preprint. https://arxiv.org/abs/2306.07458

Tailor D, Patra A, Verma R, Manggala P, Nalisnick E (2024). Learning to Defer to a Population: A Meta-Learning Approach. AISTATS 2024. https://arxiv.org/abs/2403.02683

Tampu IE, Nyman P, Spyretos C, Blystad I, Shamikh A, et al. (2026). Pediatric brain tumor classification using digital pathology and deep learning: Evaluation of SOTA methods on a multi-center Swedish cohort. Brain Pathology. https://doi.org/10.1111/bpa.70029

Tang X, Zou A, Zhang Z, Li Z, Zhao Y, Zhang X, et al. (2024). MedAgents: Large Language Models as Collaborators for Zero-shot Medical Reasoning. Findings of ACL 2024. https://arxiv.org/abs/2311.10537

Tang Y, Yang D, Li W, Roth HR, Landman B, Xu D, et al. (2022). Self-Supervised Pre-Training of Swin Transformers for 3D Medical Image Analysis. CVPR 2022. https://arxiv.org/abs/2111.14791

Tini P, Novi F, Donnini F, Perrella A, Bagnacci G, Mazzei MA, et al. (2025). Assessing ChatGPT-4 as a clinical decision support tool in neuro-oncology radiotherapy: a prospective comparative study. Journal of neuro-oncology. https://doi.org/10.1007/s11060-025-05254-z

Touhami M, Ahmad Fauzi MF, Ur Rehman Z, Mansor S (2026). Federated Learning for Histopathology Image Classification: A Systematic Review. Diagnostics (Basel, Switzerland). https://doi.org/10.3390/diagnostics16010137

Tran D, Liu J, Dusenberry MW, et al. (2022). Plex: Towards Reliability using Pretrained Large Model Extensions. arXiv preprint (Google Research). https://doi.org/10.48550/arXiv.2207.07411

Tzanis E, Klontzas ME (2025). ReclAIm: A Multi-Agent Framework for Monitoring and Correcting Performance Decline in Medical Imaging AI. arXiv preprint. https://arxiv.org/abs/2510.17004

Vasey B, Nagendran M, Campbell B, Clifton DA, Collins GS, McCulloch P, et al. (2022). Reporting guideline for the early-stage clinical evaluation of decision support systems driven by artificial intelligence: DECIDE-AI. Nature Medicine. https://doi.org/10.1038/s41591-022-01772-9

Vazhentsev A, Sviridov I, Barseghyan A, Kuzmin G, Panchenko A, Nesterov A, et al. (2026). Uncertainty-aware abstention in medical diagnosis based on medical texts. npj Artificial Intelligence. https://doi.org/10.1038/s44387-026-00097-z

Vazquez J, Facelli JC (2022). Conformal Prediction in Clinical Medical Sciences. Journal of Healthcare Informatics Research. https://doi.org/10.1007/s41666-021-00113-8

Verma R, Barrejon D, Nalisnick E (2023). Learning to Defer to Multiple Experts: Consistent Surrogate Losses, Confidence Calibration, and Conformal Ensembles. AISTATS 2023. https://arxiv.org/abs/2210.16955

Verma R, Kandoi S, Afzal R, Chen S, Jegminat J, Karlovich MW, et al. (2025). Domain-Specific Foundation Model Improves AI-Based Analysis of Neuropathology. arXiv preprint. https://arxiv.org/abs/2512.05993

Verma R, Nalisnick E (2022). Calibrated Learning to Defer with One-vs-All Classifiers. ICML 2022. https://arxiv.org/abs/2202.03673

Vermeulen C, Pages-Gallego M, Kester L, Kranendonk MEG, et al. (2023). Ultra-fast deep-learned CNS tumour classification during surgery. Nature. https://doi.org/10.1038/s41586-023-06615-2

Vijayan S, Nair AM, Dominic N, Chiramal JA, Pawar S, Goswami L (2026). Artificial intelligence-assisted chest X-ray for tuberculosis case finding in low- and middle-income countries: implementation experiences and impact. BJR open. https://doi.org/10.1093/bjro/tzag007

Wang D, Wang L, Zhang Z, et al. (2021a). "Brilliant AI Doctor" in Rural Clinics: Challenges in AI-Powered Clinical Decision Support System Deployment. CHI 2021. https://doi.org/10.1145/3411764.3445432

Wang J, Jin Y, Stoyanov D, Wang L (2024a). FedDP: Dual Personalization in Federated Medical Image Segmentation. IEEE Transactions on Medical Imaging. https://doi.org/10.1109/TMI.2023.3299206

Wang J, Jin Y, Wang L (2022). Personalizing Federated Medical Image Segmentation via Local Calibration. European Conference on Computer Vision (ECCV). https://doi.org/10.1007/978-3-031-19803-8_27

Wang J, Wang J, Athiwaratkun B, Zhang C, Zou J (2024b). Mixture-of-Agents Enhances Large Language Model Capabilities. ICLR 2025 (arXiv). https://arxiv.org/abs/2406.04692

Wang L, Ma C, Feng X, Zhang Z, Yang H, Zhang J, et al. (2024c). A survey on large language model based autonomous agents. Frontiers of Computer Science. https://arxiv.org/abs/2308.11432

Wang L, Yang Z, LaBella D, Reitman Z, Ginn J, et al. (2025a). Uncertainty quantification in multi-parametric MRI-based meningioma radiotherapy target segmentation. Frontiers in Oncology. https://doi.org/10.3389/fonc.2025.1474590

Wang W, Chen C, Ding M, et al. (2021b). TransBTS: Multimodal Brain Tumor Segmentation Using Transformer. MICCAI 2021 (LNCS). https://doi.org/10.1007/978-3-030-87193-2_11

Wang W, Zhao Y, Teng L, et al. (2023). Neuropathologist-level integrated classification of adult-type diffuse gliomas using deep learning from whole-slide pathological images. Nature Communications. https://doi.org/10.1038/s41467-023-41195-9

Wang X, Wang L, Bu H, Zhang N, Yue M, Jia Z, et al. (2021c). How can artificial intelligence models assist PD-L1 expression scoring in breast cancer: results of multi-institutional ring studies. NPJ breast cancer. https://doi.org/10.1038/s41523-021-00268-y

Wang Y, Zhang Y, Liu Y, et al. (2021d). ACN: Adversarial Co-training Network for Brain Tumor Segmentation with Missing Modalities. MICCAI 2021 (LNCS). https://doi.org/10.1007/978-3-030-87234-2_39

Wang Z, Wu J, Cai L, Low CH, Yang X, Li Q, et al. (2025b). MedAgent-Pro: Towards Evidence-based Multi-modal Medical Diagnosis via Reasoning Agentic Workflow. arXiv (preprint, under review). https://arxiv.org/abs/2503.18968

Weishaupt LL, Chen C, Williamson DFK, Chen RJ, Jaume G, Ding T, et al. (2025). Evidence-based diagnostic reasoning with multi-agent copilot for human pathology. arXiv preprint. https://arxiv.org/abs/2506.20964

Widmann D, Lindsten F, Zachariah D (2019). Calibration tests in multi-class classification: A unifying framework. Advances in Neural Information Processing Systems 32 (NeurIPS 2019). https://arxiv.org/abs/1910.11385

Williams E, Niehaus S, Reinelt J, Merola A, Mihai PG, Villringer K, et al. (2021). Automatic quality control framework for more reliable integration of machine learning-based image segmentation into medical workflows. arXiv preprint. https://arxiv.org/abs/2112.03277

World Health Organization. (2021). Ethics and governance of artificial intelligence for health: WHO guidance. WHO. https://www.who.int/publications/i/item/9789240029200

Wu J, Wang Z, al. MH, et al. (2025). Medical SAM adapter: Adapting segment anything model for medical image segmentation. Medical Image Analysis. https://doi.org/10.1016/j.media.2025.103547

Wu L, Xu Z, Wang H, Zhou F, Deng W, Zhang C, et al. (2026a). Dual- versus Single-Suggestion AI Support for Radiographic Interpretation in Residents: Randomized Multireader Study. arXiv preprint. https://arxiv.org/abs/2610.09589

Wu Q, Bansal G, Zhang J, Wu Y, Li B, Zhu E, et al. (2023). AutoGen: Enabling Next-Gen LLM Applications via Multi-Agent Conversation. arXiv 2308.08155 (COLM 2024). https://arxiv.org/abs/2308.08155

Wu X, Zhang H, Garduno-Rapp NE, Rousseau JF, Thakkallapally M, Ji Y, et al. (2026b). Orchestrator multi-agent clinical decision support system for secondary headache diagnosis in primary care. Journal of the American Medical Informatics Association (JAMIA), ocag111 (preprint arXiv:2512.04207). https://doi.org/10.1093/jamia/ocag111

Xi Z, Chen W, al. XHG, et al. (2025). The rise and potential of large language model based agents: a survey. Science China Information Sciences. https://doi.org/10.1007/s11432-024-4222-0

Xing Z, Yu L, Wan L, Han T, Zhu L (2022). NestedFormer: Nested Modality-Aware Transformer for Brain Tumor Segmentation. MICCAI 2022 (LNCS). https://doi.org/10.1007/978-3-031-16443-9_14

Xu L, He X, Zhou J, Zhang J, Mao X, Ye G, et al. (2021). Artificial intelligence-assisted colonoscopy: A prospective, multicenter, randomized controlled trial of polyp detection. Cancer medicine. https://doi.org/10.1002/cam4.4261

Xu Y, Guo W, Wei Z (2025). Selective Conformal Risk Control. arXiv preprint. https://arxiv.org/abs/2512.12844

Yan W, Liu H, Wu T, Chen Q, Wang W, Chai H, et al. (2025). ClinicalLab: Aligning Agents for Multi-Departmental Clinical Diagnostics in the Real World. NeurIPS 2025 (preprint arXiv:2406.13890v2, 9 Oct 2024). https://doi.org/10.52202/085713-2458

Yang J, Zhou K, al. YL, et al. (2024). Generalized Out-of-Distribution Detection: A Survey. International Journal of Computer Vision. https://doi.org/10.1007/s11263-024-02117-4

Yao S, Zhao J, Yu D, Du N, Shafran I, Narasimhan K, et al. (2023). ReAct: Synergizing Reasoning and Acting in Language Models. ICLR 2023. https://arxiv.org/abs/2210.03629

Ye J, Kalra S, Miri MS (2024). Cluster-based histopathology phenotype representation learning by self-supervised multi-class-token hierarchical ViT. Scientific reports. https://doi.org/10.1038/s41598-024-53361-0

Yevudza WE, Buckman V, Darko K, et al. (2024). Neuro-oncology access in Sub-Saharan Africa: A literature review of challenges and opportunities. Neuro-Oncology Advances. https://doi.org/10.1093/noajnl/vdae057

Yin S, Luo X, Yang Y, Shao Y, Ma L, et al. (2022). Development and validation of a deep-learning model for detecting brain metastases on 3D post-contrast MRI: a multi-center multi-reader evaluation study. Neuro-Oncology. https://doi.org/10.1093/neuonc/noac025

Yu F, Moehring A, Banerjee O, et al. (2024). Heterogeneity and predictors of the effects of AI assistance on radiologists. Nature Medicine. https://doi.org/10.1038/s41591-024-02850-w

Yu Y, Bates S, Ma Y, Jordan MI (2022). Robust Calibration with Multi-domain Temperature Scaling. arXiv preprint. https://arxiv.org/abs/2206.02757

Zanier O, Da Mutten R, Vieli M, Regli L, Serra C, Staartjes VE (2023). DeepEOR: automated perioperative volumetric assessment of variable grade gliomas using deep learning. Acta neurochirurgica. https://doi.org/10.1007/s00701-022-05446-w

Zawy Alsofy S, Nakamura M, Suleiman A, Sakellaropoulou I, Welzel Saravia H, Shalamberidze D, et al. (2021). Cerebral Anatomy Detection and Surgical Planning in Patients with Anterior Skull Base Meningiomas Using a Virtual Reality Technique. Journal of clinical medicine. https://doi.org/10.3390/jcm10040681

Zenk M, Baid U, Pati S, et al. (2025). Towards fair decentralized benchmarking of healthcare AI algorithms with the Federated Tumor Segmentation (FeTS) challenge. Nature Communications. https://doi.org/10.1038/s41467-025-60466-1

Zenk M, Zimmerer D, Isensee F, Traub J, Norajitra T, Jäger PF, et al. (2024). Comparative Benchmarking of Failure Detection Methods in Medical Image Segmentation: Unveiling the Role of Confidence Aggregation. arXiv preprint. https://arxiv.org/abs/2406.03323

Zhang L, Wei Y, Fu Y, Price S, Schonlieb CB, Li C (2022a). Mutual Contrastive Low-rank Learning to Disentangle Whole Slide Image Representations for Glioma Grading. arXiv. https://arxiv.org/abs/2203.04013

Zhang X, Zhao Z, Wang R, Chen H, Zheng X, Liu L, et al. (2024a). A multicenter proof-of-concept study on deep learning-based intraoperative discrimination of primary central nervous system lymphoma. Nature communications. https://doi.org/10.1038/s41467-024-48171-x

Zhang Y, He N, Yang J, et al. (2022b). mmFormer: Multimodal Medical Transformer for Incomplete Multimodal Learning of Brain Tumor Segmentation. MICCAI 2022 (LNCS). https://doi.org/10.1007/978-3-031-16443-9_11

Zhang Y, Jiao R, Liao Q, Li D, Zhang J (2021). Uncertainty-Guided Mutual Consistency Learning for Semi-Supervised Medical Image Segmentation. arXiv preprint. https://arxiv.org/abs/2112.02508

Zhang Y, Qi Y, Qi X, Senhadji L, Wei Y, Chen F, et al. (2023a). FedSODA: Federated Cross-assessment and Dynamic Aggregation for Histopathology Segmentation. arXiv preprint. https://arxiv.org/abs/2312.12824

Zhang Y, Wang S, Zhang Y, Chen DZ (2023b). RR-CP: Reliable-Region-Based Conformal Prediction for Trustworthy Medical Image Classification. UNSURE Workshop at MICCAI 2023 (Lecture Notes in Computer Science). https://doi.org/10.1007/978-3-031-44336-7_2

Zhang Z, Ai W, Wells K, Rosewarne D, Do T, Carneiro G (2024b). Learning to Complement and to Defer to Multiple Users. arXiv preprint. https://arxiv.org/abs/2407.07003

Zhang Z, Nguyen C, Wells K, Do T, Rosewarne D, Carneiro G (2024c). Coverage-Constrained Human-AI Cooperation with Multiple Experts. arXiv preprint. https://arxiv.org/abs/2411.11976

Zhang Z, Peng L, Dou W, Sun C, Aktas HE, Bejar AM, et al. (2025). Rethink Domain Generalization in Heterogeneous Sequence MRI Segmentation. arXiv preprint. https://arxiv.org/abs/2507.23110

Zhao J, Lin SY, Attias R, et al. (2025). Uncertainty-aware ensemble of foundation models differentiates glioblastoma from its mimics. Nature Communications. https://doi.org/10.1038/s41467-025-64249-6

Zhao Z, Jiang Y, Bales C, Wang Y, Fischer G (2024). Development of Advanced FEM Simulation Technology for Pre-Operative Surgical Planning. arXiv preprint. https://arxiv.org/abs/2409.03990

Zheng H, Shi Z, Yi P (2025). MedCoAct: Confidence-Aware Multi-Agent Collaboration for Complete Clinical Decision. arXiv preprint. https://arxiv.org/abs/2510.10461

Zhou T, Canu S, Vera P, Ruan S (2021). Latent Correlation Representation Learning for Brain Tumor Segmentation With Missing MRI Modalities. IEEE Transactions on Image Processing. https://doi.org/10.1109/TIP.2021.3070752

Zhou Z, Luo G, Chen M, Weng Z, Zhu Y (2025). Federated Learning for Medical Image Classification: A Comprehensive Benchmark. arXiv preprint. https://arxiv.org/abs/2504.05238

Zhu F, Cheng Z, Zhang X, Liu C (2022). Rethinking Confidence Calibration for Failure Prediction. ECCV 2022. https://doi.org/10.1007/978-3-031-19806-9_30

Zhu Y, Gu L, Wang Z, Sang H, Sui D, Tang W, et al. (2025a). Auditing medical multi-agent AI reveals risks of false consensus. arXiv preprint. https://arxiv.org/abs/2510.10185

Zhu Y, He Z, Hu H, Zheng X, Zhang X, Wang Z, et al. (2025b). MedAgentBoard: Benchmarking Multi-Agent Collaboration with Conventional Methods for Diverse Medical Tasks. arXiv preprint. https://arxiv.org/abs/2505.12371

Zhu Y, Yin X, Liew AW, Tian H (2024). Privacy-Preserving in Medical Image Analysis: A Review of Methods and Applications. arXiv preprint. https://arxiv.org/abs/2412.03924

Zhylka A, Sollmann N, Kofler F, Radwan A, De Luca A, Gempt J, et al. (2021). Tracking the Corticospinal Tract in Patients With High-Grade Glioma: Clinical Evaluation of Multi-Level Fiber Tracking and Comparison to Conventional Deterministic Approaches. Frontiers in oncology. https://doi.org/10.3389/fonc.2021.761169

Zuo M, Xing X, Zheng L, Wang H, Yuan Y, Chen S, et al. (2025). Weakly supervised deep learning-based classification for histopathology of gliomas: a single center experience. Scientific reports. https://doi.org/10.1038/s41598-024-84238-x